

First Look at E-mu's Proteus Synth Expander ■ Public Domain Software for the ST

Electronic Musician

U.S. \$2.95/Canada \$3.95
October 1989

Guitar Effects

- *Automating with MIDI*
- *Multi-Effects Roundup*

**How New Copyright Law
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**Review of the Roland
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*1989 Music & Sound Awards

**Keyboard Magazine, December, 1988.

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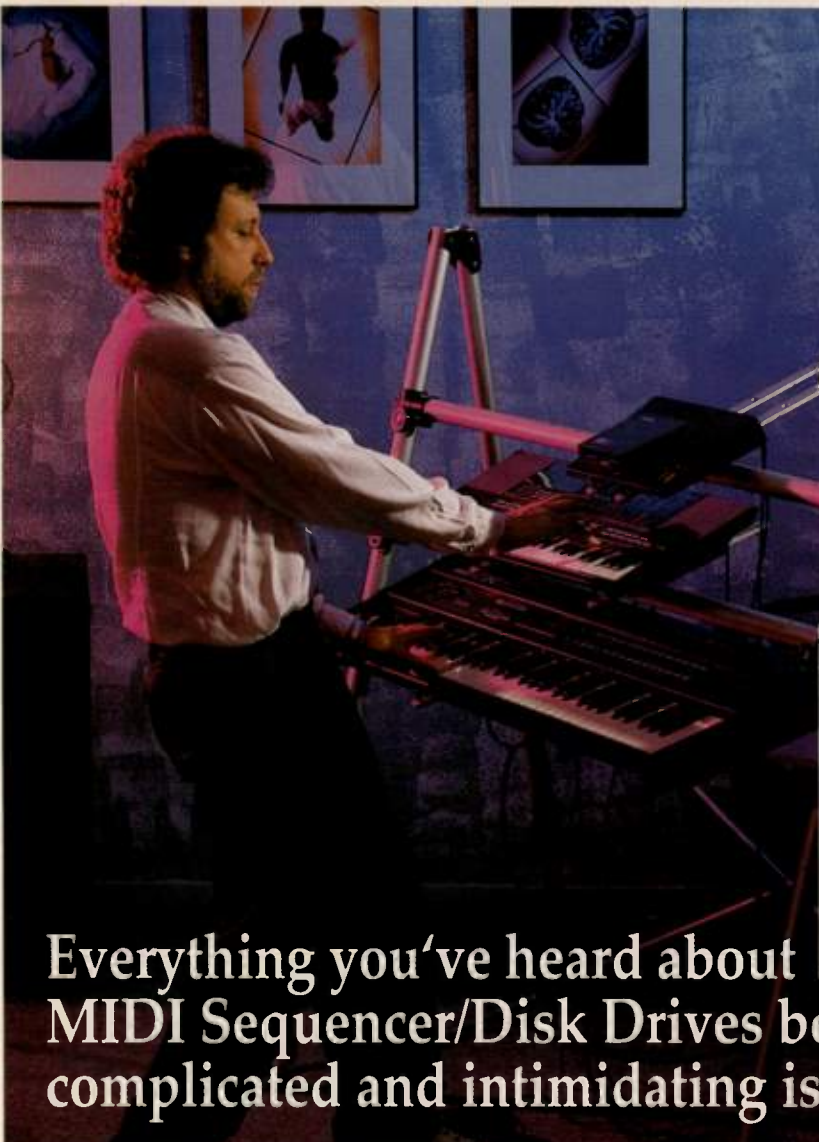
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WHAT'S NEW THIS MONTH

Flash! It appears a compromise has been reached between DAT manufacturers and the RIAA that could lead to widespread DAT product introduction by early 1990. DAT machines will be able to record CDs, but it will not be possible to make copies of tapes containing material recorded from CDs. This should not cause problems for musicians using DAT on their studio projects. We'll present more information as it becomes available.

ABOUT EM (Electronic Musician):

Since its inception in 1975 under the name *Polyphony*, EM has been a communications medium for sharing ideas, circuits, tips, and other information and is dedicated to improving the state of the musical art.

SUBSCRIPTION SERVICES:

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This mail-order distribution service (a.k.a. Mix Bookshelf) offers products (books, instructional tapes, music software, etc.) oriented toward our readership. For a free catalog, contact: EM Bookshelf, 6400 Hollis St. #12, Emeryville, CA 94608; tel. (415) 653-3307 or (800) 233-9604.

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Single/back issue price is \$3.50. For a listing of published articles, send a SASE (self-addressed, stamped envelope) to our Emeryville, CA, address and request a "Back Issue Listing."

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Occasional errors are unavoidable. We list known errors in "Letters." We compile published corrections annually for those who order back issues; to receive a copy, send a SASE to "Error Log Listing" at our Emeryville, CA, address.

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To have events (seminars, concerts, contests, etc.) listed, send dates and times *three months* prior to the event deadline to "EM Calendar Listing" at our Emeryville, CA, address.

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Manufacturers: Send press releases to our Emeryville, CA, address, attention: What's New. A release must be received three months prior to the cover date to be included in that issue. Regarding reviews, there are more products than pages available to review them. We welcome unsolicited software, books, etc., for review on a space-available basis; contact the editorial staff regarding hardware reviews.

Readers: Unless otherwise noted, EM reviews production versions of hardware/software (there are no "reviews" written from press releases). We encourage readers to scan "What's New" for new product announcements and contact the manufacturer for more

information. Note: Manufacturers constantly update products, and prices and specifications stated in EM are subject to change. EM does not make product recommendations. Reviews represent only the opinion of the author.

LETTERS:

We welcome opposing viewpoints, compliments, and constructive criticism and will consider these for publication unless requested otherwise (we reserve the right to edit them for space or clarity). All letters become the property of EM. Neither the staff nor authors have the time to respond to *all* letters, but all are read. If you are having problems with your gear, please call the manufacturer, *not* us.

PROBLEMS WITH ADVERTISED PRODUCTS:

Information in ads is the responsibility of the advertiser. EM does not have the resources to check the integrity of every advertiser. However, we try to monitor ads and ensure that our readers get fair and honest treatment. If you encounter problems with an advertiser, let us know by writing to our Emeryville, CA, address. Tell us the problem and what steps you have taken to resolve it.

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DO-IT-YOURSELF (DIY) PROJECTS:

If you do not know basic electronic construction techniques, get a good book on the subject from your local electronics supply store (or try *Electronic Projects for Musicians*, available from EM Bookshelf). EM specifies parts values following international protocol, thus minimizing the use of decimal points and zeroes. A nanofarad (nF) = 1,000 pF or 0.001 μ F. Suffixes replace decimal points. Examples: 2.2k Ω (U.S. nomenclature) = 2k2 (Intl. nomenclature). 4.7 μ F (U.S.) = 4 μ 7 (Intl.). 0.0056 μ F (U.S.) = 5n6 (Intl.).

If you detect an error in a schematic or listing, let us know. If a project doesn't work for you, contact us to see if anyone has reported any errors (wait at least a month for EM to be in circulation).

HELP US HELP YOU:

Please reference EM when asking manufacturers for product information, returning warranty cards, etc. Advertising provides our financial base, and ad purchases are based on your feedback to manufacturers about which magazines you like.

To the best of our knowledge, the information contained herein is correct. However, Electronic Musician, its owners, editors, and authors cannot be held responsible for the use of the information in this magazine or any damages that may result.

S A N S U I

How the world's first 6-track cassette system becomes the world's only 10-track system...

WS-X1 — the first all-in-one 6-track studio...

The Sansui WS-X1 is more than a multi-track cassette recorder — it's a complete 6-track studio with mixing, eq, effects and mastering all in one. The WS-X1 has a pro-featured 8 channel mixer, balanced inputs, built-in digital signal processing plus an on-board 2-track mixdown deck.

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...and, how to build a 10-track from scratch

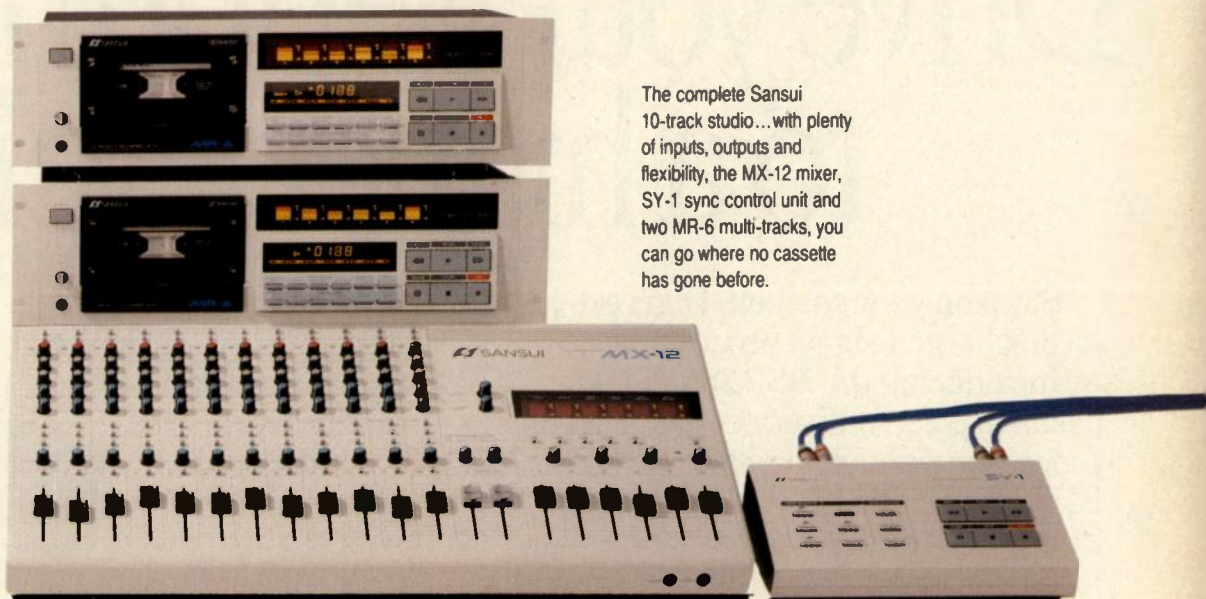
SY-1 — the 10-track sync link...

An MR-6 and a WS-X1 add up to a 10-track system when you use the SY-1 Sync Control Unit. Using one track of each 6-track deck, the SY-1 locks both machines together so you can record on up to 10 tracks simultaneously. The SY-1 also features a MIDI to FSK converter, so you can sync your entire studio to a sequencer or drum machine. And there's more.

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10



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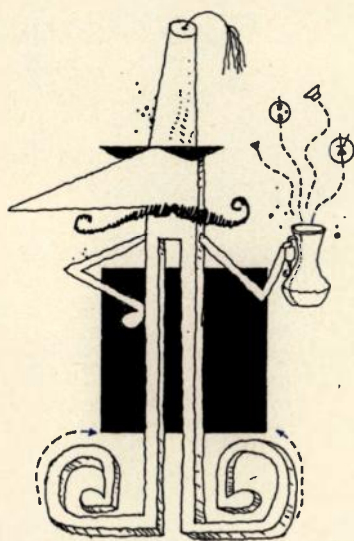
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Comments on Carlberg from Cooper, support for Southworth, and seekers of the upgrade path are part of this month's eclectic mix.



CRITIQUING THE CRITIC

As Robert Carlberg pointed out in witty fashion ("Be A Reviewer," July 1989 *EM*), any album, such as my *Soft Wave* debut album on Sona Gaia, can be reviewed from dozens of different approaches and biases, both positive and negative. Artists, of course, are likely to respond to reviews in similarly diverse fashion. For example:

(The Offended Artiste Response)

You vituperative, infantile, unschooled, prattling, uncultured buffoon. How dare you draw breath from the same planet I occupy? "Aimless tootling by some guy with a horn," is it? Your egregious misrepresentation of great art shamelessly exposes your lugubrious deficiencies in musical discernment. Rarely have I witnessed such brutal dismemberment of both syntax and logic beneath a single byline. Never again, I pray, will my musical pearls be cast before such unskilled, critical swine. For the benefit of humanity, I recommend that you and your ilk be neutered immediately.

(The Technical/Corrective Response)

Thank you for your recent review of my album, *Soft Wave*. Your detailed survey of the equipment I employed, however, did exclude a few pivotal items: Tracks 3, 6, and 14b(2) also include a prototype of the Sonic Steroid Aural Enhancer, a Korg/Gatorade joint development project. The uncredited performance was handled by Lugar Axhandle during the Red Shift Effect, 1989, at Bug-Eyed Studios in Clandestine, California, USA.

(The Kill the Messenger/Bobby Knight Response)

You slime bucket. You dirt bag. What in the #\$\$%& do you know about music, you airbrain? You don't even know how to blow the horn on your car, let alone mouth off about my #\$\$@##@ing great record, you nose-picking lump of walrus pus. So listen, dogface, the rest of us learned to write by age six, then we grew up and moved on to something useful. It's about time you get those training wheels off your brain and get a real job. You'd make a great grenade-tester at Bombs 'R Us.

(The Jessie Allen Cooper Response)

While I recognize Robert Carlberg's column was intended to be a parody of the reviewing process, I do appreciate the 3 1/2-star plug he gave to *Soft Wave*, plus numerous glowing comments: "It happened with Kenny G., and it has happened again with Jessie Cooper's debut"; "mastery of the tenor and soprano saxophones"; "high-tech outing in a sparkling new age setting"; "combining the sensuous new age sensitivity with jazz idealism"; and "a unified effort of uncommon beauty." Hey, I'm not making this up.

As an artist, it can be a frustrating experience to see months of intense studio work be casually dismissed in a critic's negative review. Of course, I am learning not to overreact to reviews, either positive or negative. As Robert's piece made clear, there are many, many ways to review music. Fortunately, there are also many ways to make music, and I'm pleased Robert found ample merit in *Soft Wave* to say plenty of positive things. So keep this in mind, folks: next time you visit a record store, remember that Robert Carlberg's words about *Soft Wave* by Jessie Allen Cooper are, "A must buy."

Jessie Allen Cooper
California

SOUTHWORTH MAY HAVE GONE SOUTH, BUT...

I have been handling tech support for Southworth Music Systems since February. The company has since closed up shop, but there are plenty of JamBox 4 and 4+ owners who will need tech support. I am licensing the sale of parts and such from the company and would simply like to keep the ball rolling.

JamBox owners who would like more information on parts, software, or service should contact me at 430 Uniontown Rd., Westminster, MD 21157; or call (301) 876-2678, 10 to 4 Eastern time.

Glenn Workman
Maryland

DAT'S THE TICKET

In response to Daniel Kumin's two-part series on DAT (April and May 1989 *EM*), I'd like to compliment Daniel and *EM* for publishing the first realistic and factual account of the



Rock at its hardest. The Roxy, L.A.

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• LETTERS

DAT scene. Mr. Kumin obviously did his homework and research and had the background in audio to assimilate the information available to him, producing two very informative articles.

I have specialized in DAT for the last eighteen months and have worked for two "gray market" importers. His warning regarding some gray market vendors is excellent advice. However, your readers should be aware that legitimate gray operations will and do provide as good (and in some instances, better) support than you would get from a USA service station (e.g., a free loaner when service is required).

To take issue on a couple points: the Sony DTC-1000ES home deck (which has been discontinued and will be replaced later this year), although the first on the market, does not have optical digital I/O, nor does it have extensive programming; but it does let you record (when modified) at 44.1 kHz. The DTC-500ES does have optical I/O and programming, but cannot record at 44.1 kHz. Finally, the Pioneer D-900 has been available here since November 1988. It is in fact very different from the Sony DTC-1000ES—not only in quality (audio and construction), but also in features.

**Scott Ikier
California**

THINKING ABOUT SYNCHING

About David Humrick's query in the July 1989 *EM*: his problems with FSK tape sync relate to the tape deck. The problem is crosstalk at the record head and is inherent in narrow-gauge multitracks. The simple solution for a 4-track: record the sync tone on tracks one and four on your first pass. Then sync to track four when you record on track two, and sync to track one when you record on three. Finally, record over one of your sync tracks, and you're ready to mix.

**Peter Elsea
California**

MIDI ROUTING SOLUTION REVEALED

Regarding Alan Campbell's "Service Clinic" (June 1989), there are at least two more reasonably priced solutions to the HP100/MT32 problem. Roland has offered the MPU-103 for many years, currently at \$360 retail.

Also, the Pelican Systems MIDI Processor (\$149.95 retail), is a half-rack box that can change, block, and pass any or all channels on two MIDI lines at once. It also provides filters for specific MIDI events and can merge two inputs.

**Tom White, President
Future Music (Reno)
Nevada**

THE FINAL WORD ON FINALE (?)

I've re-read Mr. Lehrman's review of *Finale* (July 1989) several times, and while I congratulate him on a fair and accurate review, what sticks in my mind are the words "ripoff," "flawed," "clumsy," "absurd," and "programmer's disease," not that he *did* find some "cleverness" and "brilliant features" in *Finale*.

This scares me because you guys have clout. *EM* has become the bible for a small, fickle industry where products come and go like "hit" singles, and this review may influence some potential users to give up on *Finale* before even giving it a chance.

I evaluate Macintosh software and hardware for a living and see hundreds of new products each year. As first generation packages go, I am very impressed with *Finale*. Yes, it does have problems, but of all the Mac products I've handled, the folks at Coda have demonstrated the highest commitment to keeping *Finale* an evolutionary product, with plans for improvements based on suggestions from end users. (Since we received this letter, Coda has lowered the price and made several improvements to the program—Ed.)

Finale is not for everyone, and this program reaffirms the importance of dealing with only the best service-oriented music and computer dealers who will take the time to qualify a user's needs and not push something like *Finale* onto someone whose needs are best satisfied by *Deluxe Music Construction Set*. Find a dealer who lives in the real world and work with that dealer to make the most of your hardware/software configuration.

I just started to like computers when I saw my first Macintosh in 1985. The early music programs were toys but held a lot of hope for the future. I find it incredible that in this very short time, in an industry that is still in its infancy, something with the power of *Finale* even exists! I have the feeling that ten years

Steve Miklas
Pittsburgh Com-
puter Store
Pennsylvania

OPERATION HELP

Address all Operation Help requests to "Operation Help," c/o Electronic Musician. We will print your address and phone number (if supplied).

ERROR LOG/UPDATES

regret any problems this may have caused to Alesis and our readers; the toll-free line is now available.

"The EM MIDI Channel Filter," October 1988: Terminal 4 of IC3B connects only to pin 1 of IC8 and should not connect to the junction of R2 and Y1. The oscillator circuit in Tim Dowty's "MIDI Channelizer" (May 1989) is correct. Thanks to reader James R. Davey for bringing this to our attention.

"The EM DX/TX Patch Librarian,"
April 1989: Change line 99 to:

POKE53281,0:POKE53280,0:POKE646,8

Also, in lines 260 and 270, change 19 to 18 if your main menu box has a hole in it. Thanks to Donnie Ellison for these suggestions.

"Live Film Sync Without SMPTE," August 1989: On page 69, paragraph 1, the second sentence should read, "You can even set up fractional tempos: 481, 482, 483...489 clocks/beat gives in-between tempos, in 1/4-beat/minute increments."

"Build A Fiber-Optic Guitar Cable,"
February 1989: Radio Shack no longer carries fiber optic components, including the hard-to-find Motorola MFOE71 and MFO72 emitter/detector pair for which parts numbers were quoted. However, the parts and cable are now available from Fiber Link Systems, a manufacturer of fiber optic MIDI systems. Fiber Link Systems, 2063 Ponderosa Dr., Livermore, CA 94550; tel. (415) 455-4457.

"Random1: A Patch Generator/Librarian for the Korg M1," August 1989: Unfortunately, LDW BASIC is no longer available. *Random1* author David Snow believes it may be possible to use GFA BASIC (available from Antic Software, 544 2nd St., San Francisco, CA 94107), although this may require some code modifications. **EM** readers can purchase the compiled Random1 program and source code from Mr. Snow for \$6 postpaid (Maryland residents add 5% sales tax), at 9824 Maple Leaf Dr., Gaithersburg, MD 20879.

"What's New: New Gear from the Summer Music Expo," September 1989: The correct phone number for Cannon Research, manufacturers of the Frontal Lobe and the PCM Channel, is (916) 272-8692.

Equipment credits for photographs, September 1989: Thanks to David Holman for supplying the Minimoog and Prophet-5, and to Marcus Ryle for the Oberheim SEM and ARP 2600. ■



HitMan is a film scoring tool bringing full SMPTE event editing to the ST for the first time. HitMan is based around a SMPTE event editor, that specifies SMPTE times for each event in a cuelist. A sound effects macro editor lets you assign names to specific MIDI events and times, for easy inclusion of sound effects in the list. A complete tempo map is provided, with multiple time signatures and MPE hooks to our KCS sequencer.

Special features include ability to lock to SMPTE and tap in "hit points", an analyzer to search a range of tempos for the best match to a set of specified hits, tap-tempo for real-time tempo entry, variable SMPTE offset, and tempo calculator for precise timing of musical interludes between cue points.

Disk File Output		Disk DAS		Time		Time	
Disk File Output		Disk DAS		Time		Time	
0	00121517.14 Lim on	1	1.00	0	1.00	175	175
1	00121518.14 Lim on	2	0.63	0	0.63	175	175
2	00121519.14 Lim on	3	0.63	0	0.63	175	175
3	00121520.14 Lim on	4	0.63	0	0.63	175	175
4	00121521.14 Lim on	5	0.63	0	0.63	175	175
5	00121522.14 Lim on	6	0.63	0	0.63	175	175
6	00121523.14 Lim on	7	0.63	0	0.63	175	175
7	00121524.14 Lim on	8	0.63	0	0.63	175	175
8	00121525.14 Lim on	9	0.63	0	0.63	175	175
9	00121526.14 Lim on	10	0.63	0	0.63	175	175
10	00121527.14 Lim on	11	0.63	0	0.63	175	175
11	00121528.14 Lim on	12	0.63	0	0.63	175	175
12	00121529.14 Lim on	13	0.63	0	0.63	175	175
13	00121530.14 Lim on	14	0.63	0	0.63	175	175
14	00121531.14 Lim on	15	0.63	0	0.63	175	175
15	00121532.14 Lim on	16	0.63	0	0.63	175	175
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17	00121534.14 Lim on	18	0.63	0	0.63	175	175
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24	00121541.14 Lim on	25	0.63	0	0.63	175	175
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26	00121543.14 Lim on	27	0.63	0	0.63	175	175
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28	00121545.14 Lim on	29	0.63	0	0.63	175	175
29	00121546.14 Lim on	30	0.63	0	0.63	175	175
30	00121547.14 Lim on	31	0.63	0	0.63	175	175
31	00121548.14 Lim on	32	0.63	0	0.63	175	175
32	00121549.14 Lim on	33	0.63	0	0.63	175	175
33	00121550.14 Lim on	34	0.63	0	0.63	175	175
34	00121551.14 Lim on	35	0.63	0	0.63	175	175
35	00121552.14 Lim on	36	0.63	0	0.63	175	175
36	00121553.14 Lim on	37	0.63	0	0.63	175	175
37	00121554.14 Lim on	38	0.63	0	0.63	175	175
38	00121555.14 Lim on	39	0.63	0	0.63	175	175
39	00121556.14 Lim on	40	0.63	0	0.63	175	175
40	00121557.14 Lim on	41	0.63	0	0.63	175	175
41	00121558.14 Lim on	42	0.63	0	0.63	175	175
42	00121559.14 Lim on	43	0.63	0	0.63	175	175
43	00121560.14 Lim on	44	0.63	0	0.63	175	175
44	00121561.14 Lim on	45	0.63	0	0.63	175	175
45	00121562.14 Lim on	46	0.63	0	0.63	175	175

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RolandCorp US
7200 Dominion Cir.
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PERIPHERALS

The Atari ST's MIDI port contains both thru and out signals in the same jack, causing inadvertent combination of the signals with some types of cables and necessitating an additional thru box to use MIDI thru. Practical Solutions' MIDI Thru/Out Adapter cable (\$14.95) provides separate, standard, female MIDI thru and out connectors.

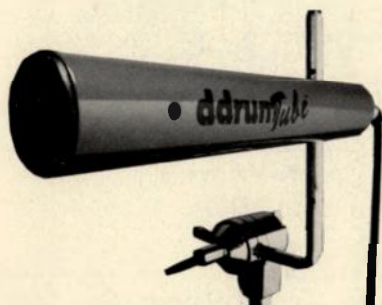
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1135 N. Jones Blvd.
Tucson, AZ 85716
tel. (602) 322-6100

Antex Electronics introduces the Series 2/Model SX-10 (\$1,995) stereo digital audio processor board for IBM PC/AT, PS/2, Model 25/30 computers and compatibles. The processor can receive both analog and digital data (including audio), store it on hard disk, and manipulate the digitized data. Video can be displayed in the foreground while audio playback occurs in the background. The SX-10 offers up to a 50 kHz sampling rate with 16-bit resolution.

Antex Electronics, Inc.
16100 S. Figueroa St.
Gardena, CA 90248
tel. (213) 532-3092

Atari Computer introduced the Megafire 44 (under \$1,200; additional cartridges, \$150) removable-cartridge hard disk for ST and MEGA computers. The 8 x 5 x 1.6-inch drive offers 25 ms access time, and the 4-inch, 44 MB cartridge features a new graphite overcoating to prevent media contamination, allow dense data-packing, and reduce the risk of head crashes.

Atari Computer
1196 Borregas Ave.
Sunnyvale, CA 94088
tel. (408) 745-2000



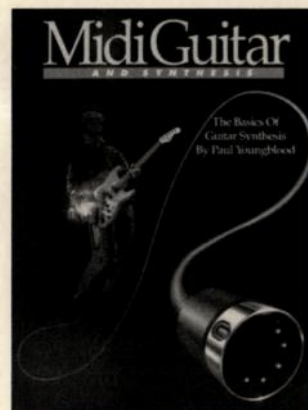
ddrum The Tube

Current Music Technology is now shipping the Mac'n Rak Plus (\$3,495; retrofit your Plus, \$1,995), a 4U rack-mount unit that consists of a Macintosh Plus; a 20 MB hard disk; Opcode Studio Plus MIDI interface; and an optional Opcode Timecode Machine. The Mac'n Rak SE (\$4,595; retrofit, \$2,295), with two floppy disk drives and a 40 MB hard drive, and a single-floppy SE30 (retrofit, est. \$2,000) version are also shipping. Custom configurations are available.

Current Music Technology
146 Paoli Pike
Malvern, PA 19355
tel. (215) 647-9426

PUBLICATIONS

Hal Leonard Publishing has released *MIDI Guitar and Synthesis* (\$14.95) by Paul Youngblood. Topics include comparisons of various types of guitar converters, how synths and samplers work, phrasing and articulation for



Midi Guitar and Synthesis

certain sounds, how to buy equipment, connecting and troubleshooting MIDI systems, and more. Available from EM Bookshelf; see FYI page for more information.

Hal Leonard Publishing Corp.
7777 West Bluemound Rd.
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1-10FFTK Econo-Tier Kit 45° angle and mounting hardware.
- Individual parts can be ordered for custom displays.

STANDTASTIC

1-800-876-6923
1325 Meridian St.
Anderson, IN 46016

T A S T I C

WHAT'S NEW SIGNAL PROCESSORS

The Rocktron Allan Holdsworth Juice Extractor (\$439) is a 1U rack-mount multiple processor that includes 3-band parametric EQ, Rocktron's HUSH II single-ended noise reduction, reactive load emulator, three filter curve responses (footswitchable), and six outputs. The unit can pad up to a 100-watt amplifier output for use with line-level processing.

Rocktron Corp.
2870 Technology Dr.
Rochester Hills, MI 48309
tel. (313) 853-3055

Scholz R&D's Rockman XPR (\$849), the rack-mount, upgraded version of the XP 100, sans speakers and amp, is a MIDI-programmable multi-processor and guitar preamp with 100 user-programmable settings. The XPR offers six discrete distortion circuits, 5-band output EQ with separate 3-band pre-distortion EQ, compressor, simultaneous stereo chorus and stereo echo or reverb, an effects loop, and more.

Scholz Research & Development
1560 Trapelo Rd.
Waltham, MA 02154
tel. (617) 890-5211

SOFTWARE

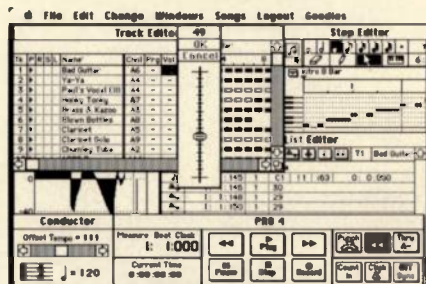
Magnetic Music introduced Prism (\$99), a graphics-based, mouse/menu-driven 16-track sequencer for IBM, C1, and compatibles. The program features Texture file compatibility, a custom windowing environment, and Graphic Mapper, which allows the user to draw a graph and hear the shape applied to any type of music data.

Magnetic Music
6 Twin Rocks Rd.
Brookfield, CT 06804-1910
tel. (203) 775-7832

Passport has announced Pro 4 (\$495), a sequencer for the Macintosh that provides an integrated track editor

for recording and playback, a song playlist for live performance, complete graphic editing, event list-based editing, a global change filter, graphic faders, and the ability to sync directly to SMPTE via MIDI time code with Passport's MIDI Transport.

Passport Designs, Inc.
625 Miramontes St.
Half Moon Bay, CA 94019
tel. (415) 726-0280



Passport Pro 4

Hybrid Arts introduced GenEdit, a universal patch editor/librarian for the Atari ST (\$249) and the Macintosh (\$349). GenEdit utilizes sysex commands and a Template Editor that lets the user create a virtual control panel for any MIDI instrument, including drum machines, effects devices, etc. Twelve preset templates are included.

Hybrid Arts, Inc.
8522 National Blvd.
Los Angeles, CA 90232
tel. (213) 841-0340

SOUNDS

Sweetwater Sound introduced the SW800 Soundblock (\$995) for the Kurzweil K250, Expander, and RMX. The SW800, which consists of ten ICs that install like the factory soundblocks, adds 55 new keyboard setups. Featured sounds include Harmonica, Glockenspiel, Wurlitzer Electric Piano, Orchestral Hits, L/A-type sounds, and more.

Sweetwater Sound, Inc.
4821 Bass Rd.
Ft. Wayne, IN 46808
tel. (219) 432-8176



Rocktron Allan Holdsworth Juice Extractor

The reason to buy an Otari is because it's *your* music.

If you have a personal studio, Otari lets you choose exactly the right audio machine for *your* kind of music. Because no one audio machine can be everything to everybody, we build a complete

quarter-inch 2-track mastering machine with 3 speeds and 10.5" reels at a price that will astound you!

Then there's the multitracks, from the supremely affordable half-inch 8-track MKIII-8 with both 7.5 and 15 ips speeds, to the two-inch 32-track MX-80 with its full-function remote controller, 30/15/7.5 ips speeds, spot erase, and gapless, seamless, punch-in/out.

The MX-80, (and its little brother, the one-inch, 16-track MX-70) share a constant-tension, servo-controlled transport, plus

the Otari 37-pin standard parallel interface for SMPTE synchronizers, editors, and controllers.

The MX-70 multitrack and the MX-55N 2-track. High performance and enhanced features to fit your creative energies.

We're convinced that somewhere in this line of great sounding audio machines there's one that will fit your music to a note.

Call us at (415) 341-5900 for complete information on any one, or all. Better yet, call your nearest Otari dealer for a demo!

The 8-track MKIII-8 and the MX-50 2-track. Professional sound and features at an affordable price.

line of superb machines that, alone or working together, will fit your application perfectly.

For example, our MX-55NM quarter-inch 2-track delivers gapless, seamless, punch-in, punch-out for flawless edits, and HX-Pro* for increased headroom.

And if you're on a *really* tight budget, check out the new MX-50, a

Otari's MX-80 multitrack and the new MX-55NM 2-track. Alone or together, the ultimate choice for your personal studio.

OTARI

Great sound inspires great music. Once again, Ensoniq helps you capture that moment of inspiration.

The new VFX^{SD} combines the heralded VFX sound, dynamic effects processing and a powerful new 24-track sequencer in one keyboard.

The all-new 24-track sequencer has 60 sequence/song structure and 25,000 note capacity standard, expandable to 75,000 notes. A variety of recording modes and editing features guarantee you'll be comfortable making music with the VFX^{SD}.

Our new synth offers the bold sound first introduced in the VFX, the keyboard that redefined state-of-the-art synthesis. The 24-bit effects are dynamically integrated to become a

true part of the sound. And exclusive Ensoniq features like Patch Selects and Poly-Key[™] Pressure let you add expression to your music.

The VFX^{SD} is 21-note polyphonic, and has four stereo outputs and an onboard disk drive to store sounds, sequences, even MIDI Sys. Ex. information. It features many new drum and percussion sounds while remaining fully compatible with the VFX sound library.

Rich, animated sounds combine with a 24-track sequencer to give you the only complete music production synthesizer, the Ensoniq VFX^{SD}.



THREE MORE SOUND REASONS TO BUY AN ENSONIQ.

The Ensoniq VFX

If you already have a sequencer, the VFX is the ideal sound source and MIDI master controller. It offers a wide range of unique sounds and expressive controllers. As *Electronic Musician* put it, "...this is a Synthesizer, with a capital S."

The Ensoniq EPS

If you need a composing environment based on the sonic versatility of a sampler, look into the EPS. Optional memory expansion, multiple outputs and SCSI compatibility allow you to configure the EPS to your needs and budget. "The price is right, the features are phenomenal, and sampling is made about as easy as anyone can make it..." said *Stage & Studio*.



INTRODUCING THE ENSONIQ VFX^{SD}, THE FIRST MUSIC PRODUCTION SYNTHESIZER.



The Ensoniq EPS-M

If you want all the power of the EPS in a rack-mount module, choose the EPS-M. The expansion options of the EPS come as standard equipment in the EPS-M, "a serious professional piece of gear," according to *Sound On Sound*.

Whether you need a synthesizer or sampler, look to the award-winning American

company that makes technology more musical—Ensoniq. Call 1-800-553-5151 for the dealer nearest you.

Yes, I want more information about the following:
(Check as many as you like.)

☐ VFX^{SD} ☐ VFX ☐ EPS ☐ EPS-M

Name _____

Address _____

City _____ State _____ Zip _____

Phone Number (____) _____

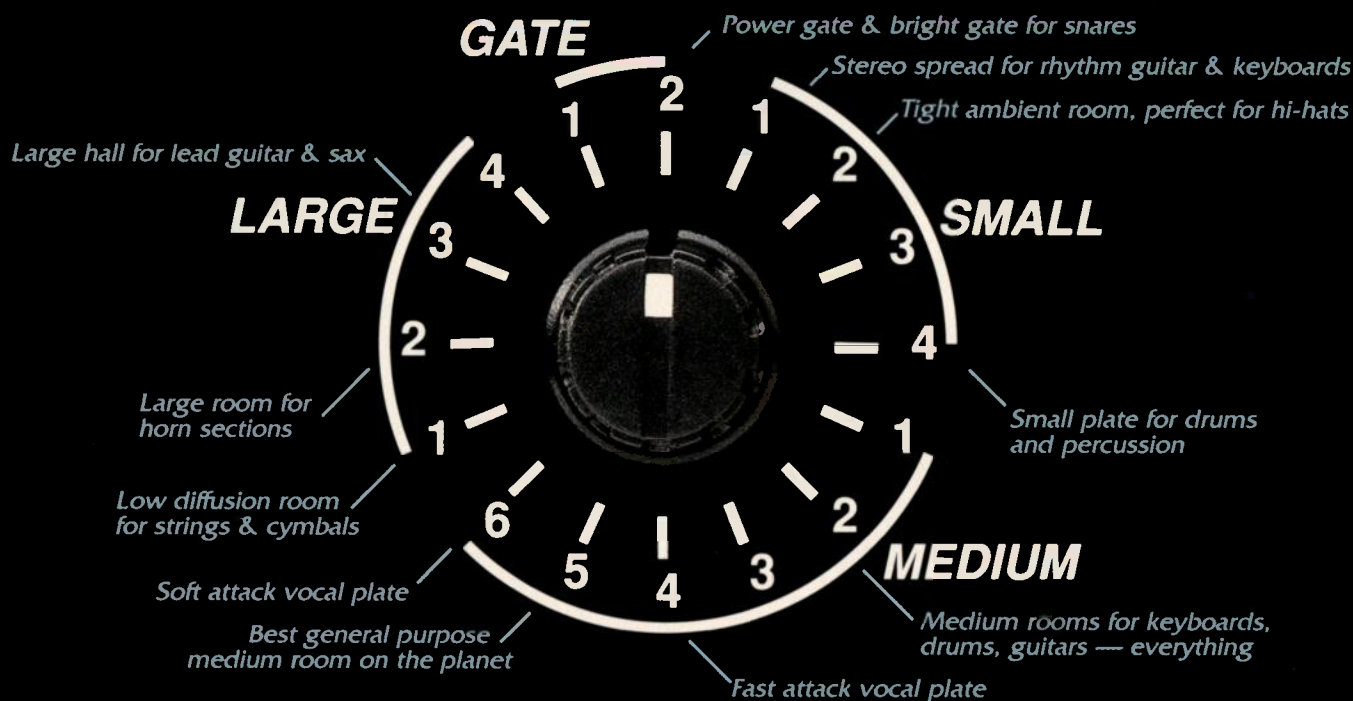
Mail To: Ensoniq
155 Great Valley Parkway
Malvern, PA 19355



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"Professional reverb, \$200... Unbelievable!"



That's what professional studio owners, record producers and experienced engineers are saying about the Alesis MICROVERB® II Digital Reverb. They're used to spending \$2000 to \$10,000 on a digital reverb. But it's not expensive reverb they're after...it's great *sounding* reverb. Deep, rich, dense. Full of character. The sound that MICROVERB® II delivers in dead quiet 16 bit stereo.

One simple knob chooses MICROVERB® II's 16 classic reverb programs. Each one carefully crafted. Each one worth the full price alone. This is not only one of the greatest all purpose reverbs ever made, it's also the greatest value. Simple, direct, timeless. MICROVERB® II is a front line, master quality digital processor that will dramatically improve the sound of your music.

Don't be fooled. There's plenty of cheap reverbs out there. But they *sound* cheap. Don't trash your music and waste your money

with a wimpyverb, or try to make do with a captured reverb in a keyboard, when you can have MICROVERB® II. Use it on your next demo tape. Wrap it around your next hit record. Isn't it great when price isn't an issue for the very best?



MICROVERB® II is the first and last word in professional reverb. Believe it.

We can prove it. Call 1-800-5-ALESIS toll free, and we'll send you "The MICROVERB® II Demo Tape" so you can hear a classic professional reverb in action.

**Then see your
Alesis dealer.**



• WHAT'S NEW



Tannoy NFM-8 Near Field Monitor

SPEAKERS

The Tannoy NFM-8 (DMT) near-field monitor (\$998/pair) is the only primary mixing reference speaker of its (diminutive) size that boasts an 8-inch Dual Concentric driver in a ported, mediate enclosure, offering solid bass response. Frequency response is said to be flat (± 3 dB) up to 25 kHz.

Tannoy North America Inc.
300 Gage Ave., Unit 1
Kitchener, Ontario
N2M 2C8 Canada
tel. (519) 745-1158

STANDS

The MIDIdesk (\$799), from Standtastic, is a stand for the home studio, designed to be attractive as well as efficient. It can hold rack-mount units, keyboards, computers, drum machines, etc. Its removable rack-mount bays enable the user to quickly remove rack gear for transporting. A cord management system and a locking door assembly are included.

Standtastic
1325 Meridian St.
Anderson, IN 46016
tel. (317) 642-5205

TOOLS

SGW presents the Solderstat (\$8, plus \$2 s/h), a wrist-strap solder dis-



SGW Solderstat

Some things just make you want to SCREAM!



The Multi-function All-tube **SOUND MACHINE!**

- All-tube (tube preamp *and* tube power amp!)
- Overdrive from subtle to screaming saturation!
- Maximum overdrive **even from single coils!**
- Speaker Out: plug straight into a guitar cabinet, and wail!
- Built-in power soak and Cabinetulator Out
- Captures the **SOUND** of a hot-rodded classic British tube-amp stack right onto tape, or direct into a PA, without the hassles of miking
- **Screams** with creamy sustain **even at low volume**
- **Transforms** every existing guitar amp into an ultra-high-gain rig; for example, turns a sterile transistor amp into a Rock Monster
- Fits in your gig bag, only 9½" wide; rack-mount kit available
- Comes fully loaded with the world's best tubes. *Groove Tubes™*

DOUBLE STACK IN HALF A RACK

\$349 suggested retail

For more information, contact your local dealer or

Hughes & Kettner

Hughes & Kettner, Inc. Dept. 8M
35 Summit Avenue, Chadds Ford, PA 19317
Tel: (215) 558-0345 FAX: (215) 558-0342



● WHAT'S NEW

penser that attaches to any elastic or velcro static-control wrist strap and dispenses up to ¼ lb. of solder directly from the wrist. An elastic strap is included (a static-control elastic strap or velcro strap is optional).

SGW Co.

6414 Hallee Rd.

Joshua Tree, CA 92252

tel. (800) 537-1535 (in USA)

or (619) 366-2661

REV UP

Dr. T's Music Software (tel. [617] 244-6954) released V. 1.44 of *Keys!* for the ST. The company also is shipping V. 1.7 of *KCS* and *Level II* for the ST, adding full GEM compatibility, additional editing and composition features, a clock resolution of 1,536 ppm, and support for up to eight MPE modules...**Voyetra Technologies** (tel. [914] 738-4500) released V. 3.0 of *Sequencer Plus* for the IBM, integrating an enhanced sequencer with *Patch Master Plus* universal librarian/MIDI data analyzer, and *Conversion Plus*,

a *Sequencer Plus-to-Personal Composer* song file converter. Voyetra has also slashed the program's price to \$349...**Twelve Tone Systems** (tel. [617] 273-4437) released V. 2.1 of *Sound Globes* for the IBM. The new version requires 30 KB less memory than previous versions and adds several improvements, including a better conversion utility for file transfer to *Cakewalk* and better bend and mod

major new features, including three new modules (Time Compressor, Noise Oscillator, and Pitch Envelope). The price remains \$349, and upgrades are available for \$35...**Passport Designs** (tel. [415] 726-0280) announced *NoteWriter II* for the Mac. New features include an expandable, user-configurable symbol library, use of the numeric keypad for note entry, and multiple-page scores.



Akai S1000HD

functions. The company also announced a free *Cakewalk*-to-standard MIDI file conversion utility and certified their *Cakewalk 2.0* and *Cakewalk Professional 2.1* for OS/2 compatibility...**Digidesign** (tel. [415] 327-8811) released *Turbosynth V. 2.0* for the Mac, with ten

Upgrades are \$99...The **Akai** (tel. [817] 336-5114) S1000/S1000HD adds SCSI capabilities with the V.1.3 operating system. The S1000 requires installation of the IB103 SCSI interface card (\$159.95), which comes standard on the S1000HD. Version 1.3PB is a similar update for the



S1000PB, which also requires the IB103 interface card...Tascam (tel. [213] 388-5200) has added a third sampling time, 44.1 kHz, to its DA-50 DAT recorder...Optical Media International (tel. [408] 395-4332) announced an upgrade, at just above cost (\$1,300) plus trade-in, to registered CDS3 owners who own an Emulator III. The new Pro CDP Universal Model 422/SCSI rack-mount drive allows use of the new "Master Studio Collection" and "Digital Sound Series" sound disks...Interval Music Systems (formerly Drumware, Inc.; tel. [213] 478-3956) has released *GenWave/16*, an upgrade



Voyetra Sequencer Plus Version 3.0

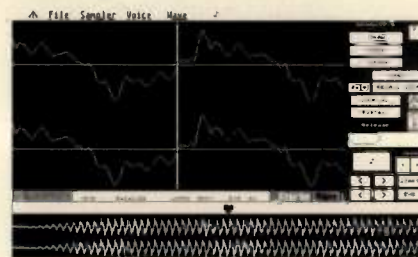
from its Gen/Wave 12 waveform editor for the Atari ST. The new version supports 16-bit, stereo samplers.

CALENDAR OF EVENTS

September 23 Electronic Music Symposium at the University of California-Irvine. Several well-known lecturers will contribute, including *EM* editor-in-chief Craig Anderton, and an evening concert will feature well-known artists. Admission is \$75. University of California-Irvine Extension, Electronic Music Dept.; tel. (714) 856-5414.

September 25 MIDIcourse 101. Course about the MIDI specification and applications. Ten one-hour sessions, \$95. The Ivory Palace, 205 S. State St., Ukiah, CA 95453; tel. (707) 462-8863. Also offered, starting September 27, at Computech, Lakeport, CA; tel. (707) 263-5504.

October 2-6 Sony Professional Audio Training Group course: Sony MXP-2000 Technical Service. Contact Raymond Callahan or James Gayoso; tel. (305) 491-0825, ext. 186.



Interval Music Systems GenWave/16

October 18-21 Audio Engineering Society 87th Convention, New York City. AES, 60 E. 42nd St., New York, NY 10165; tel. (212) 661-8528.

October 19 *Mix* magazine's fifth annual Technical Excellence & Creativity (TEC) Awards, honoring the people, products, and institutions that have made outstanding contributions in professional audio. Manhattan Center Studios, 311 W. 34th Street, New York City. Contact Hillel Resner or Karen Dunn; tel. (415) 653-3307.

If you have an event you want listed, see the FYI page for more information. ■

YOU CAN JUDGE A KEYBOARD MIXER BY THE NUMBER OF PLUGS IT RECEIVES.

Twenty eight. Count 'em, twenty eight different inputs into the MM-1 keyboard mixer from Tascam.

That includes 12 mono channels, 4 stereo channels, 4 effect sends and 4 stereo returns. You're not going to run out of channels any time soon with the MM-1.

Once you've got the input inside, the rack-mountable MM-1 gives you full EQ on all channels. That's the kind of fine tuning control that other keyboard

mixers try to tell you isn't necessary. Probably because they don't have it.

In addition, the MM-1 can memorize up to 100 channel-muting settings, or "scenes," which are then MIDI-addressable for instant recall right from your keyboard.

See the rugged, compact MM-1 at your Tascam dealer.

And then take a look at the incredible suggested retail price of \$1095.

At that price the only question is, when will you plug in the MM-1?



TASCAM

In the Public Eye: Free Atari Software

Your phone and computer can access a wealth of free public domain programs that will turbocharge your Atari ST—and here are some of the very best.

By Jim Johnson

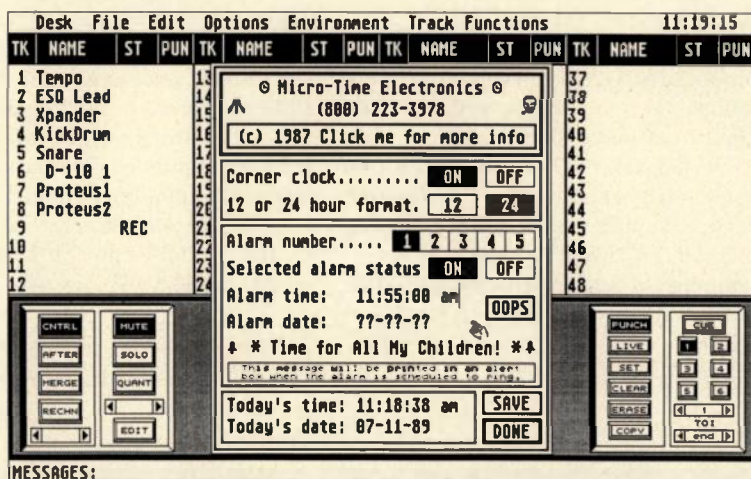


FIG. 1: ALARMCLK.ACC is a combination clock/alarm clock with custom alarm messages.

My first articles about becoming an ST power user appeared last year, and since then, several things have happened. Of course, the Earth has moved a good portion of the distance around the sun, but more importantly, the ST has, too. Software and hardware designers have unleashed a plethora of new goodies to our MIDI studios, with the expected results—the ST is currently a strong contender, if not the strongest contender, for the title of “most powerful MIDI computer.”

Although one of my previous articles covered hot public domain (PD) programs available from telecommunications networks, plenty of new and useful programs for the ST have shown up on numerous bulletin boards since then. I've spent probably more time than I should have downloading them, testing them, tossing out the bad, and keeping the good; so in this article, I'll share a few more of these goodies with EM readers, and tell you where they can be found.

UPDATES TO SOME OLD FAVORITES

George Woodside's *Turtle*, now up to Version 3.0, has several new convenience features that you'll appreciate if you've been backing up your drives as often as you should—like the ability to format disks in the middle of a backup session, or to exclude certain file types from a backup. Also included is TERRAPIN.PRG, a program for backing up and restoring large data files that won't fit on a single floppy (such as archive files and uncompressed samples).

Turtle will now format disks with an extra sector per track, for a disk capacity increase of ten percent. Although I used to distrust extended-capacity formatters, letters from readers and a bit more research have changed my mind. Increasing the number of sectors per track from nine to ten seems to be fine, but going to eleven is risky, as is extending the number of tracks to 81 or more. All of the extended disk formatters I've seen will let you select the number of sectors and tracks to format, so it's possible to

squeeze that extra storage out of your floppies with any of several programs. I use DCFMT30.PRG, which can also format disks with an MS-DOS boot sector. (Although it is possible for STs and PCs to read one another's disks, thanks to their identical disk formats, unfortunately, a PC can't read a disk that was originally formatted on an ST, because the ST's desktop formatter doesn't write an MS-DOS boot sector on the disk. DCFMT30 solves this problem.)

SuperBoot, which is currently on Version 5.1, can now program up to 30 different configurations, accessed through the function, control, and alternate keys. You can also automatically run STARTGEM.PRG from *SuperBoot*, without all the extra steps involved in setting up STARTGEM.INF files, but in my view it doesn't work the way it

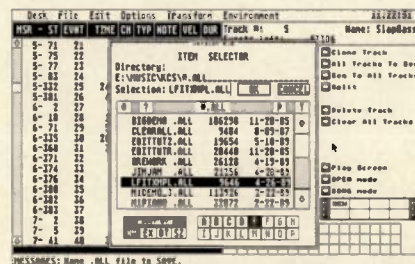


FIG. 2: The GEM file selector program FSELV60.PRG is a very useful AUTO folder program.

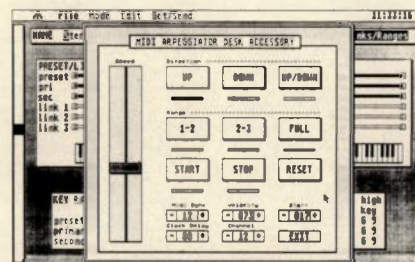


FIG. 3: Arpeggiators of the dinosaur era live again with ARPEG.ACC.

Play
with
a
Pro

Creativity is the most dynamic element in music. A truly powerful sequencer won't inhibit creativity by frustrating you with its technology.

Pro 4™ is Passport's new sequencing system for Macintosh, designed to give you innovative and advanced tools to inspire you, without puzzles to challenge your patience. It provides new ways to picture your music, without cluttering your vision.

Pro 4's main advantage over competitive sequencing software is its extreme ease of use. It commands your attention if you're looking for the next generation in MIDI sequencing.

Inspired by Master Tracks Pro™, it provides an integrated track sheet for recording and playback, note pad, song editing and a complete visual editing system. Edit multiple sequences, perform most operations in real time during playback, view your music in a variety of ways and lock everything to SMPTE.

MASTER TRACKS PRO 4

Pro 4™ is completely compatible with Encore™, our new composing system that will print your sequences in standard music notation.

For more information see a Passport dealer near you or call (415) 726-0280 for details on the Next Generation of Music Software.

PASSPORT®

625 Miramontes St.
Half Moon Bay, California 94019
(415) 726-0280



Graham Nash

Member, Rock & Roll Hall of Fame

Fostex Equipment: E-16 MTR, 4050 MIDI Remote, T-20 Headphones.

Recent Projects: Graham's recent solo album, "Innocent Eyes" (Atlantic) was recorded in Los Angeles, Hawaii and Massachusetts. Most of the compositions and arrangements were worked out at home on his E-16. The hit "Shadowland" from CSN&Y's latest album, "American Dream" (Atlantic), was recorded on the E-16.

TIMBUK3

Pat & Barbara K. MacDonald

Fostex Equipment: D-20 Digital Master Recorder, E-8 MTR, 4030/4035 Synchronizer / Controller, 4010 SMPTE Time Code Generator, RM-865 and RM-900 Near-Field Reference Monitors, T-20 Headphones.

Recent Projects: TIMBUK3's first album, "Greetings from TIMBUK3" (I.R.S.), was recorded on a B-16 (forerunner of the E-16). Their third and newest album, "Edge of Allegiance" (I.R.S.), is the first digital project for the group. They use a D-20.

Tom Scott

Musician/Composer

Fostex Equipment: E-16 MTR (2), E-2 Master Recorder, 4030/4035 Synchronizer / Controller, 4010 SMPTE Time Code Generator, T-20 Headphones.

Recent Projects: Nationally recognized band leader of "The Pat Sajak Show", Tom's latest albums are "Streamlines" and "Flashpoint" (GRP Records). He also scored the NBC TV Movie "American River", and the film "Sea of Love" starring Al Pacino features his distinctive sax.



PROJECT STUDIOS.

Throughout the audio production cycle—from composition/demo to multitrack master to digital stereo master—you'll find Fostex products in use by professional composers, arrangers, engineers and producers. You hear the results of Fostex Project Studios every day on TV, film soundtracks and, of course, radio & records. Trust Fostex for great products at great prices. Trust in yourself for the sound of success.

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Stewart Levin
Musician/Composer

Fostex Equipment: B-16 MTR, 4030/4035 Synchronizer/Controller, 4010 SMPTE Time Code Generator, T-20 Headphones.

Recent Projects: Stewart writes the music for several popular TV shows — "thirtysomething", "The Wonder Years", "The Dick Van Dyke Show '88" — and he scored the feature film "Heathers."

Top Performer®



Performer Version 3 is here!

What could we add to the program that is already the pre-eminent sequencer for professional recording?

How about graphic editing that displays note events simultaneously with *all* continuous controller data. Filters that let you view just the events you want to see. And zoom controls for precise editing.

Want to compare tracks? Simply re-size the editing display and open up more windows. Notes and phrases play when you point at them. And there's a Conductor Track for editing key signatures, tempo and meter changes.

We've also added a powerful feature that lets you conduct real-time tempo changes before, during *or* after you record. And unlike the current crop of clunky sequencers, Performer works smoothly and intuitively—something you can't tell from spec sheets.

So whether you're just getting into sequencing software or ready to trade up, check out the one that professionals depend on: Performer.

Developed for Apple Macintosh and Atari ST computers by Mark of the Unicorn, Inc. 222 Third Street, Cambridge, MA 02142 (617) 576-2760



**After using a
mouse accelerator
for about ten
seconds, I
couldn't go back.**

should, so I still do it the old way. And speaking of which, a new AUTO folder GEM boot program called *HeadStart* is a nice alternative to *StartGEM*, especially for mouse addicts. (An AUTO folder, by the way, is a place where you store programs that automatically run when the ST is first started. See "ST Power User, Part II" in the December 1988 issue of **EM** for more information.) This one lets you choose the program to boot with the mouse, or will execute a default program after five seconds.

NEW STUFF

MACCEL.PRГ, a mouse accelerator, worked its way out of Atari's software labs early this year and has been enthusiastically put to work by just about everyone who comes across it. With MACCEL, if you move the mouse quickly, the mouse pointer will fly across the screen, but if you move slowly, the mouse pointer will track the mouse in the usual precise way. After using the program for about ten seconds (literally), I was comfortable with it, and the original "unenhanced" mouse now feels like it's moving through molasses! MACCEL works with every program I've tested it with so far and has become a permanent fixture of my AUTO folder.

Also, I've finally found a combination clock/alarm clock that is absolutely fantastic—ALARMCLK.ACC (Fig.1). Not only does it update the display every second, it provides five programmable alarms, each of which also displays a custom message on the screen. Alarms can be set to occur every day, every hour, or just at specific times and dates. This is a must if you tend to lose track of time when you are making music, as I do.

One of the best AUTO folder programs I've seen so far is an enhanced GEM file selector called FSELV60.PRГ (Fig. 2). The ST's standard file selector (the little window that pops up every time you load or save a file in a GEM program) is not bad at all, but does leave

a lot to be desired. FSELV60 provides all of the features of the standard file selector, along with buttons for changing the disk drive, a directory printing function, and the ability to change any attributes of a file, including its name. It also lacks a few of the standard file selector's more annoying traits—for example, it lets you change the extension in the path, without automatically showing you all the files in the current directory. This is mundane stuff, yet the program is definitely worth having around.

The most exciting program I've found is a desk accessory, *DC Stuffer*, that allows you to load up to 32 different desk accessories, all of which are instantly available from the desk menu!

The PD version of *DC Stuffer* only allows DAs to be loaded when booting up, but the shareware version (which must be ordered directly from the authors) allows DAs to be loaded and deleted at will anytime during a session, without rebooting the computer, so you don't need to waste a lot of precious memory on programs that you only use occasionally.

This DA works well in conjunction with SuperBoot, so there's no need to load every DA you own each time you boot the machine. *DC Stuffer* is the final element needed to turn the ST into the astounding productivity tool we all knew it could be. If you use DAs at all, this program is an absolute must.

FOR THE BEGINNER

Telecommunications Basics

By translating computer data into audio tones, computers can communicate with other computers over the phone. The translating device, called a *modem*, connects between your computer and the phone line. You also need *telecommunications software* so that your computer can properly control and interact with the modem.

Getting online (the process of hooking up your computer to a computer at the other end of the phone network) is initially very intimidating—and don't let anyone tell you otherwise. Modems send and receive data at various speeds, using various transmission/reception protocols, and these must all be set correctly in the telecommunications software to ensure proper data transfer. To make life even more difficult, many telecommunications programs are more complex than most people need, and lack documentation designed for the neophyte. However, the silver lining is that once you start to figure out the telecommunications process, everything falls into place. For best results, get some tutelage from an experienced telecommunicator and learn by doing.

Once you're online, you can do something as simple as transfer files to a friend who lives across town, or access a

database across the country. Although the latter might seem like an invitation to bank-breaking long distance bills, many telecommunications services (e.g., CompuServe, PAN, Esi, Delphi, etc.) use networks (e.g., Tymnet, Telenet) that provide local access numbers in most major metropolitan areas. Cost is still a factor, though. In addition to a per-minute rate charged by the services themselves, there will also be a per-minute access charge for the network. All in all, an hour online can cost \$10 or more (or less, depending on the service, time of day, and many other factors). It can easily cost you \$50 to \$100 of connect time to explore and learn about a system; however, some services offer a couple of hours of free connect time upon signing up to help you along the (sometimes very steep) learning curve.

Still, you get what you pay for, and you get a lot: free programs like the ones Jim describes; forums on subjects from MIDI music to science fiction; patches for your synths, and samples for your samplers; informational databases and online encyclopedias; and travel information (you can even make your own hotel and plane reservations). For more on telecommunications, see the September 1986 issue of **EM**.

—Craig Anderton

Where to Find the Programs

The programs mentioned in this article can all be downloaded from GEnie, General Electric's telecom network. The program locations on GEnie are listed below. Many of these can also be found on CompuServe, or any local or national ST bulletin board. Because most of these programs are shareware, I've also listed each author's name and address and the requested contribution for each program; if you don't have a modem, you can order them directly from their respective programmers. (You should also include a formatted, blank disk with your order.) All of the programs listed below have an .ARC extension, which means they have been archived, or compressed, into a smaller file. They must be "de-archived" by a program such as ARC.TTP (which is

available from numerous sources, including bulletin boards) before they will run.

- **ALARM.ARC** Alarm clock desk accessory. File 8555, Atari library, or send \$5 to Micro-Time Electronics, 837 NE Sixth St., Grants Pass, OR 97526.
- **ARPEG.ARC** MIDI Arpeggiator. File 9614, Atari library, or send \$15 to Richard Homme, Phonix Productions, 83 Holyrood Ave., Oakville, Ontario, L6K 2V4 Canada.
- **HEADST10.ARC** HeadStart auto-boot program. File 9528, Atari library, or send \$10 to CodeHead Software, Box 4336, No. Hollywood, CA 91607.
- **DCFRT301.ARC** Extended disk formatter. File 9380, Atari library, or send \$10 to Double Click Software, 10101-O Blisssonnet, Suite 1181, Houston, TX 77036.

- **DCSTUFR.ARC** Desk accessory loader. File 8306, Atari library, or send \$15 to Double Click Software, same address as above.
- **FSELV60.ARC** Enhanced file selector. File 7075, Atari library, or send \$10 to Gene Sothan, 9395 North Wall, Portland, OR 97203.
- **MACCEL.ARC** Mouse accelerator. File 9278, Atari library.
- **SUPERBT5.ARC** SuperBoot version 5.0. File 8796, Atari library, or send \$15 to Gordon W. Moore, 2300 Beach St., Ashland, KY 41101.
- **SB_PATCH.ARC** SuperBoot 5.1 upgrade. File 8887, Atari library.
- **TMONEY.ARC** Time Is Money calculator DA. File 660, MIDI library.
- **TURTLE.ARC** Turtle 3.0 hard disk backup. File 9397, Atari library, or send \$15 to George R. Woodside, 5219 San Feleciano Dr., Woodland Hills, CA 91364.

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AND MIDI, TOO!

Lest anyone think that the only freebie software to be found is disk formatters and other boring utilities, here are two little desk accessories that were written for the same reason that we bought the ST—to help make music. ARPEG.ACC (Fig. 3) is a keyboard-controlled MIDI arpeggiator that cycles through notes held down on your controller in various ways: up, down, up and down, over several ranges, etc. It's a fairly stripped-down little program, and I have some very definite complaints about it (e.g., it stops playing whenever a button is clicked). Nonetheless, if you miss the kind of arpeggiators that were once found on instruments like the Korg Polysix and Roland Jupiter-6, and you have an external sequencer to capture the MIDI output, it's definitely worth the download time.

Another, more serious program is a French DA called *Time Is Money* (Fig. 4). This hip little tool calculates times from musical quantities like tempo, note type, and song length. Changing any value on the screen will update any other related

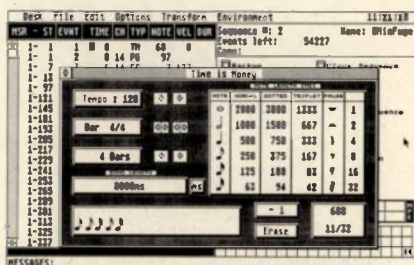


FIG. 4: The desk accessory *Time Is Money* calculates times from musical quantities like tempo, note type, and song length.

value—for example, if you change the tempo, all of the note times will change, etc. This is especially handy if you have non-MIDI programmable digital delays and find yourself faced with a fifteen-minute session with the calculator each time you want to change the tempo of a song. I haven't been able to figure out all of the features of this program—the document file is in French, which I don't speak—but most of it is pretty easy to use, even without instructions.

And so ends another look at tools that can help make you an ST power user. The number of good shareware and freeware programs floating around out

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there can only be counted as a blessing, yet I'm beginning to see it as a bit of a curse, because the type of power that these programs afford almost demands that I increase my ST's memory beyond 1 meg! And to think I used to be satisfied with a paltry 64K...

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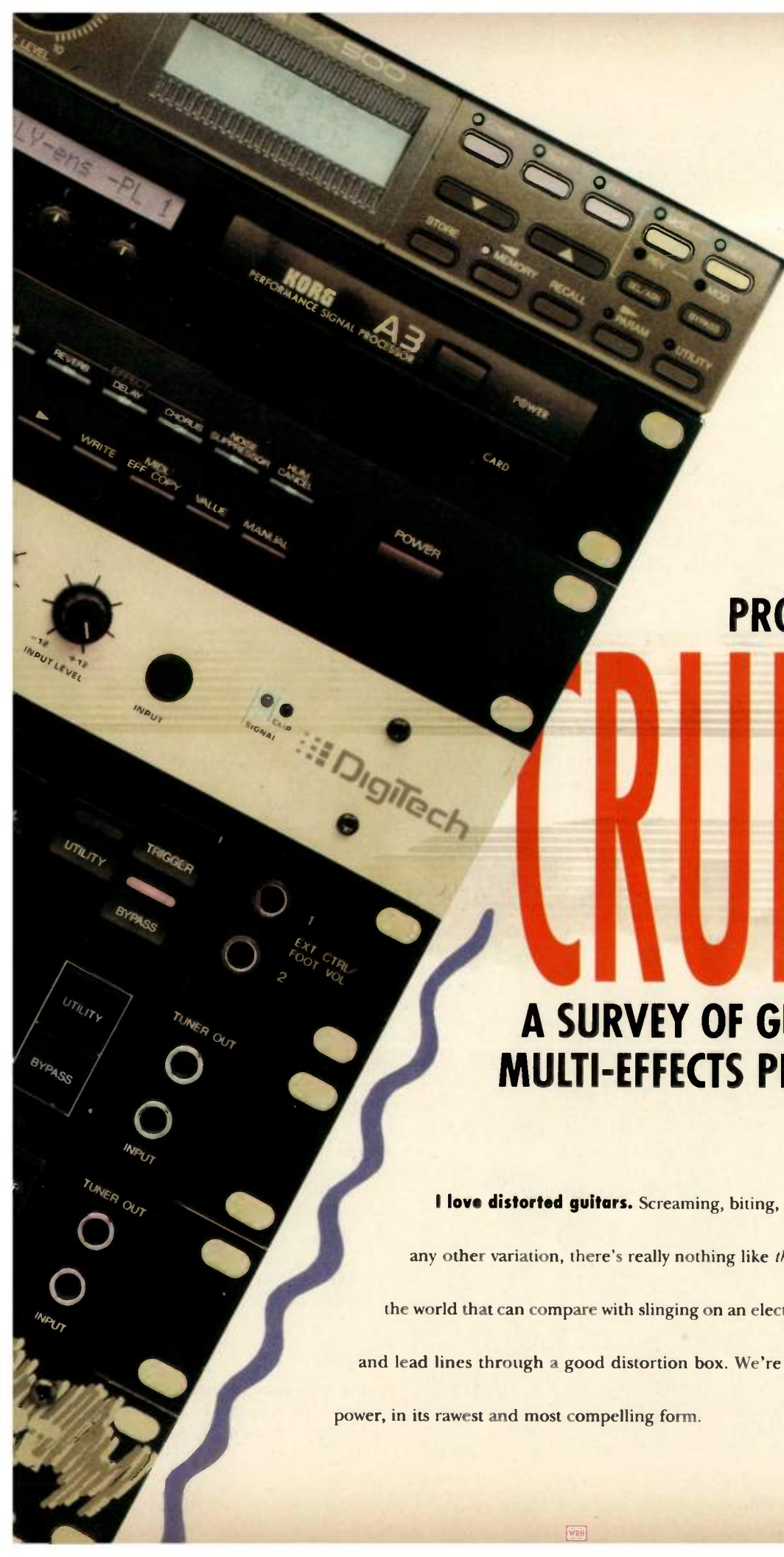
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BY BOB O'DONNELL

● GUITAR PROCESSORS

The art and science of creating those good distortion sounds has progressed rapidly in recent years. The order of the day for guitar-oriented signal processors is instant digital distortion (though some companies defiantly swear by analog distortion), combined with numerous other effects. At last, the guitar now has the kind of sonic flexibility—and instant gratification—formerly associated with synthesizers. Thankfully, many manufacturers with strong signal processing credentials have turned their attention to today's guitarist. ART's SGE,

DigiTech's GSP-5, Korg's A3, Roland's GS-6 and Boss ME-5, and Yamaha's FX500, GEP50, SPX50D and SPX900 are all currently available, MIDI-controllable, multi-effects boxes that include distortion, and at least two others are on their way from Roland (the GP-16) and Scholz R&D (the XPR; see "What's New" in this issue). It's a tantalizing, sophisticated bunch, with high-quality sounds and numerous features, including the very expressive option of real-time control over the sound.

If you're somewhat new to the field,

however, it can also be a confusing choice. So, I'll explain what you need to know when choosing a guitar processor, include a chart showing the differences between the existing units, and provide a quick lowdown on their respective sound quality. (For a few tips on how you can use the MIDI ports on these magic boxes to give you more immediate and expressive control over your sound, see Craig Anderton's companion piece on automating guitar effects.)

CONCEPTUALLY SPEAKING

Most modern guitar sounds rely on a combination of different effects—just turning distortion on and off is not going to give you the big, fat, bright, active sounds you hear on records. Instead, you'll need to combine several different effects into one complete timbre, a task for which multi-effects boxes were expressly created.

The most common additions to distortion are compression, delay, chorus/flange, EQ (equalization), and reverb. Not surprisingly, nearly all the processors included in the survey include these basics, as well as a few other effects that can have a powerful effect on the sound (such as a pitch shifter or harmonic exciter; see the sidebar for details on how different effects alter a signal). Obviously, one factor when choosing a guitar processor is to find out what types of effects it offers—bearing in mind that you may use it for purposes other than guitar, such as vocals or drums.

However, just because a processor has all the effects you want doesn't mean it can produce them all at once; that depends on the maximum number of simultaneous effects it can create and the particular combinations it allows. For example, even though the Yamaha SPX900 can perform pitch shifting effects, it cannot do so at the same time it produces distortion.

The reason for this and similar limitations in other boxes is that each unit has a finite amount of processing power (yes, there's a computer inside these boxes, too), which is generally less than would be required to run all of its effects at once. Also, certain effects require a

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Effects shown in previous page (top to bottom): Yamaha FX500, Korg A3, Roland GS-6, DigiTech GSP-5, Yamaha SPX900, Yamaha GEP50, Yamaha SPX50D, ART SGE.

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• GUITAR PROCESSORS

FOR THE BEGINNER Why Rack-Mounts?

To get the sound they want, guitarists have turned to a number of different products over the years. Footpedal effects ("stomp boxes") have been a mainstay for decades, and these mighty midgits have improved tremendously over the last few years, offering better sound quality and more programmability than ever before. If you're looking for a cost-effective means of getting good distortion sounds, these are definitely the way to go.

The next wave of units, the famous Scholz Rockman and its derivations, offer improved sound quality and more options in a slightly larger package. These are still in wide use and are particularly useful for practicing (several can also drive headphones directly).

As guitarists have become more sophisticated, though, these formats have begun to be eclipsed by rack-mount signal processors. Rack-mountable units (which are designed in 19-inch-wide boxes to fit into racks specially designed to hold them, hence their name) usually offer the benefits of higher sound quality, more extensive programming options, more extensive control options (such as MIDI), memory for storing different sounds (or patches), durability, and more convenient packaging, albeit at a higher price. (The exception in this article is the Boss ME-5, which combines rack-mount sound quality in a floor-pedal package—kind of like a stomp box on steroids.)

The latest generation of rack-mount units can function as multi-effects processors, allowing them to recreate many different types of effects, often several at a time. Buying one of these saves you from having to buy multiple individual effects and concern yourself with the problems of interconnecting patch cords, plugging in AC adapters, or replacing batteries. With rack-mount multi-effects units, everything can be controlled from a single source.

lot more processing power than others, so manufacturers make decisions based on what they think will be the most useful combinations.

Directly related to this is the order in which these effects can be placed. Though individual effects may be a pain to work with when trying to create a system, one advantage is that you can hook them up in any combination and in any order. Unfortunately, the same is not true of most multi-effects boxes (although the as-yet-unreleased Roland GP-16 reputedly offers this capability). What they offer instead are preset algorithms, or chains, which offer a certain combination of effects in a useful, "normal" order. In many instances, these algorithms may prove to be more than you'll ever need, but if you want to do something unusual this may not be possible.

One potential solution is to use additional effects in conjunction with your guitar multi-effect of choice (if you have a few favorite existing processors, you may want to do that anyway). With the help of a feature called a "programmable effects loop," found on a few of the multi-effects boxes, you can even automate these external effects, whether they are MIDI-equipped or not. The loop automatically switches the connected effect in or out for each program (different sound), depending on how you set it up. The only boxes that currently offer this function are Yamaha's very similar GEP50 and SPX50D and the Boss ME-5, but this feature is supposed to be included on Scholz' XPR and XPR100 (which is the same as the XPR but with a built-in 50 watt/side power amp and speakers).

CHARTING NEW TERRITORIES

Now that we've covered the basic principles, let's look at the table accompanying this article. Along the top is a row of available products, and along the left, a column of important features. The first row lists the type of sound generating method used to create distortion. Note that only the ART SGE and the Boss ME-5, two interesting combinations of analog and digital processing, use analog distortion (Scholz's XPR, which is entirely analog, also uses this technique).

The next row displays the number of distortion settings available and whether or not separate equalization (independent of any other EQ effects) is included

GUITAR-ORIENTED MULTI-EFFECTS

Products	ART SGE	Boss ME-5	DigiTech GSP-5	Korg A3	Roland GS-6	Yamaha FX500	Yamaha GEP50/SPX50D	Yamaha SPX900
Type of Distortion	Analog	Analog	Digital	Digital	Digital	Digital	Digital	Digital
# of Distortion Algorithms (Additional EQ?)	12 (No)	3 (No)	3 (No)	4 (Yes)	8 (No)	1 (Yes)	1 (Yes)	1 (Yes)
Compression	Yes	Yes	No	Yes	No	Yes	Yes	Yes
Delay (Max. time in milli-sec's)	Yes (500)	Yes (500)	Yes (1800)	Yes (800)	Yes (999)	Yes (740)	Yes (500)	Yes (1480)
Mod. Delay (Chorus, Flange, etc.)	Yes	Yes (Analog)	Yes	Yes	Yes	Yes	Yes	Yes
Exciter	Yes	No	Yes	Yes	No	No	No	Yes
EQ Type	3-band Fixed	3-band; Sweep-Mid	2-band Semi-Para., or 5-band Fixed	2-band Para., or 3-band Semi-Para.	3-band; Sweep-Mid	3-band Fixed; Sweep-Mid.	2-band Para. plus Shelf	2-band Para. or Dyn. Shelf
Reverb	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Noise Suppression	Yes, Expander/Gate	Yes	No	Yes	Yes	Yes, Dist. Gate	Yes, Dist. Gate	Yes, Dist. Gate
Pitch Shifting	Yes (2 Octaves)	No	No	Yes (2 Semi-tones)	No	No	Yes (2 Octaves)	Yes (4 Octaves)
Other Effects	Panner, Wah	No	No	Speaker Simul., Rotary Speaker, Panner, Noise Gate, Wah *	No	No	Noise Gate, Panner	Noise Gate, Panner, Wah, Freeze (Sampling)
# of Simultaneous Effects	9	5	5	6	5	6	2	5
# of Memory Locations (RAM)	200 (200)	64 (64)	99 (99)	100 (100), Additional 100 w/ RAM or ROM card	64 (64)	90 (30)	99 (49)	99 (49)
# of Preset Order Combinations	6	2	13	20*	4	4	3	4
Limitations Imposed by Multi-effects	Yes	Yes	Yes	Yes	No	No	Yes	Yes
Responds to MIDI Continuous Controllers?	Yes (8)	No	Yes (58)	No	No	Yes (2)	No	Yes (2)
Other Real-Time Control?	Yes, MIDI Note #	Yes, Volume w/ EV-5 or EV-10	No	Yes, Volume & one preset parameter (w/ KVP-001s)	Yes, Volume w/ FC-100 Mk. II & EV-5 or EV-10	Yes, Delay Time w/ Footswitch	Yes, MIDI Note # & Key On	Yes, MIDI Note # & Key On; Any two parameters (w/FC7s)
Options or Accessories?	N/A	EV-5 (\$79); EV-10 (\$140)	FS 300 (incl.); PDS 3500 MIDI Pedal (\$199); FX 17 (\$129)	FC6 Foot Controller (\$225); KVP-001 (\$60); SPC01, 02 and 03 ROM Cards (\$70 each)	FC-100 Mk. II Foot Controller (\$350); EV-5 (\$79); EV-10 (\$140)	MFC1 MIDI Foot Controller (\$310)	MFC1 (\$310)	RCX1 Remote (\$295); MFC1 (\$310); FC7 Foot pedal (\$45)
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*Additional effects and algorithms may be added with ROM cards

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• GUITAR PROCESSORS

within the distortion effect program. Nearly all the processors have drive controls that control how much distortion actually occurs, so even though there may be only one basic setting, you can produce a wide degree of variations with all the boxes. The addition of dedicated distortion EQ allows for even more variety, because the tonal colors created by equalization have an enormous effect on distortion settings.

The next four rows give simple yes/no answers concerning the availability of effects, with a few details thrown in for good measure. All the units covered in the survey have some type of EQ program, so the EQ Type row offers more detail on the choices. One box that deserves special mention in this regard is the Yamaha SPX900, which offers 2-

band parametric EQ on every program. As you can see, all the boxes offer digital reverb (though some have more reverb programs and parameters than others).

Distortion effects are notoriously noisy, but fortunately, most units offer some sort of noise reduction, ranging from the Yamaha units' simple noise gate within the distortion effect to Roland's very sophisticated and very successful digital noise suppressor and hum canceler on the GS-6. However, when working with any gate-type noise reduction, the threshold level must be set carefully, particularly if you're also using a compressor. With too high a threshold, low-level sounds (such as long sustained notes) can annoyingly disappear and reappear as both effects try to work at the same time.

FOR THE BEGINNER

What Do the Effects Do?

Here's a quick explanation of how the various effects alter an input signal:

Compressor: Continuously adjusts the gain, or volume, of a sound to maintain a fairly consistent level. As a signal gets weaker, a compressor will automatically raise the level; with stronger signals, the compressor lowers the level. The sonic result is that compressed signals have a more limited dynamic range, giving more punch and sustain.

Delay: Adds a series of discrete echoes to a sound, each of which occurs a programmed amount of time after the previous echo (or in the case of the first echo, after the onset of the signal). Most units let you set independent delay times for the right and left channel for stereo effects.

Equalizer (EQ): Sophisticated tone controls that affect the color of a sound. Fixed equalizers let you adjust the level of preset frequencies, while sweepable or semi-parametric types permit you to adjust the frequencies as well as the levels.

Exciter or Enhancer: Brightens a signal, in theory without the stridency associated with simply boosting treble EQ. The method involves emphasizing high har-

monics, yielding a richer sound.

Modulated Delay: Adds life and movement to a sound by combining the direct signal with a delayed signal whose delay time is constantly changing (i.e., being modulated). Chorus, flange, ensemble, and symphonic effects all fit into this category.

Noise Gate: Turns a sound off after its level passes below a low, programmed volume level, or threshold. It is often used to cut out low-level noise caused by distortion.

Pitch Shifting: Alters the pitch of a signal by a programmed amount and, when combined with the direct signal, produces "bigger" sounds. Small amounts of pitch shift (less than a semitone) make a sound "fatter," while large amounts allow you to play harmonies with a single note. Most units (in fact, all those profiled here) offer only parallel harmonies, although some specialized devices track the key and mode of your playing to produce "intelligent" multipart harmonies.

Reverb: Makes the signal sound as if it's being played in a room of a particular size. By combining thousands of different discrete echoes, or reflections, it creates a wash of sound that adds ambience to the input signal.

The Pitch Shifting row contains the maximum range of the effect, and the Other Effects row lists other effects the box can produce. The Korg A3 is a standout here because its open architecture allows additional effects programs and combinations to be added via ROM cards, which is tremendously cool (and forward thinking). Also note that for all boxes capable of shifting pitch more than a semitone, MIDI note on messages can alter the pitch shifting interval in real time (see the Other Real-Time Control row).

The maximum number of simultaneous effects, the question of effects order and combinations, and the concept of limitations to using multiple effects at once were discussed earlier. The # of Memory Locations row lists how many different sounds can reside in memory at one time. Only the Yamaha units have certain factory presets that are not in RAM and cannot be overwritten.

The next three rows relate to an important, but often overlooked capability of these multi-effects processors: real-time control. The MIDI Continuous

Controllers row lists how many controllers, such as mod wheel or any continuous controller messages, can alter parameters of a given sound in real time; the Real-Time Control row mentions other possibilities (and, if necessary, the required hardware); and the Options row lists the hardware available to change programs and alter parameters for each of the different units.

Remember that any MIDI-controllable processor can receive MIDI data from any source; it need not be from the same company's product. Also, you should know that Roland's foot controller uses a non-MIDI protocol to communicate with the GS-6. With the addition of Roland's optional RRC-to-MIDI converter, however, you can use the FC-100 Mk II as a general-purpose MIDI controller. Korg's FC-6 is somewhat similar, but it has both a company-specific remote jack and a MIDI out.

As a rule, all the units respond to MIDI patch changes; the SGE and all of the Yamaha processors also include a MIDI program change table (each of the Yamaha units actually has four), so

you can assign the different effects patches to whatever MIDI program change numbers you want. (For example, you could set it up so that MIDI program change 5 calls up patch 43 on, say, the FX500.) In addition, all but the Yamaha FX500, GEP50, and SPX50D can send and receive patch data via MIDI system exclusive (sysex) messages. In conjunction with a hardware- or software-based MIDI sequencer, you can use this feature to store and recall an unlimited number of effects patches.

A few units have limitations when it comes to real-time control—assuming they even offer it in the first place. In Korg's A3, the one alterable parameter available per patch (in addition to volume) is preset and determined by the effect chain in use. Similarly, the Digi-Tech GSP-5's MIDI controller settings are global and affect every patch in a similar way. Thus, unlike other units, you cannot program different controllers per patch (or assign the same controller to affect different parameters in different patches). On the other hand, the GSP-5 permits a lot more different

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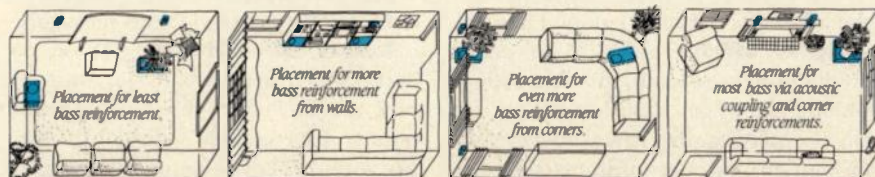
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THE LISTENING

Digesting the specs for a given product is one thing, but when it comes to distortion (and other effects), sound quality is really all that counts. So, here are some subjective capsule comments regarding the sound quality of each available unit, with a particular emphasis on distortion.

ART SGE: The SGE blends analog compression, distortion, EQ, and expansion with digital delay, chorus, reverb, pitch shifting, etc., to produce a combination that seems to offer the best of both

worlds. Wow, does this thing sound huge, particularly when you use the pitch-shifting an octave down. I have to admit there are little things I don't like, such as the coarseness of the parameter settings, the occasional glitches while changing programs, etc., but I found myself coming back to this one quite a bit as I went through the tests. The distortion sounds big, bright, and punchy, with enough tear in it to satisfy even the heaviest metalhead. I'd say it's guaranteed to do some seriously pleasing damage to your raw guitar signal.

Boss ME-5: The ME-5's unusual packaging combines the functionality of a foot controller with the sound of a multi-effects processor, a point that may endear it to many people by itself. Like the SGE, the ME-5 combines analog distortion, compression, and EQ with digital effects (the chorus and flange are also analog, in contrast with the other units surveyed). The resulting sound is big, fat, and very crunchy (not to mention extremely quiet). There's a certain metallic buzziness that seems to be a characteristic of analog distortion, and the ME-5 offers it in full glory. I like it.

DigiTech GSP-5: The GSP-5 has an incredibly warm, smooth, tube-like sound, which I found particularly attractive for rhythm playing. There's a distinct lack of high end in many of the factory presets (despite its 20 kHz frequency response) due to a low-end emphasis in

many sounds, but its warmth makes it sound completely unlike any of the other boxes. I would've preferred a bit more bite and edge, but if you like a really crunchy, singing sound, it's unbeatable.

Korg A3: The A3 has a more sophisticated sound than most of the other boxes and offers a wider variety of effects than any other unit (it also has the slickest user interface). The clean sounds on it are particularly effective, with a bright, airy character. The distortion sounds have a real richness and buzziness to them, while at the same time offering enough edge to satisfy those who want it. As hard as I tried, I



The BOSS ME-5 puts rack-mount sound quality in a floor pedal package.

just couldn't find any weak spots; this thing is monstrous.

Roland GS-6: The GS-6 has a big, full sound, with a lot of movement. It doesn't seem as refined as some of the other boxes, offering instead a more brute force sound. The distortion offers a nice crunch, but the overall effect isn't as punchy as I would have liked. In its favor, the GS-6 is amazingly quiet, and it's the only unit that offers balanced XLR outs for direct, in-line studio work.

Yamaha FX500: This half-rack unit (the only one I've ever seen that has an LCD) packs a tremendous punch in its tiny package, offering a good variety of patches and a clean, bright sound. Even more important, the distortion sounds can absolutely scream. At times, that screaming approaches brittleness, but overall, it's got a great crunch to it. And for the money, I think it's a tremendous value. I only wish it had more room for user presets and/or a MIDI out jack for dumping programs via sysex messages.

Yamaha GEP50/SPX50D: These two units, which are virtually identical except the GEP has more guitar-specific presets, have a slightly dirty, workmanlike sound. They're definitely not as refined as the other Yamaha products, but I think this helps give their distortion

sounds a bit more grit. Unfortunately, they're limited to a maximum of two effects at once. Also, because of the FX500's introduction, these two boxes are apparently going to be discontinued, so you should be able to get them for a good price. If you're purely interested in a Yamaha guitar processor, however, my advice is to save your money and buy an FX500, unless you can find a real bargain.

Yamaha SPX900: Unquestionably the most refined Yamaha product, the SPX900 also has a good deal of kick and crunch to it. Like the FX500, its distortion can sound overly bright, but I really like a sharp, cutting sound. If you're looking for a guitar processor that can do excellent double duty as a general multi-effects unit for the home studio, the SPX900 should be high on your list of preferences. (By the way, if you have the money, Yamaha's RCX1 remote is pretty cool.)

The sound of all these boxes will be influenced by your amplification system. All of them can work in conjunction with normal, one-channel guitar amps,

but doing so will color the sound to a certain degree, rob them all of their stereo effects, and slightly decrease the sound quality. A decent stereo sound system, on the other hand, will let all of these devices sing (or scream) in full, unfettered throat.

Another important point, particularly if you're tweaking the factory programs or creating your own from scratch, is level matching. Whether you're using these effects on stage or in the studio, you should try to match their overall volume level so that when you shift to a different effect, you don't get an abrupt change in the sound (unless, of course, that's what you want).

'SCUSE ME, WHILE I SAY GOODBYE

Creating unique timbres with a guitar and a sophisticated effects box (or boxes) is an area that few people have really explored. Of course, it's nothing new—people like Robert Fripp, Adrian Belew, Andy Summers, Bill Frisell, David Torn, and others have been producing incredible sounds from their guitars and effects racks for years. But with guitar-

oriented multi-effects, anyone has the opportunity to create and instantly recall 50 or 100 or so of their own, unique timbres with a single box. On top of that, with most units, you can make subtle or outrageous changes to the sound in real time. Want to change the distortion level, EQ setting, delay time, chorus rate, or reverb type as you play? It's all possible.

The units currently available each have their own strengths and weaknesses; you'll have to decide for yourself which features and qualities are the most important to you. What they share in common, however, is an amazing potential for creating great guitar sounds. If all you want is a big, crunchy rhythm sound and a few different screaming lead sonorities, then you certainly can wrench those timbres from these boxes. But if you want to explore even more territory, the doors are open.

Bob O'Donnell, EM's associate editor, is a guitarist and MIDIophile who secretly wishes he played lead in a heavy metal band.

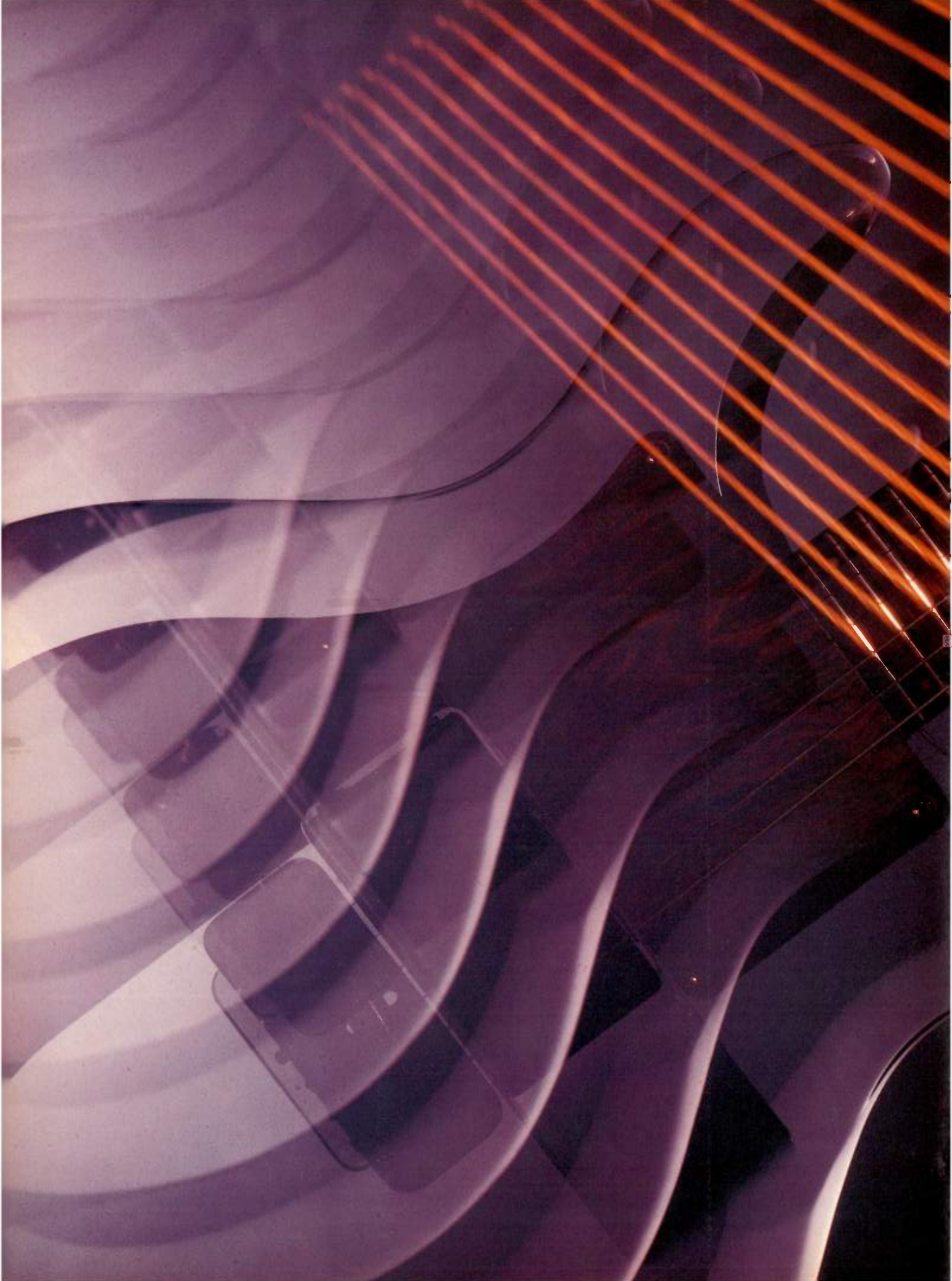
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The incredible growth of sophisticated signal processors has made getting great guitar sounds a snap. Maintaining control over those effects, however, is a different story. More boxes means more possibilities, but also more problems: having the right effects come in at the right time, at the right volume, with the right sound often seems like an impossible dream.

In fact, until MIDI came along, that goal seemed as elusive as world peace. But now there are several ways to make effects systems easier to use than ever before, so let's dive into the world of automated effects.

● AUTOMATED EFFECTS

THE FIVE-MINUTE MIDI COURSE

Although MIDI is a deep and multifaceted language, very little of it applies to effects systems, which simplifies learning what we need to know.

A MIDI *receiver* listens for MIDI data at its MIDI in connector and does what this data tells it to do; the data is identified as belonging to one of sixteen individual channels. A MIDI *transmitter* sends data out its MIDI out connector over one of MIDI's sixteen channels; this data tells the receiver(s) what to do. In most cases you want to set the receiver and transmitter to the same channel. (Some units can transmit *and* receive, but that is seldom pertinent to signal processing setups in live performance.)

The most important part of the MIDI language for signal processing is the *program change command*, which specifies one of 128 programs. As a simple example of how this works, consider a MIDI-equipped, multi-effects unit that stores 128 programs. You can either select programs from the unit's front panel or, when you have your hands full

FOR THE BEGINNER

What's a Sequencer?

A sequencer is conceptually similar to a tape recorder, except it records MIDI data instead of audio and stores this data in computer memory (or computer disks) instead of on tape. This data can be anything from notes, to program change messages, to continuous controller data.

Devices capable of synchronization can be linked electronically so they all play together. Thus, at the same time a drum machine plays measure 65, beat 3, a sequencer synchronized to the drum machine also plays measure 65, beat 3. This is the heart of an automated effects system: by programming the changes you want to occur into the sequencer and syncing it to a master timekeeper (or using it as

the master timekeeper), all changes will occur in time with the music at the correct places.

Sequencers usually allow several ways to enter data. For example, to record a program change, you would simply connect the MIDI out of an effects unit to the sequencer's MIDI in, put the sequencer into record mode and, at the appropriate place, hit the program change command on the effect. This is transmitted to the sequencer and recorded. You may also be able to enter program changes by typing them in on a computer keyboard, among other methods.

On playback, the sequencer's MIDI out goes to the effect's MIDI in. As soon as the sequencer plays back the program change message, it will change the effect's program.



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The LXP-1. All the reverb power you need, all the quality of Lexicon.

playing live, select sounds with individual switches on a footswitch unit. If both the effects unit and footswitch are set to the same MIDI channel, they can "talk" to each other. Then, when the footswitch transmits a command, such as "go to program 56," the multi-effects unit receives and obeys the command (Fig. 1).

It may seem unnecessary to use MIDI; after all, pre-MIDI multi-effects units included footswitches. However, these were usually dedicated footswitches that worked with only one particular unit.

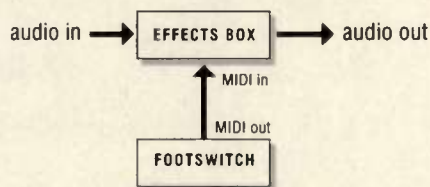


FIG. 1: How to connect a MIDI transmitter to a MIDI receiver. Be sure the effects box is set to receive on the same MIDI channel on which the footswitch is set to transmit on.

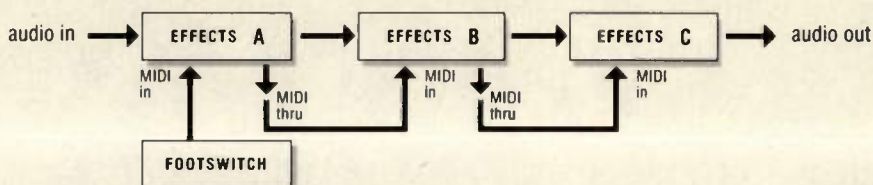


FIG. 2: "Daisy-chaining" multiple MIDI units for control by a single MIDI footswitch.

With MIDI, you can pick the MIDI footswitch that best suits your needs and use it to drive *any* signal processor that responds to program change commands. Plus, trading in your effects unit for a more sophisticated MIDI model won't make your footswitch obsolete.

USING MORE THAN ONE EFFECT

Thanks to guitar-oriented multi-effects units that include fuzz, compression, delay, equalization, and so on (see Bob O'Donnell's companion survey of guitar multi-effects), the simple MIDI setup just described may be all you'll ever need. Using more than one effects box at a time, however, is only slightly more difficult to handle.

Most MIDI devices include a *MIDI thru* jack that transmits a copy of the data appearing at the MIDI in. A single MIDI footswitch can address multiple units by "daisy-chaining" one unit's MIDI thru connection to the next unit's MIDI in (Fig. 2).

Now it gets a little complicated, and it requires a bit (or maybe even a byte) of forethought. For example, suppose you get a great sound by combining program 12 from effects unit A, program 43 from effects unit B, and program 77 from effects unit C. If you can send out only a single program change, you're clearly not going to assign all three units to the desired program number. There are several ways around this problem:



POWERS

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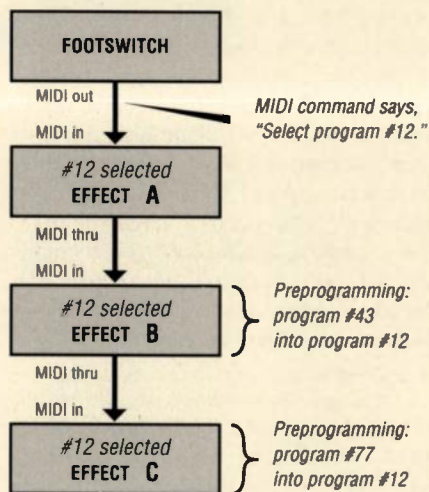


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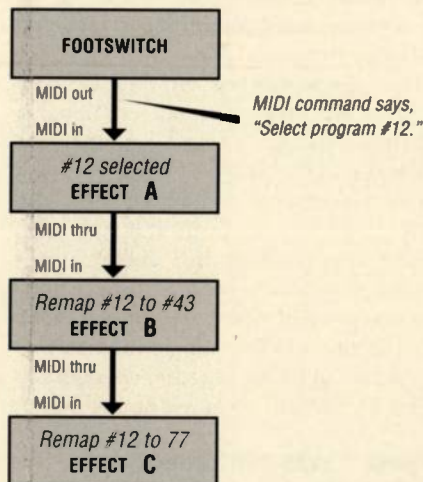
● AUTOMATED EFFECTS



EXAMPLE 1

Example 1: Copy all programs intended to work in combination into the same program number on all units. For example, copy program 43 to location 12 on effects unit B, and copy program 77 to location 12 on unit C. Sending out program change 12 from the footswitch then selects the appropriate effect on all three units.

Example 2: Redirect, or "remap," the programs so they respond to *different* program change messages. Some units



EXAMPLE 2

let you remap program change numbers so that upon receiving a particular command, the effects unit itself will select a *different* program number. For example, you could remap unit B so that receiving program change 12 calls up program 44, and remap unit C so that receiving program change 12 calls up program 77.

Example 3: Set each unit to a different MIDI channel so it receives program changes on its own channel. For best results, this requires a footswitch that can "stack" multiple commands on one switch. For example, suppose effects unit A was set to channel 1, unit B to channel 2, and unit C to channel 3. You would program a switch so when it sends out program change 12 on channel 1, it also issues commands to send program change 43 on channel 2 and program change 77 on channel 3. Pressing the switch issues the appropriate commands in rapid enough succession that it seems simultaneous.

Example 4: Use a hardware accessory (such as the Axxess Mapper, Oberheim Systemizer, etc.) to translate an incoming program change received over one channel into multiple program changes on multiple channels. You can think of these boxes as MIDI data processors that modify MIDI data in a manner conceptually similar to the way a standard signal processor modifies audio. This is like the previous example, except it requires another device, although the footswitch can be less sophisticated.

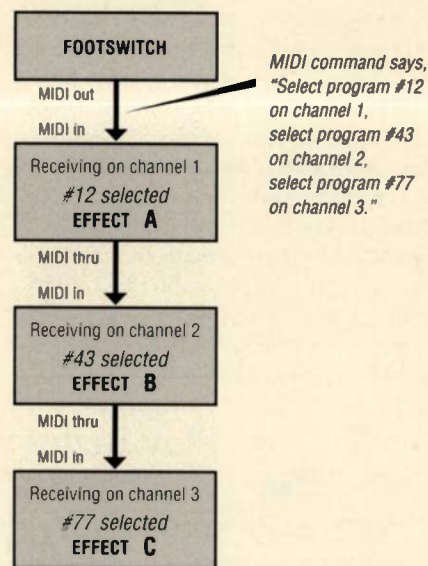
Each of these methods has advantages and disadvantages; the method you find most pertinent will often be dictated by your hardware.

If your MIDI footswitch can only send out data on one channel at a time, methods 1, 2, or 4 will work fine. If your footswitch can send multiple program changes over multiple channels with a single button press, method 3 is the way to go. You'll probably find this easier than remapping programs or copying programs over to new memory locations, and you won't need an external box. However, you'll have to spend more time programming your footswitch.

This may sound complicated, but it's not really that bad. Multi-effects units certainly simplify life; in many cases, you'll need to add effects from additional boxes only for a few special sounds. In any event, time spent programming is a small trade-off for a smoother live performance.

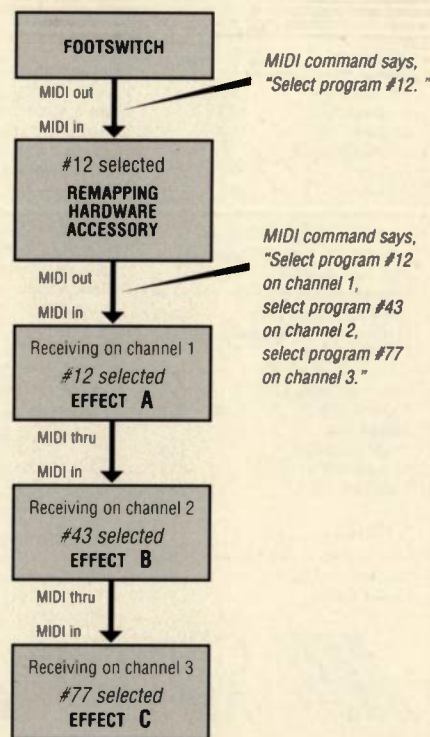
PROGRAM CHANGE PROTOCOLS

So you have a MIDI footswitch and some MIDI effects—great. What's the most logical way to call up individual programs for live performance? Here are a few suggestions.



EXAMPLE 3

If you work with a relatively small collection of programs (twenty sounds or so, either on one unit or spread over several units), it's probably best to assign these to individual switches (e.g., one for fuzz, one for compression, one for fuzz with compression, etc.) on one unit and call them up as needed—providing you don't have to press too many switches. I say this because many footswitch units work on a "bank/pro-



EXAMPLE 4

● AUTOMATED EFFECTS

gram" principle and include a bank select switch as well as several program switches. As long as you remain within the same bank, each sound is only a single switch press away. Selecting a sound from a different bank requires two or more presses—one to select the bank and one to select the program. If you have to step through several banks to get to the one you want, that requires even more switch presses.

If you use multibank footswitches, you could dedicate each bank to a single

song, which simplifies the amount of button pushing you have to do with your feet while playing. For example, suppose a device allows for sixteen banks of eight programs. Eight variations per song and sixteen songs should be enough to carry you through a set, particularly if there is some duplication between the sounds required by different songs.

Some footswitches let you increment or decrement through programs, where each press of the increment switch selects the next higher-numbered patch,

and each press of the decrement switch selects the next lower-numbered patch. If your set list is fixed, the total number of program changes that need to occur during the set is 128 or less, and you don't mind doing lots of programming, this arrangement is really easy live—whenever you need a patch change, press a switch and the next patch comes up. You don't have to think about which switch to select. Just expect that changing your sequence of songs will be time consuming.

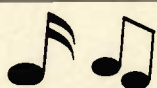
The easiest method of all, which works only if your band uses a MIDI drum machine or sequencer for time-keeping (see "For the Beginner: What's a Sequencer?"), is to have the sequencer issue program change commands to your effects units as needed instead of a footswitch. This has to be experienced to be believed: no more switch presses! You simply play along and the sequencer changes patches for you. Plus, programming is easy because you can call up programs in just about any numerical order you specify.

If the sequencer is playing a lot of parts, you'll probably be able to use only one channel for effects switching. This will necessitate copying programs to new memory locations, remapping programs, or using an external box if you use combination effects.

If your band doesn't use a sequencer, but the drummer plays to a drum machine, it's probably worthwhile to invest in a small hardware sequencer (Yamaha, Roland, Alesis, etc.) and sync it to the drums so it can provide your system with program change commands. You can take this idea even further: if your band uses a sequencer but there aren't enough channels to feed your effects setup, you can sync a hardware sequencer to the band's main sequencer and program in all the program changes you like, on as many channels as necessary.

THOSE CLEVER CONTINUOUS CONTROLLERS

In addition to program change messages, some units respond to continuous controller data (see "For the Beginner: Continuous Controllers"). The reason for this is to make certain important signal processor parameters—delay time, echo, feedback, equalization cut/boost, and so on—accessible to commands over MIDI, which lets you alter signal processing parameters as easily as you



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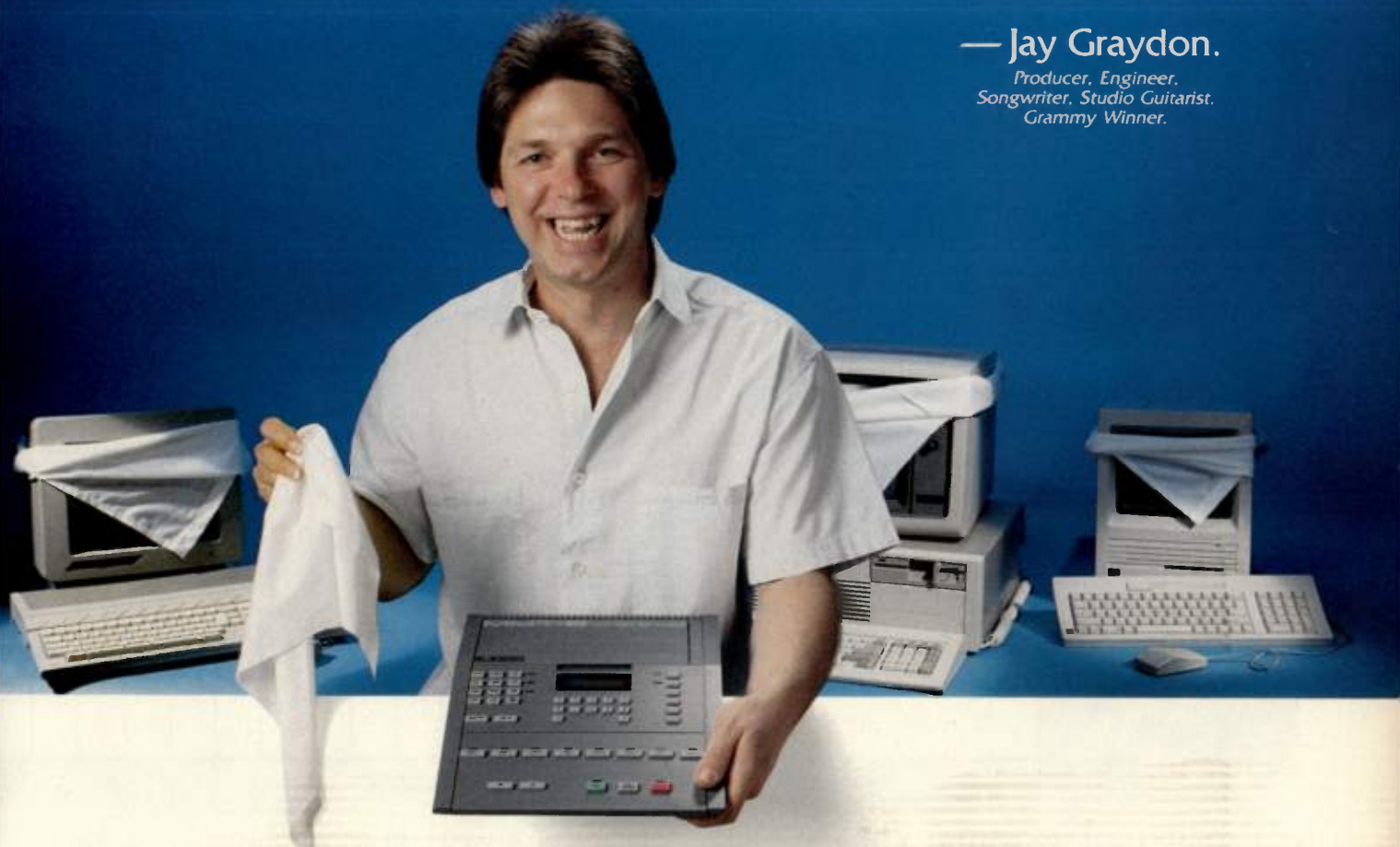
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Forgive us if we sound a little pompous. It's just we find ourselves in a curious position. Over the past several years, everyone and their brother has introduced a so-called "work station." When, to our way of thinking, they really aren't work stations at all.

To us, a work station should have the most sophisticated sequencer available. And in fact, our new W-30 does. It features 16 tracks, microscope editing, full compatibility with both Roland MicroComposers and Directors "S" Sequencing software, to say nothing of the friendliest user interface there is.

To us, a work station should also be designed around a sampler rather than a synthesizer. What this does, more than anything else, is make the system remarkably versatile. It's a whole lot easier to make a sampler sound like a synthesizer than the reverse. And speaking of

sounds, those from the W-30 can be processed through either eight polyphonic outputs or a mix output.

To us, a work station should possess an excellent memory. Which is why we've equipped the W-30 with a one mega-



Because the W-30 uses the same disks as the S-50 and S-550, you won't need to build a sound library. It already exists.

n the market, the first.

Nor does its versatility end here, because the Roland W-30 not only puts



If you squint you can probably make out the fact that the new Roland W-30 has eight polyphonic individual outputs which allow any sound to be routed individually to a mixer.

a 3.5" floppy disk drive at your disposal, it also gives you the ability to access additional data by using either a CD-ROM or a hard disk connected to an optional SCSI interface.

Of course, a work station should be able to express itself too. Which is why we've made our 61-note keyboard sensitive to both velocity and after-touch.

And it should be easy to use. Hence, the W-30 uses a large, state-of-the-art 240 x 60 dot LCD display that's capable of providing more useful information at one time than ever before.

But before we go, let us take this moment to pose a hypothetical ques-

tion. Let's just say that all of the other so-called work stations found a way to include these very same features. They'd be better, of course, but still not comparable to the re-

markable new W-30. Because they'd still be missing the most persuasive and motivating feature of all.

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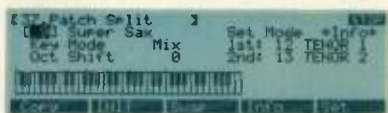
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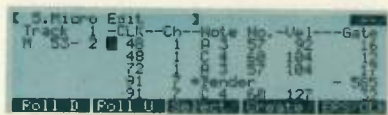
byte, user-accessible memory (ROM).

And because it comes with the most frequently-used sounds, you won't need to load in a sound disk to begin working.

The sampler section's 512k (RAM) memory is no less impressive. It's actually equal to that of a Roland S-330, and can be used for creating new sounds, or for playback, or for manipulating any of the S-Series disks. As a result, you'll not only be in a position to work with the sounds that are currently hot, you'll be in just as good a position to capture the sounds that will become hot.



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● AUTOMATED EFFECTS

FOR THE BEGINNER Continuous Controllers

Continuous controllers cover a continuous range of values, unlike a controller such as an on/off switch, which only selects between two possible values. In addition to allowing for 32 different continuous controllers, MIDI also makes provisions for 32 on/off switches and 26 as-yet-undefined controllers.

To accommodate sending controller information between devices, each of the sixteen MIDI channels has 128 available controller numbers (0 to 127). Controller numbers 0 through 31 handle continuous controller data; numbers 32 to 63 handle additional data for these continuous controllers, if required; numbers 64 to 95 carry on/off switch data; and controllers 96 to 121 are currently undefined. Controllers 122 to 127 affect various channel modes and are not pertinent to this discussion.

Some de facto standards have appeared that assign particular functions to particular controller numbers. For example, controller 1 carries synthesizer modulation wheel data, controller 4 carries footpedal data, and controller 7 carries master volume data. With switch controllers, 64 carries sustain switch data, and 65 is for portamento on/off.

Many devices allow for assignable controllers. For example, the delay feedback function may be assignable to any controller number you desire. So, you could assign it to controller 4 to vary it from a footpedal, to controller 7 to have it track master volume messages generated by a MIDI transmitter, or controller 1 if you wanted to control delay feedback from a synthesizer's modulation wheel.

For more on continuous controllers, check out Chapter 4 in my book *MIDI for Musicians*, the *MIDI Resource Book* by Steve De Furia and Joe Scacciaferro (both are available from EM Bookshelf; see FYI page), or the official MIDI specification, available from the International MIDI Association (5316 W. 57th St., Los Angeles, CA 90056).

would alter programs (well, almost).

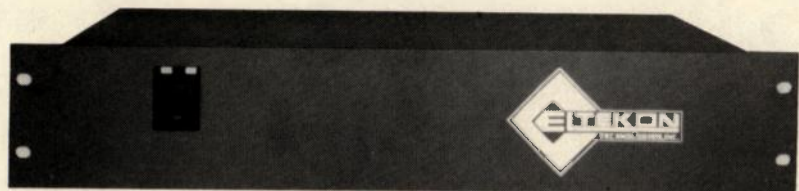
The main difference between continuous controllers and program change commands is that a program change is a single event, whereas most continuous controller changes are a *series* of events, such as a volume fade (each message lowers the volume a bit more than the previous message) or change in delay time. There are two main ways to create continuous controller changes: manual control or automated.

The manual method requires a pedal that can generate continuous controller

messages, such as the CFC-4 from Lake Butler Sound. This unit contains four pedals that can be assigned to various continuous controllers. Some synthesizers can translate volume pedal motion into continuous controller data; another option is Anatek's Pocket Pedal (reviewed May 1989), a dedicated box that converts motion from just about any pedal into continuous controller commands. You use any of these pedals the same way you use a conventional wah-wah or volume pedal.

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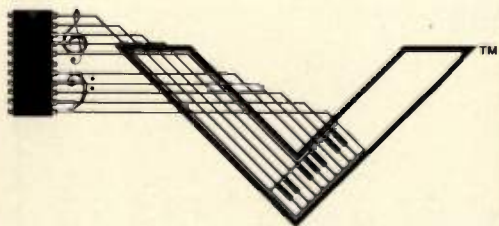
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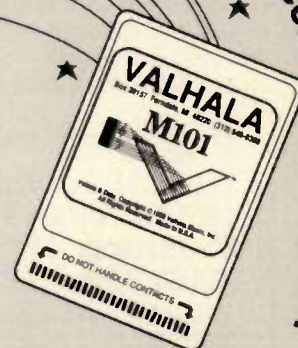


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● AUTOMATED EFFECTS

you use a sequencer. You can program continuous controller changes into a sequence right along with program changes, so that the sequencer can, for example, fade out a sound or change the amount of delay time feedback.

To program this data into the sequencer, you can either use the motions of a pedal that generates continuous controller information, or use a synthesizer's mod wheel to create the appropriate messages. Some sequencer software lets you "draw in" controller values with a mouse or enter values from a computer keyboard.

THINGS THAT GO BUMP IN THE CHORUS

Perfection is elusive, and automated effects systems are no exception. One problem is that some units react to program changes in unpredictable ways. A delay may mute itself for a few milliseconds while switching from one program to another, create audible glitches, or transition in a totally glitch-free manner. Furthermore, some units will glitch on

converter (June 1989 EM) to switch a VCA on and off.

Parameters controlled by continuous controller messages are also subject to glitching, sometimes to an even greater extent than program changes. The only way to get a handle on this is to try varying different parameters and seeing what happens. Take note of which parameters change smoothly and which ones don't, for future reference.

ADDITIONAL HINTS AND TIPS

Here's a grab bag of tips you may find helpful when checking out automated effects systems.

■ Beware of unorthodox program change numbering schemes. Units may respond to the 128 possible program change numbers as 0 to 127, 1 to 128, or as banks (e.g., eight banks of sixteen programs). So if you send out program change 11, some units might respond as 10, some as 11, and some as 23 (assuming several banks of eight programs, and assuming that you call bank 2, program 3). Check to see how your unit responds and, if necessary, make up a conversion chart for your various effects.

■ Learn how to read the MIDI implementation chart that should be included with each MIDI-controlled unit (usually located toward the rear of the instruction manual). This will provide information on how the unit reacts to program changes and continuous controller commands.

■ To retrofit older effects for MIDI control, check out the Scholz R&D MIDI Octopus (September 1989 EM) and the Peavey effects control system reviewed in this issue.

■ Make sure you back up any programs in your effects unit or footswitches. If you don't have a computer to do this, bring your units to someone who does; it's a real drag to come up with a great set of programs and lose them because you didn't make backups.

Automated effects systems can take the guesswork and tap dances out of playing guitar live, especially if you can sequence all your changes. Once you try it, you won't be able to go back to conventional setups.

Craig Anderton is, among other things, editor-in-chief of this magazine. His latest album, *Forward Motion*, uses a lot of MIDI guitar and was released in July on the *Sona Gaia* label (distributed by MCA).

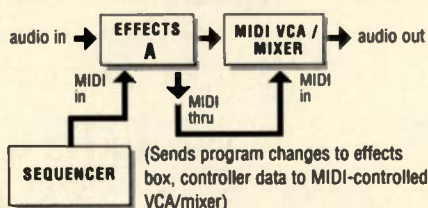


FIG. 3: Using a sequencer and MIDI-controlled VCA to shut off a signal momentarily while changing programs.

some types of programs but not on others. (Usually delay is most problematic; EQ and other effects are less so, if at all.)

There are very few ways to minimize glitches. If you primarily use a straight guitar sound (or one with some fuzz) and these are on all the time with effects added in the background, try running the normal sound into one mixer channel and the effect through another one. This will allow you to keep on playing while the effect changes.

If you play in a band that uses a sequencer, and you experience serious glitching, connect a MIDI-controlled VCA (voltage-controlled amplifier), such as the J.L. Cooper MixMate or Iota MIDI-fader, after the effect and program it to shut off your signal momentarily while the program is changing (Fig. 3). Or, if you want to save some bucks, use Thomas Henry's MIDI-to-trigger

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arch 1, 1989, was an important day for musicians, composers, and other owners of copyrightable works. This was the effective date for

amendments to the Copyright Act of 1976 (hereafter called "Act"), the federal statute that governs copyright protection in this country. These amendments came out of the Berne Convention Implementation Act of 1988, signed by President Reagan last October.

Among changes in the Act, the most significant are:

- ♦ As before, copyright takes effect the moment the work is created. However, it is now no longer necessary to place a copyright notice on publicly distributed copies of protected material.
- ♦ The U.S. is now a party to the Berne Convention for the Protection of Literary and Artistic Works. This is the oldest, most important, and effective multilateral treaty governing international copyright policy. U.S. citizens now automatically receive legal protection for their works in

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• COPYRIGHT LAW

many foreign countries without doing anything more than simply creating those works.

♦ Prior to March 1, a person who acquired ownership from someone else was required to record the transfer document in the Copyright Office as a prerequisite to filing an infringement lawsuit. This requirement has now been eliminated.

♦ Compulsory licensing for jukeboxes is eliminated.

♦ Definitions for "Berne Convention works" and "country of origin" have been added; these are used to determine whether a work is eligible for national treatment under the Berne Convention.

In most other respects, the Act remains unchanged.

ESTABLISHING COPYRIGHT

Including a copyright symbol on publicly distributed works used to be necessary to avoid the loss of copyright protection. However, now that the notice requirement has been eliminated, creators of protectable material do not have to be concerned about losing their rights if a notice is not used. This is good for copyright owners, but may be of concern to those who use protected works. As in the past, creators of copyrightable works are immediately entitled to legal protection for their material on the date it is created, without doing anything more. As a general rule, this protection will continue for a period equal to the life of the creator, plus 50 years. However, after March 1, since a notice may not be included with copyrighted works, it will be much more difficult for users of material created by others to know whether that work is copyrighted or not. Appearance of a copyright notice on artwork, books, motion pictures, records, and other material is usually the only way to readily tell whether it is protected.

Persons who use material created by others, without checking its copyright status, risk incurring liability for infringement. An unauthorized use of protected material may infringe on one or more of the owner's exclusive rights. Unless granted a license, only the copyright owner is entitled to copy, vary, publicly perform, display, and distribute protected material.

Although use of a notice is not required to obtain copyright protection or to avoid losing it, owners should con-

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● COPYRIGHT LAW

tinue to place this notice on their works so they can defeat claims of *innocent infringement*. This occurs when someone copies, publicly performs, or otherwise exercises an owner's exclusive right, without authorization, for a protected work that does *not* feature a notice. When a claim of innocent infringement is defeated, the copyright owner can obtain all available remedies for infringement, including injunctive relief (an order by the court to do or not do something) and damages, which would not be possible if a claim of innocent infringement is allowed to stand.

FOREIGN RIGHTS

For many years, U.S. citizens have been able to protect their works in some foreign countries. This protection, called "national treatment," means a country will give the same protection to the copyrightable works of foreign nationals that it gives to its own nationals. For example, under national treatment a U.S. author's literary works receive the same legal protection in France as those created by a French citizen. This is not nec-

**Though not
required, owners
should continue
to place the
copyright notice
on their works
to avoid claims
of innocent
infringement.**

essarily the same legal protection given in the U.S. because the protection in a foreign country is determined by that country's laws.

Prior to March 1, U.S. citizens were entitled to national treatment in the 76 or so countries that signed the Berne Convention only under conditions of

simultaneous publication. They were also entitled to national treatment in more than 40 of those countries through another multilateral copyright treaty known as the Universal Copyright Convention (UCC). Some countries are parties to both treaties, while others (such as the U.S.S.R.) have signed only one. Before March 1, the U.S. was a party only to the UCC.

To receive national treatment in UCC countries, both before and after March 1, U.S. citizens must use a specific form of copyright notice of their works. This is necessary to satisfy any formalities set by a UCC signatory country as a condition of its granting national treatment.

Regarding Berne Convention countries, prior to March 1, U.S. citizens were not required to use a notice to receive national treatment in these countries; however, as nationals of a nonsignatory country they were eligible for protection under the Berne Convention only if copies of their works were first publicly distributed simultaneously in the U.S. and in a Berne Convention country (e.g., Canada). Simultaneous publica-



ES-50 synchronizer and RTS 14 rack sold separately.

tion was interpreted to mean first publication in the U.S. and a Berne Convention country on the same day, or first publication in a Berne Convention country within 30 days from the first publication date in the U.S.

Simultaneous publication is no longer necessary for U.S. citizens, whose works now automatically qualify for national treatment under the Berne Convention. Eligibility is effective immediately on the date a work is created; nothing more need be done. It is not necessary to place a notice on works, and there is no waiting period. Furthermore, published and unpublished works are eligible for national treatment. This was not the case under the Berne Convention or the UCC prior to March 1.

Amendments to the Act also change the registration requirement, but only as it affects foreign nationals. Prior to March 1, both U.S. citizens and foreign nationals were obligated to register a copyrighted work to enforce their exclusive rights. Now, foreign nationals do not need to register.

However, to encourage registration

by everyone, the statutory damage amounts in the Act (those specified in the statutes and awarded without having to prove the loss) have been doubled. After March 1, the minimum award increased from \$100 per infringement to \$200, and the maximum increased from \$50,000 to \$100,000.

OWNERSHIP TRANSFERS

If a person acquired copyright ownership from someone else, it is no longer necessary to record a transfer document with the Copyright Office as a prerequisite to filing an infringement lawsuit. However, filing a transfer is still necessary for other purposes. It can safeguard an ownership transferee in the event of a conflicting transfer; should an ownership dispute arise because ownership of the same work has been transferred to other transferees by the same person, the first recorded transfer may prevail.

Of course, all of the above is intended only to acquaint you with these important changes. The copyright law has many other facets, and explaining the law in its entirety is beyond the scope of

this article. If you need to protect your work, by all means consult an attorney versed in copyright law. You might also want to read the following books for additional background information:

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David Weinstein is special counsel to Bourke & Jacobs, P.C. Attorneys At Law in Denver, Colorado, and practices copyright and trademark law. He is the author of *How to Protect Your Creative Work: All You Need to Know About Copyright*.

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Tips

from EM Readers

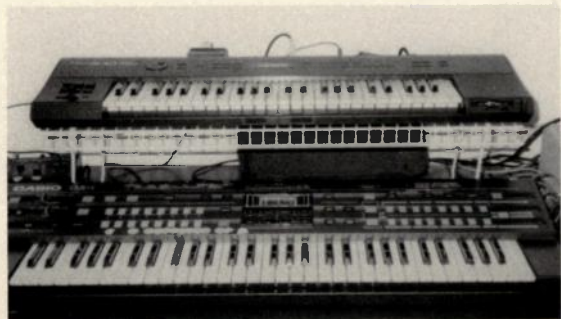
By Bruce Tovsky

DX7 BANK- AND PATCH-CHANGING FROM A SEQUENCER

By Dan Titus

LOW-COST KEYBOARD STAND

Try using wire shelves as keyboard stands. They make it a snap to pass cables through and around; it's easy to see where cables are in the spaghetti jungle; the shelves break down fast; and they don't collect a lot of dust. Wire racks and shelves come in many styles and configurations; I use a 4-foot shelf designed to either stack on an existing shelf, hang, or stand alone (Fig. 1). You can even use a wire basket for a printer stand. Wire products are sold at discount and hardware stores.



DAN TITUS

FIG. 1: Wire closet racks are an economical and ingenious alternative to keyboard and printer stands.

If you, like me, were frustrated with the inability of a DX7, updated with E!, to change banks as well as patches from a sequencer command, fear not. Here's an absurdly simple and easy way to do it without ever having to recite the dreaded "patch map" words.

Before we start, if there are any DX7 owners out there who haven't updated their synths with E!, go out right now and do it. With the update, you have instant access to ten banks of 32 patches, not including the external cartridge. On top of this, there are additional MIDI commands that turn the DX7 into a nifty master keyboard for any MIDI setup.

The only hardware requirement to implement this tip is a sequencer that allows you to insert continuous controller messages (e.g., mod wheel, breath control, pedal, etc.) into a sequence. What we will do is redefine the controller to scroll through banks of patches on the DX. In the four years I've had my DX, I've never once used the breath controller, so I chose that controller to redefine.

Begin by accessing the function pages on the E! DX. Go to page 5 (the "physical" page) and press membrane switch 11. The LCD will show Keyboard Control at the top and n-s

underneath ("n" being the controller to be redefined, "s" being the new designation; for example, Breth=Timbre). Pressing 11 repeatedly will step through the "n" choices: breath, mod, or foot. Using + and - will step through the "s" choices; select Data Slide, the data entry slider, which is the default controller for banks on the E! DX. The DX is now set up to receive breath controller data as a change in data slider information. (Those with sophisticated MIDI patch bays or mappers can sometimes use controller remapping to accomplish the same function, but overall, the E! update is faster and easier to program.)

Now all you have to do is enter a continuous controller event into your sequence to call up the appropriate bank, followed by a program change message to call up the desired patch. The DX will instantly switch from bank to bank and patch to patch (give or take a MIDI clock or two, so make sure the command precedes a note on message). I use Dr. T's KCS for the Commodore 128 and remap the breath controller to controller #6 (data entry). The following breath controller values call up the following banks:

Value	Bank#	Value	Bank#
1-12	0	13-25	1
26-38	2	39-51	3
52-63	4	64-77	5
78-89	6	90-102	7
103-115	8	116-127	9

As is evident, the 127 value range of the breath controller is spread through the ten banks of the E! DX.

A side note: the E!quipped DX doesn't remember the redefined controller unless you store it with a program, which is a drag considering this is a more-or-less global command.

COMPACT MUSIC STUDIO STAND

Being a relentless do-it-yourselfer, I've created a relatively inexpensive, easy-to-build rack from 2-inch PVC pipe and shelving. The pipe is cut with a hacksaw, assembled using PVC glue, cleaned with lacquer thinner, and painted. Then, appropriate shelves are cut to fit as needed and fastened into place. Presto, a compact ministudio that will hold everything.

You need to plan accurately, on paper, where everything will go, taking into consideration the tubing size and elbow dimensions and the dimensions of all gear that will go in it. Then draw it on graph paper, with front and side views, to get the right depth offsets. I wasn't sure how the height would work out, so I simply built it upward, a step at a time, by placing the instruments in place before deciding on the next shelf level (see Fig. 2).

The materials are available at hardware stores and plumbing shops. You might find different qualities of Ls and Ts, and some may need grinding or touching up. I used about 30 feet of pipe.

You only have a short time during assembly, maybe fifteen to 30 seconds, to get the angles right before the glue sets, so be sure to work with a good-sized carpenter's square and preassemble each stage before gluing. It is helpful to clean each joint with lacquer thinner first. I placed the lower keyboard on the shelf, then kept cutting vertical and horizontal tubes until the fit was right before actually gluing it together. This method can be used all the way to the top (see Fig. 3).

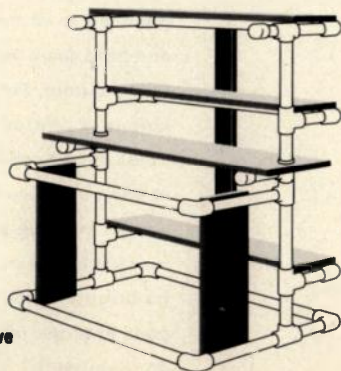


FIG. 3: Perspective of studio stand.

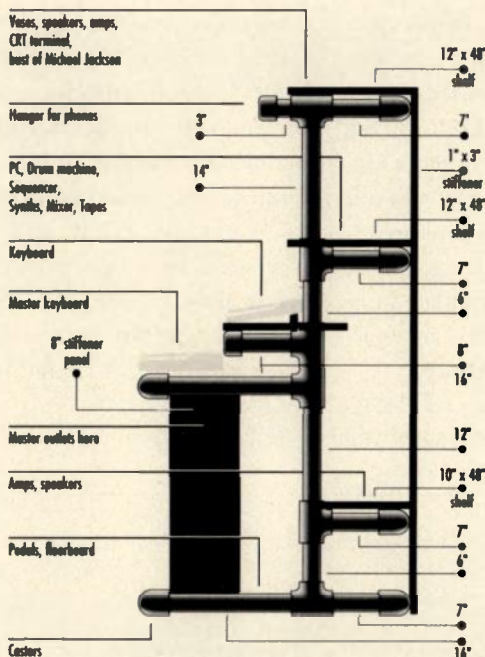


FIG. 2: Side view of studio stand. Dimensions equal actual length of tubing.

At first, I was fearful the assembly would be shaky and weak, but once the rear and two side stiffeners were in place, everything became surprisingly rigid. Before painting, wipe everything down with lacquer thinner and install the rear stiffener. Flat black spray looks great. If you make a serious mistake, it's not the end, because the pipe can be cut back and spliced again with a straight coupler. This will also be your course of action should you wish to expand or modify your stand.

The shelves are the most expensive part of the project, costing about \$10 for the 12 x 48-inch, white, vinyl-coated ones, which look sharp, though there are also less costly options. Predrill the shelves in place and secure them with self-tapping screws. Countersunk drywall screws work fine. To finish up the project, install one multi-outlet strip (with a single switch) somewhere convenient and secure all wires with velcro strips. You can go hog-wild and install a floor board with pedals and casters or a side shelf for a computer keyboard. Let your imagination roll.

By Dan Titus

COIL THOSE CABLES

It's always a struggle putting away microphone cables after a gig. It takes time to coil each cable and place it in its proper place, they get twisted, and your hands and arms wind up smelling like what was on the floor that night.

I now carry all my cords on a light-weight, plastic spool. Connect the microphone cables in series, end-to-end; you can get six 20-foot mic cables on one spool. Wind the



cords onto the spool by turning the attached handle crank, just like reeling in fishing line. You can get spools at K-mart for about \$8 each.

TABLE THAT GEAR

If you're looking for a nice table for your gear, or a cost-effective desk, get a collapsible utility table. I have my entire MIDI setup on two tables. You can't beat the price (\$35 to \$45), and they fold up easily if you need to move them for any reason.

By David Snow

ST BASIC TRANSLATION OF "BEAT-IT"

If you've been honing your computer programming chops on the Atari ST, you've probably come up against the frustrating quirks and limitations of ST BASIC. Many published programs are written in more efficient languages, which is great as long as you have the required compiler or interpreter. Unfortunately, many novices lack that luxury, but it is sometimes possible to translate from one language to another without an unacceptable compromise in performance.

A reader recently requested an ST BASIC translation of the software for "Beat-It," the

drum sensor interface project that appeared in the December 1988 issue of *EM*. Because ST BASIC is too sluggish for time-critical tasks, the program incorporates a machine code module that handles all interface and MIDI business. This version works in the same manner as the LDW BASIC version; refer to the original article for more information. Simply type in Listing 1 (use an ASCII editor, such as 1st Word with WP mode off; the ST BASIC editor is a pain), save the file, load BASIC, and run the program.

By Dan Titus

SHINE WITH CHAMOIS

Chamois (soft pieces of animal hide) are used primarily for washing cars, leaving a clean, streak-free finish without scratching the car; they will do the same for your gear. Use a chamois that has been washed several times to remove any tanning oils, cut it into 5 x 5-inch pieces, and soak the pieces in distilled water (which doesn't contain any minerals or salts found in normal tap water). You can buy chamois at any auto parts store or discount store.

MANUAL MANIA

I've got a number of owner's manuals. The problem is, they're never in the same place when I need them. The answer is to get holes punched in all the manuals and bind them in one large, 3-ring binder. For a nominal charge, a printer will punch holes in any paper stock and in any configuration you desire, or you can buy a single-twist hole punch designed to go through thick booklets or pads of paper (about \$15), and do it yourself.

ST BASIC Version of "Beat It"

LISTING 1

```

10 'BEAT IT - Drum Sensor Interface for the Atari ST
20 'by David Snow: ST BASIC version 1/29/89
30 '
40 defint a-z
50 dim play(149)
60 play(0)=0:           play#=varptr(play(0))
70 progarray(0)=0:      progarray#=varptr(progarray(0))
80 notearray(0)=0:      notearray#=varptr(notearray(0))
90 '
100 def seg=play#
110 for index#=0 to 296
120 read bytes: poke index#,val("&h"+bytes)
130 next
140 data 60,1A,00,00,01,0C,00,00,00,00,00,00,00,00,00,00
150 data 00,00,00,00,00,00,00,00,00,00,FF,FF,48,E7,FF,FE
160 data 20,6F,00,42,2C,50,2A,68,00,04,49,FA,00,F8,30,3C
170 data 00,80,61,00,00,DC,30,3C,00,7E,61,00,04,30,3C
180 data 00,04,61,00,00,CC,3E,3C,00,03,3C,3C,00,06,19,87
190 data 70,00,30,3C,00,CC,00,00,47,61,00,00,86,30,36,60,00
200 data 61,00,00,AE,55,46,51,CF,FF,E6,42,47,42,46,3F,3C
210 data 00,02,3F,3C,00,01,4E,4D,58,8F,4A,40,66,00,00,8C
220 data 3A,39,00,FA,00,00,02,45,00,FF,4A,45,67,00,00,76
230 data 8A,46,67,00,00,70,38,3C,00,07,3A,04,E3,4A,09,05
240 data 67,00,00,5E,36,3C,00,03,88,3A,30,00,67,00,00,2A
250 data 51,C8,FF,F6,19,84,70,00,36,07,52,47,0C,47,00,04
260 data 66,00,00,04,42,47,30,3C,00,00,00,43,61,00,00,42
270 data 30,36,20,00,61,00,00,3A,30,3C,00,90,00,43,61,00
280 data 00,30,30,35,20,00,61,00,00,28,30,3C,00,40,61,00
290 data 00,20,30,35,20,00,61,00,00,18,42,40,61,00,00,12
300 data 51,CC,FF,9B,3C,05,60,00,FF,66,4C,DF,7F,FF,4E,75
310 data 3F,02,3F,00,3F,3C,00,03,3F,3C,00,03,4E,4D,5C,8F
320 data 34,1F,4E,75,00,00,00,03,00
330 ' INITIALIZE PROGRAM/NOTE ASSIGNMENT
340 for drum=0 to 7
350 progarray(drum)=drum*32: notearray(drum)=60
360 next
370 fullw 2
380 '
390 display:
400 clearw 2: gotoxy 0,0
410 print "DRUM:      PROGRAM:  NOTE:"
420 for drum=0 to 7
430 print drum+1;tab(10);progarray(drum);tab(20);notearray(drum)
440 next
450 print: print " ENTER DRUM NUMBER TO EDIT, OR PRESS 'RETURN' TO START:"
460 '
470 getinput:
480 input " ", drums
490 if len(drums)<>0 then goto getdata
500 print
510 print " READY TO PLAY. PRESS ANY DRUM NUMBER TO STOP (CTRL-C TO EXIT). "
520 call play#(progarray#,notearray#)
540 goto display
550 '
560 getdata:
570 drum=val(drums)-1
580 if drum < 0 or drum > 7 then goto display
590 input " ENTER NEW PROGRAM NUMBER OR PRESS 'RETURN': ", prog#
600 if len(prog#)=0 then goto getnote
610 progarray(drum)=val(prog#)
620 '
630 getnote:
640 input " ENTER NEW NOTE VALUE OR PRESS 'RETURN': ", note#
650 if len(note#)=0 then goto display
660 notearray(drum)=val(note#)
670 goto display

```


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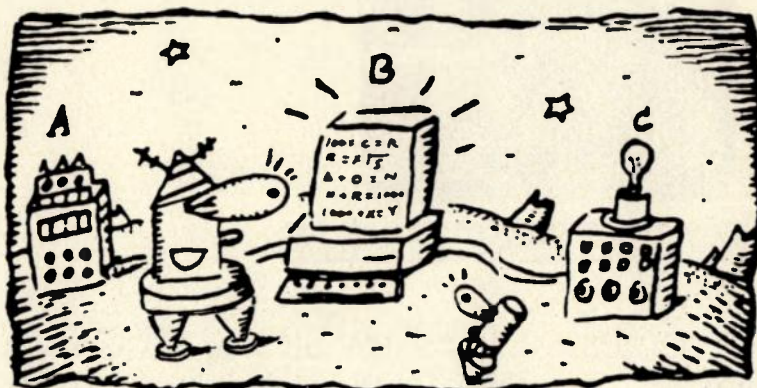
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MIDI Programming in C, Part Two: MIDI Data Debugger

Said the IBM-clonehead who wanted to view the MIDI data arriving at its MPU-type MIDI interface, "Take me to your header."

By Jim Conger



This month's C program, *testmidi.c*, for IBM-clone computers (see Part One in the September 1989 *EM*) allows you to see, onscreen, any MIDI data that reaches the MPU-401 type of interface (the most common MIDI interface for IBM compatibles). You can also send out MIDI data by typing the MIDI commands on the computer's keyboard. These abilities make *testmidi.c* ideal for experimenting; you can try out MIDI messages and see what they do. Later, when you have a group of MIDI messages worked out, you can "hard-wire" them to make your own utility programs. The *mididemo.c* program listed last month is an example of a "hard-wired" utility (see the FYI page for information on ordering back issues).

One warning: *testmidi.c* shows data in hexadecimal notation. Almost all MIDI documentation presents MIDI codes in hex, so you might as well get used to it. See the sidebar if hex is new to you.

USING THE PROGRAM

Let's get *testmidi.c* running. The program uses the *putcmd()*, *putdata()*, and *getdata()* functions presented in the *io401.c* listing last month. Type in *testmidi.c* (see Listing 1) and compile/link it along with *io401.c*.

Testmidi.c is just one big loop that

continues forever until the user hits the ESC key. During each pass of the loop, the program reads any MIDI data from the MPU's buffer into an array called *mdata[]*. After the data is read, all of it is displayed on the screen, in hex, by *printf()*. The program then checks to

see if one of the computer's keys has been hit with *kbhit()*. If so, the keystroke is read by *getch()*. The ESC key causes *testmidi.c* to exit back to DOS. The "c" key causes *testmidi.c* to prompt for an MPU command (like FF hex for reset). Any other key prompts for entry of two hex digits to be sent out the MPU's MIDI out port as MIDI data.

With *testmidi.c* running, try hitting a key on your synth. You will see something like:

```
90 3C 24 <- middle C key hit here
90 3C 00 <- middle C key
              released here
```

If you have your MIDI 1.0 Specification handy (reprinted in several books from the EM Bookshelf or directly from the International MIDI Association, 5316 W. 57th St., Los Angeles, CA

FOR THE BEGINNER

Hexadecimal Base-ics: The Versatility of Sixteen

In everyday life we use decimal numbers. This means we count in base 10. For example, we know the number 127 means one 100 + two 10s + seven 1s (where 100 = 10², 10 = 10¹, and 1 = 10⁰); each digit is some multiple of a power of ten. Computers work in base 2 partly because electronically it's most simple to be either on or off (two choices). Hexadecimal (hex) is base 16 math, and computer programmers use it as a shorthand for writing in base 2 because it's easier to comprehend quickly than a string of ones and zeros (which is all there is in base 2). For example, the number 100, in decimal, can be expressed as:

$$100_{10} = 0110\ 0100_2 = 64_{16}$$

Hex may look complicated, but it's not. Base 16 requires sixteen digits just as regular base 10 has ten digits (0 to 9); the last six digits of base sixteen are made up with the letters A to F.

Decimal:

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Hex:

0 1 2 3 4 5 6 7 8 9 A B C D E F 10

So A₁₆ = 10₁₀ and F₁₆ = 15₁₀. To convert a decimal number to hex, treat each digit in hex as a multiple of some power of 16. (Remember that anything raised to the power of zero equals one.)

90056), you should be able to decode all this data. 90 hex is MIDI for note on, channel 1. 3C hex is the MIDI note number for middle C. 24 is the key velocity for a medium-soft keypress. You get the idea.

Try sending your synth some data. Press the spacebar, then enter 90<return>. Continue with 3C, 24. You should hear your synth play middle C (like magic!). Shut down the note with 90 3C 00. Try the pitch bender, the synth's data dump button (if it has one), and changing the synth's MIDI channel.

One thing you may find confusing is MIDI running status, which improves the efficiency of MIDI data transfers. All MIDI status messages have numerical values above 7F hex. If the same status message is transmitted repeatedly (running status), save some time by transmitting only data bytes after the first status message. For example, our note on/off message pair above may show up as 90 3C 24 3C 00 if your synth uses running status. The second 90 hex is "assumed."

Fool around with testmidi.c and your MIDI setup. If you get really adventurous, you can even put the MPU into the record mode. Send commands FF (reset) and 22 (start record). The MPU will start putting out F8 (time out) markers and F9's (measure ends). MIDI messages will be tagged with timing

Let's convert 8023₁₆ into a decimal number.

$$\begin{array}{r} 8 \times 16_{16}^3 = 8 \times 4,096 = 32,768 \\ + 0 \times 16_{16}^2 = 0 \times 256 = 0 \\ + 2 \times 16_{16}^1 = 2 \times 16 = 32 \\ + 3 \times 16_{16}^0 = 3 \times 1 = 3 \\ \hline 32,803_{10} \end{array}$$

Try these other examples (note: if there is no base value on a given number, assume it is decimal): 1C₁₆ = (1 x 16) + 12 = 28; F1A₁₆ = (15 x 256) + (1 x 16) + 10 = 3866 decimal.

MIDI data is usually represented by groups of two hexadecimal digits. In the C language, hex numbers are written with a "0x" before the digits. 0x4F = 4F hex = (4 x 16) + 15 = 79 decimal.

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● C PROGRAMMING

bytes. This is the beginning of sequencing on the MPU. Take a look at your MPU manual for other commands and try them out.

FINAL COMMENTS

If you work with MIDI a lot, you end up doing math in hexadecimal. You can save time by using *Sidekick's* calculator, which will switch back and forth between decimal, hex, and binary. I recently switched to *PC Tools Deluxe*, which has the same abilities plus many more. *PC Tools Deluxe* also has the best user interface of any program I've run on any machine. That's subjective, I admit, but

with a mouse, *PC Tools* takes on the best applications ever put on a Mac, *Meta-phor workstation*, or whatever.

Next month we will use *testmidi.c* to experiment with sending and receiving patch data from a synth. This is the first step in writing your own patch librarian.

(*Jim Conger's C programming books are available from EM Bookshelf; see FYI page for more information.—Ed.*)

Jim Conger is the author of *C Programming for MIDI* and *MIDI Sequencing in C*. His current ambition is to reduce the U.S. birth rate by popularizing *MIDI programming*.

C Language MIDI Data Debugger

```
/* testmidi.c MIDI data input and output via the mpu-401 */
/* link with io401.c - Jim Conger 1989 */

#include <stdio.h> /* library function header files */
#include <conio.h>
#include <ctype.h>
#include "io401.h"

#define BUFSIZE 100 /* how big to make data buffer */
#define ESC 27 /* ASCII code for ESC char */

main()
{
    int c, i, j, k, x, result;
    int mdata[BUFSIZE]; /* midi data buffer */

    printf("\n **** READMIDI - MIDI Data Input And Output ****");
    printf("\n\nThis program sends and receives MIDI data.");
    printf("\nAll data is displayed in hexadecimal (base 16).");
    printf("\n\nTo enter MIDI data, first hit the space bar.");
    printf("\n\nTo enter a MPU command, first hit the C key. Then");
    printf("\n\nenter MIDI data/commands by typing in two hex digits.");
    printf("\n\nExit this program by hitting the ESC key.\n\n");

    set_delay(50); /* slow down i/o for 20 mhz computer */
    putcmd(UART); /* put MPU-401 in UART (dumb) mode */

    while (1){
        j = 0;
        while ((i = getdata()) != -1){ /* read MIDI data until none left */
            mdata[j++] = i; /* put data in the buffer */
            if (j > BUFSIZE - 1){
                printf("\n*** MIDI data overflowed buffer ***");
                break;
            }
        }
        if (j){ /* if there is some data */
            for (k = 0; k < j; k++){ /* print all there is in one shot */
                if (mdata[k] >= 0x80) /* skip line on each MIDI message */
                    printf("\n");
                if (mdata[k] != 0xFE) /* don't print active sensing bytes */
                    printf(" %02x", mdata[k]);
            }
        }
        if (kbhit()){ /* if key pressed, prompt for MIDI data */
            c = getch();
            if (c == ESC) /* quit if ESC key pressed */
                break;
            printf("\nInput two hex digits, then <ret>. -> ");
            result = scanf("%x", &x);
            if (result == 0)
                printf("\nUse only two 0-9,A-F digits. ESC to exit.");
            else{
                if (toupper(c) == 'C'){
                    i = putcmd(x); /* send command to MPU */
                    printf("\nputcmd() returned %02x", i);
                }
                else{
                    putdata(x); /* send data to MIDI out */
                }
            }
        }
        putcmd(RESET); /* put MPU-401 back to startup state */
    }
}
```

Listing 1: MIDI Data Debugger

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The J.L. Cooper PPS-1 Hot Rod Mod

A single resistor change and a few minutes work will improve your life—or at the very least, get more level out of your PPS-1 sync box.

By Robert Damiano

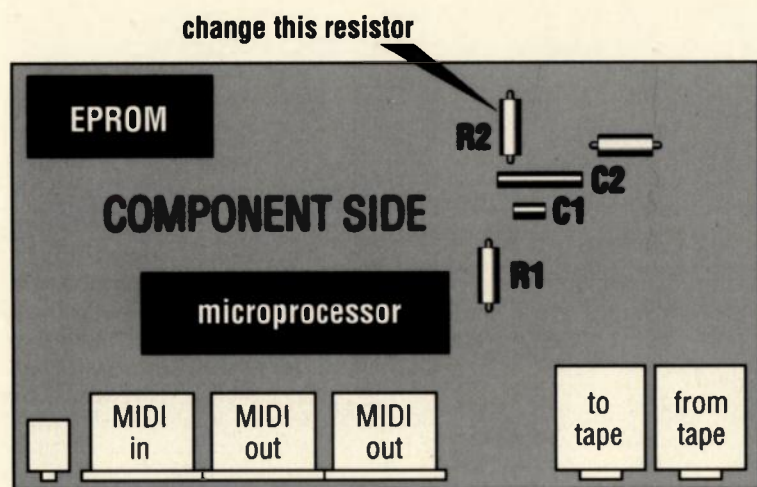


FIG. 1: How to locate R2 on the PPS-1 circuit board.

The J.L. Cooper PPS-1 sync box is a convenient and cost-effective way to synchronize most MIDI sequencers to tape, but there is a problem: the output is barely enough (and sometimes not enough) to drive a -10 dBm recorder (the operating level for most low-end pro tape recorders). Although you can always use a preamp or your board to boost the signal, there's a better way. This simple modification changes a single resistor inside the PPS-1, producing a hotter output (Fig. 1). (Note that this mod is only for early PPS-1 units; later units have the resistors mounted in single, in-line packages, which makes modification more difficult. Although J.L. Cooper does not endorse this mod, the company foresees no obvious problems if it is installed with care. Of course, performing any mod on a piece of equipment will void the warranty.—Ed.)

As is, the PPS-1 will print (record) a level of about -10 to -12 VU (Volume Units) on my Fostex E-16, and that's just a little too weak to be 100% reliable. On the other hand, recording time code at

too high a level can cause "bleeding" to the adjacent track(s). With my setup, the optimum time code output level is about -2 to -4 VU, which this mod achieves. (By the way, its manual doesn't explain this, but if you have a Fostex E Series recorder, you can switch the Dolby off on the outside tracks to record time code. Space limitations prohibit a complete discussion, but the switch to do this is right under the calibration pots on the two outside tracks: 1 and 8 [E8], or 1 and 16 [E16].)

Fig. 2 shows the PPS-1's output—a 5V_{pp} (volts peak-to-peak), TTL level pulse, which is much too strong for my tape recorder. Attenuator R1/R2 reduces the signal level to about 0.2V_{pp} (volts peak-to-peak) with R2 equal to 1k. Changing R2 from 1k to 3k3 (3.3k) increases the signal to about 0.6V_{pp}, which registers about -3 VU on a -10 dBm machine.

Trying to get more level for a +4 dBm deck (the operating level for high-end pro recorders) requires making R2 much larger, which implies making C1 correspondingly smaller since it pro-

vides a filtering effect in conjunction with R2. I didn't pursue this, but I mention it so that if you ever need to drive a +4 dBm machine with a PPS-1, you have a place to start.

WHAT'S REQUIRED?

You'll need a soldering iron, some solder, a 3k3 resistor, wire cutters, solder wick or solder-sucker, a screwdriver, flux solvent, static wrist strap, and a clean workbench. Before doing anything, put on the static wrist strap and ground it to earth ground. Straps are available at Radio Shack and other computer/electronics stores. The PPS-1 has some very complex circuitry inside that's easily damaged by electrostatic discharge (ESD; see sidebar).

First, disassemble the PPS-1 by removing the four screws on the bottom and the single screw on top. Then remove the bottom of the box. Second, remove the nuts from the To Tape and From Tape ¼-inch jacks. Now, with a little wiggling (and luck) the circuit board should come out of the enclosure. Your wrist strap is on, isn't it?

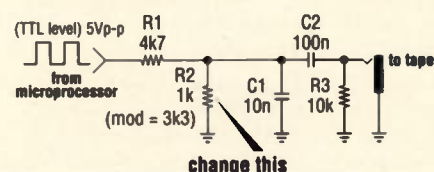


FIG. 2: PPS-1 output stage schematic.

Now place the circuit board on your workbench and put the box and screws in a safe place. Referring to Fig. 1, locate R2, the resistor to be changed.

Use either a solder-sucker or piece of solder wick (copper braid that absorbs molten solder) to remove the solder holding R2 to the board. Once the solder is removed, it should be easy to gently



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● PPS-1 HOT ROD MOD

pull R2 from the board with a pliers, especially if you heat R2's connections, but *be careful and patient*. Fortunately, the PPS-1 uses a high-quality, double-sided

board that is relatively difficult but not impossible to damage. Excessive heat is the main culprit to beware of here, so if at first you don't succeed, let the board

cool before you try again.

Replace R2 with a 3k3 resistor, solder it in, and you're done. Although it's usually considered good practice to

FOR THE BEGINNER

What's Electrostatic Discharge (ESD)?

Electrostatic electricity is what happens when you run your shoes across a rug on a dry winter day: your body accumulates an electrostatic charge that can reach tens of thousands of volts. Touching metal discharges the accumulated charge (an electrostatic discharge —ESD), and if that something metal is a complex integrated circuit, it might not survive thousands of volts rammed down its little semiconductor throat.

ESD damage can be obvious (e.g., a blown chip), or it can manifest itself over time by weakening the IC, but not destroying it. ESD can scramble

semiconductor memory data and cause other problems far too numerous to list.

While much gear is protected from ESD (especially devices in grounded, metal cases), card connectors, I/O ports, and other areas that expose the internal circuitry are very vulnerable.

When working on electronic gear, you can minimize ESD damage by always making sure your body has not built up any static charge. The easiest way to do this is to wear a conductive wrist strap attached to an earth ground. The strap drains off any charge from your body. Better-quality soldering stations have

grounded tips so that the iron itself doesn't cause static damage.

There is one serious caution, though: if you're grounded, *do not, repeat, do not touch any voltage sources whatsoever, even batteries*. Make sure all units you come in contact with, except for the soldering iron, are unplugged. Any current entering your body and flowing through you to ground can cause ventricular fibrillation, a dangerous heart trauma caused by electrical current flowing through the heart.

Working on electronic gear is fun and productive, but exercise caution at all times. —Craig Anderton

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clean the PC board with a solvent to remove solder flux (*which can absorb water in humid environments—Ed.*), this creates an environmental dilemma, as these agents use chlorofluorocarbons (CFCs), the chemicals that are eating holes in our ozone layer. Either use just a little bit of flux remover, or forego it altogether until an environmentally acceptable substitute is developed.

TESTING 1, 2, 3

Reassemble the unit before testing, because the box provides needed shielding. I had a brief panic attack when I tried the PPS-1 with the board completely outside of the box—glitch city.

Connect the unit to your tape recorder as normal, and set the switch to To Tape. Now monitor the PPS-1 signal on your deck. If everything is on and your machine is set to the correct channel, you should see a level somewhere around -2 to -4 VU. If you have an oscilloscope, view the signal with no MIDI clock going in. It should be a square wave of about 0.6V_{pp}.

Now, before you get too comfortable, try recording a sync track on tape and synchronizing your sequencer to it. This mod only affects the operation of the PPS-1 when recording a sync track, so testing it with a previously recorded track won't change anything.

PROBLEMS (?)

I didn't have any problems; the only possible source of error is C1, whose value would need to be lowered in order to keep the same frequency response for the R1/C1 filter. If you experience difficulties and you're sure you didn't do any damage or misread a resistor color code, try substituting a 30 to 50 nF (0.03 to 0.05 µF) capacitor for C1 before you give up. Since mine worked fine, I didn't bother changing the cap, and you probably won't need to either.

So there you are: for the price of a resistor and a little time, you just made a product of exceptional value even better! I hope this mod benefits your studio as much as it has mine.

By day, Robert Damiano works in a robotics factory, striving for better component insertion through science. By night, he operates his 16-track studio, Neo-Sync Labs, slowly breaking into the music-for-video scene and producing solo projects. He also designs music software and 16-input, trigger-to-MIDI converters.

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FIRST TAKE: Capsule Comments

This autumn, our reviewers catch the (Acoustic) Wave, interact with OvalTune, and take command with programmable footswitches.



All EM reviews include 11-step "LED meters" showing a product's performance in specific categories chosen by the reviewer (such as ease of use, construction, etc.) and a "VU meter" indicating an overall rating. The latter is *not* a mathematical average, since some categories are more important than others. For example, if a guitar synth has great documentation and is easy to use, but tracks poorly, it could have several high LED meters and a low overall rating.

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Please remember that these are opinions, and, as always, EM welcomes opposing viewpoints. We urge you to contact manufacturers for more information, and, of course, tell them you saw it in EM.

the EM rating system

**Ensoniq/Bose
Acoustic Wave Piano
(AWP-5000, \$2,995-
\$3,495; AWP-4000,
\$2,795-\$3,295; AWP-
3000, \$2,495-\$2,995)**

By Daniel Kumin

The Ensoniq/Bose Acoustic Wave Piano represents a genuine step forward in electronic piano design. Ensoniq's sampler and VLSI expertise, combined with Bose's intimate knowledge of speaker design and acoustics, has produced an instrument blessed with both a rich supply of onboard musical raw material and the acoustical reproduction system needed to deliver it into the room.

The AWP-5000 is a 24-voice, polyphonic, 88-key instrument finished in a fine imitation of Steinway black. A pair of loudspeakers fire from the left and right extremes of the top, which also presents a 29-key, two-slider control panel (with headphone jack) and a traditional music rack. There are two ped-

als; the right one provides sustain, while the left does double duty for both *una corde* (soft pedal) and *sostenuto* functions. The back panel includes MIDI in/out/thru jacks, pairs of RCA audio jacks for stereo in and out, and a cable for the pedal controls. The line inputs mix with the piano so you can play along with other sound sources (CD or tape, drum machine, another musician, etc.). The back panel is rounded out with a three-position velocity curve selector switch.

Overall, the AWP-5000 provides a very close simulation of the "real" piano experience, albeit in stereo, and a good part of the credit goes to the Bose sound system. It employs two minuscule "switching-mode" amps, a pair of full-range Bose 901-type 4.5-inch drivers, and the firm's patented Acoustic Wave technology. This last is a labyrinthine tube enclosure, computer-tuned to permit a small driver to reproduce full-range sounds with limited cone motion. The drivers are mounted such that much of their sound reflects from the facing wall, thus—like a real piano—providing a blend of direct and reflected sound at the performer's ears.

The AWP-5000's sound quality is pretty terrific. The instrument has 1.5 megabytes of 13-bit samples, shared among ten different sounds. The bottom two-plus octaves of the main "Steinway" sound, Grand 1, are nothing short of amazing; they're good enough to fool almost anyone and, more important, satisfying enough to play for hours at a sitting. There's lots of guts, brassy ringing, hammer thud, and dynamic tonal changes. The top two-and-a-half octaves are equally good, with uncanny hammer ping and harp ringing. The middle two—seemingly the soft underbelly of sampled pianos in general—are more disappointing. They are very good, but lack the extremes of realism and impact that the outer regions offer, and they

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can sound a little "phasey" on sustained tones. (There's also more background noise across the range than one would expect from the claimed equivalent of 24-bit resolution.)

The Grand 2 patch, which is of about the same quality, sounds more like a Yamaha grand piano. The remaining eight sounds are: Ragtime, a detuned piano sound (okay, but who needs it?); Electric 1/Electric 2, fairly generic Rhodes and Yamaha sounds; Harpsichord, excellent; Vibes and Marimba, superb. A good acoustic bass sample and

a mediocre electric bass round out the sonic complement. These last two, when selected, cover the bottom three octaves in a split mode.

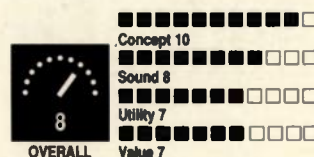
As a MIDI musician's main keyboard/piano module/controller, the AWP has both strengths and weaknesses. Among the former are terrific sound, piano-like feel, and the fact that it nurtures compositional creativity by encouraging you to sit and play for long periods, without constantly fiddling with buttons and knobs and without growing bored or aggravated by the instrument. Also keep

in mind that this is one instrument (unlike the average synthesizer) that others in your household may find useful.

The weaknesses are fairly simple: lack of elaborate MIDI mapping or control parameters (the AWP-5000 transmits and receives only on channel 1), a rather bulky, non-tabletop design, relatively high cost, and an onboard sequencer that's only useful as a scratch pad. To be fair, it's handy to store an idea without booting up your computer, but there's only so much you can do with a very basic, 4-track sequencer with 2,700 notes of memory that can only play internal sounds. (Since this review was written, a local on/off feature has been added so the AWP can control external MIDI sound sources. —Ed.)

For anyone much beyond the *Für Elise* level of pianistic accomplishment, I still feel compelled to recommend the real thing. But for beginners, occasional players, families, and general hackers, the AWP-5000 (or its siblings, the AWP-3000, which deletes the sequencer, four voices, and an "enhance" effect for about \$500 less, and the AWP-4000, which offers the AWP-5000's features in the less-expensive AWP-3000 casing) is an instrument that should be of interest to electronic musicians everywhere.

Daniel Kumin is technical editor of *CD Review magazine*. He lives in New Hampshire.



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Intelligent Music
OvalTune (\$145)
By Bob Tarte

Intelligent Music's *OvalTune* stakes out a middle ground between the company's formidably complicated *M* and Aesthetic Engineering's immediately playable *Music Mouse*. *OvalTune* draws upon features from each of these programs, adds an impressive batch of sampled sounds—you

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don't need a synthesizer to use the program, just a Macintosh—then tosses in graphics capabilities intended for packaging, fabric design, or video. Try *that* on your typical synth.

Unlike a sequencer, OvalTune isn't a real-time composing environment (although you edit as you listen). Instead, you input patterns of notes a single note at a time, using the Mac or your MIDI keyboard to create loops. Each pattern may be up to 32 notes long—no chords allowed—and eight patterns can play at once.

Each pattern receives its own rhythmic treatment. Building a rhythm cycle up to sixteen steps long, you program two sets of values that repeat as your pattern plays. The Probability Cycle decides whether a note falling on a given step will play or not (five values from 0% to 100%). The Accent Cycle lets you choose one of five velocity settings (including zero) for each step.

Since rhythmic treatments are pretty basic, they're seemingly better suited to drum patterns than melodic elements. You can't speed up or slow down a single



OvalTune's Rhythm window depicting a seven-step Probability Cycle and an eight-step Accent Cycle.

step of your pattern as you can in M, yet the interplay between probability and accent allows a surprising degree of variation.

Besides loops that play automatically, OvalTune lets you perform one or more patterns in real time using the mouse. Mouse movements control note order, velocity, duration, and tempo, as they simultaneously generate kaleidoscopic graphics. But the mouse play is disap-

pointing and a far less effective means of controlling sequences than M's conducting grid or Music Mouse's onscreen keyboards. Wheeling the mouse in manic circles to perform a tune, I felt as if I were cranking a jack-in-the-box. I expected the top to pop off my Mac.

To make mouse control meaningful, graphics should supply feedback as to what is happening musically instead

of simply providing an end unto themselves. Sure, they're dazzling, but they hinder rather than aid composition by preventing multiple windows from opening at a time. Jumping from patches to rhythms, for example, demands a trip to the keyboard or menu bar, in mid-song.

OvalTune's split personality is puzzling. Do graphic artists need to whistle while they work? Do synth players long

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to bitmap Mondrian? This either/or approach verges on neither/nor for the MIDI musician. You can't step-edit a pattern or see the notes you've entered. Nor can you cut, paste, or delete a pattern; you have to overwrite the notes.

Yet, for all of this, I had a ball with OvalTune. The neatest composition feature is the ability to program macros, called "Presets." These hold up to nine independent blocks of eight patterns, each with corresponding melodies, rhythms, patches, samples, tempos, and other goodies. A seamless flow from verse to chorus to drum solo is a few key clicks away, and an entire performance can be recorded in real time as a "video-tape" (with attendant graphics) if you wish to carve your song in stone.

One of OvalTune's best touches is a disk containing 52 samples to be used with or in place of your MIDI keyboard patches. Sound quality of the 8-bit samples varies, depending on how many samples you want to play at once and the type of computer you have, but overall

they sound pretty good. OvalTune's SamplePlayer component also accepts sampled sound files from *Sound Designer*, *Turbosynth*, *Alchemy*, and other sources.

Like M, OvalTune can be a rewarding composing environment if you're willing to share songwriting credits with your Mac. It won't replace a sequencer, but it will provide a kind and gentle entry into interactive music software.

Bob Tarte is a writer/graphic artist who writes the "Technobeat" column for *Reggae* and *African Beat* magazine. He lives in Michigan and wishes he had a pet crow named Roger.



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Meico MIDI Commander (\$369) and Patch Commander (\$299) Programmable Footswitches

By Alan Gary Campbell

Once my system became complex enough to provide the capabilities I wanted, I promptly discovered I didn't have enough hands/feet/noses to control it all. This called for desperate action (i.e., buying more gear) to see if I could solve the problem.

First, I tried out Meico's MIDI Commander, a solid-feeling, multi-footswitch gizmo that records and plays back MIDI note and program change data. It has three footswitches up top for programming/bank-memory access, a 2-digit LED display, and six data playback footswitches that do the work of eleven. The six switches, labeled 1, 3, 5, 7, 9, and 11, call up like-numbered programs; hitting two adjacent switches at once calls up the "in-between" program (e.g., switches 1 and 3 call up program 2, 3 and 5 call up program 4, etc.). This saves switches (and money) and works pretty well, at least with my size-10 foot. The MIDI Commander also sports MIDI in and out/thru jacks, a power switch, and

a nondetachable line cord.

Programming is easy: select a bank (my terminology; the manual refers to a bank of eleven virtual switches as a "Song"), press the switch or switch combination of your choice, and, from your MIDI controller, send the program change number and/or MIDI note or chord that you want to record. You can store a different program change number and a ten-note chord (no sequences, though) per switch, on any channel. You could put a bass drone on channel 2, switch 1; an orchestra hit on channel 12, switch 2; and so on. If your hands are busy, and you need to play more notes on lots of keyboards, this will do it.

When you first enter Program mode (or Play mode), the MIDI out acts as a MIDI thru. Unfortunately, selecting a program number disables this function, so you can't monitor the effect of data as it's stored or merge the MIDI Commander output with that of your controller.

To call up banks when playing, you use the Units and Tens switches with the Song Direction (increment or decrement) switch—not a bad arrangement, as multi-footswitches go. It's fast enough to change banks between tunes on a gig. MIDI Commander stores 99 banks and remembers the program data when powered down.

Next, I tried the Patch Commander. It's also a sturdy, multi-footswitch, MIDI program change device. Like the MIDI Commander, it has three footswitches for programming/bank-memory access, an LED display, six footswitches for program recall (these only work singly), MIDI in and out/thru jacks, a power switch, and a non-detachable power cord.

Unlike the MIDI Commander, the Patch Commander is *not* easy to program. You can store up to ten program change numbers per footswitch, per bank, each on a different MIDI channel. This is great for controlling lots of effects and keyboards without having to rearrange patches, but entering that much data requires 64 presses per switch! (The unit won't record program change data from an external controller.) You have to convert from decimal to hex as you go; the display mnemonics and somewhat confusing manual don't help much, either.

Like most program change footswitches, the Patch Commander merges note on/off and performance control

data appearing at its input with its internally generated data, but it occasionally had stuck-note problems when merging lots of data, whether it was sending program change commands or not.

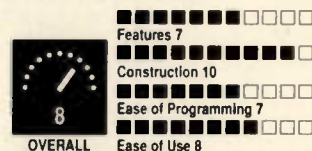
Patch Commander stores 60 banks and remembers the data when powered down. But banks are accessed solely by increment and decrement switches. These scroll when held, but not very quickly. When the club's best customer puts a fifty in the tip jar, and the drummer breaks rhythm and shouts "One, two, ..., " I don't think you'll be able to get from bank 4 to bank 58 before about bar three. (*According to the manufacturer, newer units have been updated for a faster scrolling rate, and any bank can be accessed in five seconds or less.—Ed.*)

The MIDI Commander is fun but I wish it would merge input data, at least in Program mode. The Patch Commander is more utilitarian, but the merge function needs work, and if you want "user-friendly," *caveat emptor*. Both may seem a bit pricey, but they're aimed at a specialized market, and—quibbles aside—their steel enclosures, sturdy

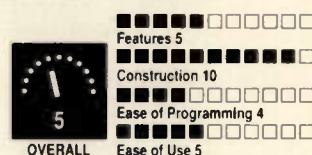
footswitches, and built-in power supplies are some of the most rugged I've seen.

Alan Gary Campbell is owner of *Musitech™*, a consulting firm specializing in electronic music product design, service, and modification; author of "Service Clinic"; and contributing editor for *EM*.

MIDI Commander



Patch Commander



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S T E A L T H

Peavey MIDI Effects Automation System

Peavey has embraced the world of MIDI—and the result is affordable, onstage MIDI and audio control.

By Robert Rich



C.R. KING

Guitarists have had programmable pedal boards for several years now, but most stomp-box programmers aren't capable of handling multiple instruments, let alone multiple functions. In today's live environment, where you want to reroute effects, mute instruments that aren't in use, change presets of a MIDI-controlled reverb, tell your sampler to load a new sound, run a tape recorder to capture your live performance, and ignite some fireworks stage left, a simple stomp-box programmer just doesn't cut it.

Of course, anything that would be cool enough to do all these things would cost a fortune and require two engineers to program it, right?

Wrong. The Peavey AMR (Audio Media Research) division has set out to prove that cool things can be simple and

fairly inexpensive, and their innovative MIDI control system seems well-suited to the needs of small studios and performing MIDI musicians. Peavey's solution for some very specific problems related to automated remote control involves four separate devices, each with a small and well-defined set of functions. These devices are easy to learn and use and fairly inexpensive, with an elegant sense of simplicity and a good solid feel.

SYSTEM OVERVIEW

The four devices in Peavey's system can be used separately or together, but they make the most sense treated as a single system since each unit seems specially suited to the needs of the others. At the hub of the system sits the MIDI Manager, which sends various kinds of setup information when it receives MIDI pro-

gram changes. Connecting to a 9-pin port at the back of the MIDI Manager, the Event Automation Controller (EAC-8) has eight relays that can open or close on command, allowing remote control of tape machines, slide projectors, guitar amps, or even pyrotechnics. The Map 8x4 is a programmable patch bay that can assign eight mixer inserts to any of four effects. Finally, there's a cute little hand-held remote, the MIDI Director, which simply sends MIDI program changes and song start/stop commands.

In a typical system setup (Fig. 1), the MIDI Director might sit next to your mixer or master keyboard, sending MIDI program changes to the Manager, whose outputs control the Map 8x4, the EAC-8, and perhaps a few other MIDI boxes.

Three of the units in Peavey's system

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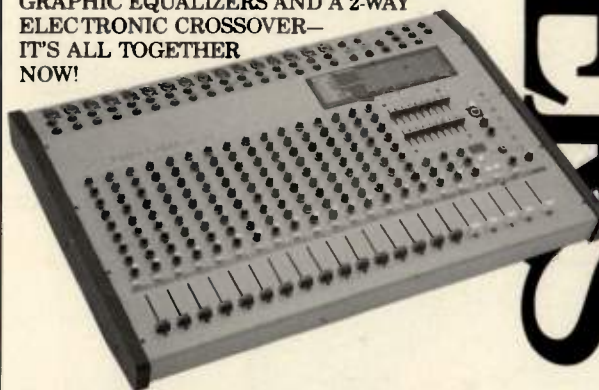
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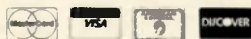


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• PEAVEY AUTOMATION

take their power from 16 VAC adapters, which are fairly big and hog a lot of space on a power strip. They may save money and make for easier UL approval, but from a user's view, they're a pain. In most every other way, though, this system looks good and the components work well together. Let's examine each product separately.

MIDI DIRECTOR

This small metal box runs off a single 9V battery, although it has a jack for an optional external adapter. Its top face holds three 7-segment LED displays, along with buttons labeled 0 to 9, Preset, Chan, Song, Start, and Stop. Small LEDs next to each of the command buttons light up when you press the associated command—very handy. Sending a program change over a particular channel is easy, as is sending song select commands.

The MIDI Director always defaults to channel 1 on power up; you can send messages over any one channel, but to send changes over multiple channels requires sending a MIDI song select command to the MIDI Manager, causing it to chain a number of presets together.

For sequencer and drum machine control, the Start button sends MIDI start messages (which of course restart

the sequence or song from the beginning), while the Stop button sends stop messages. Press Stop again during a stopped sequence, and it sends out a continue message.

MIDI MANAGER

The MIDI Manager is one of the most straightforward mapping devices I have ever seen. Although limited in its range of functions, it is a breeze to program and, despite its simplicity, boasts some unique and powerful features. Unlike some mappers that can convert any MIDI message to any other, the Manager responds only to program changes. Sending a program change to the Manager calls up one of its presets, which will in turn send a small set of data to other devices.

The MIDI Manager talks to the rest of the binary world via a single MIDI in, MIDI thru, and four MIDI outs. These four outputs can simply spew preset data from the Manager or merge preset data with input data. The Manager also has a 9-pin (DB9) auxiliary port, which can send 8-bit parallel messages to the EAC-8 (as described later).

The front panel sports a friendly, 80-character, backlit LCD display, which definitely helps make programming easier. The display has no contrast control, but luckily it doesn't really need one. A



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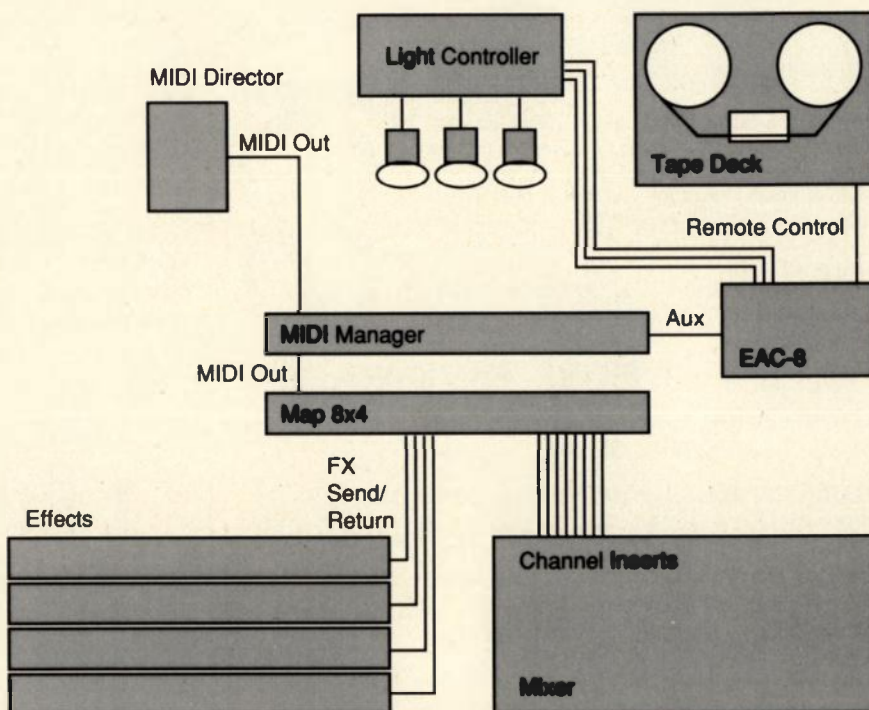


FIG. 1: A typical hookup for the Peavey MIDI Director, MIDI Manager, MAP 8x4 and EAC-8.

mode button selects between Control and Program modes; corresponding green and red LEDs indicate whether you are using the Manager or programming its presets. A Store button saves your new presets, a Param button cycles through the displayed parameters, and a pair of increment/decrement buttons change values one step at a time. Ten number buttons (0 to 9) let you select preset numbers (you can use these while editing instead of the inc/dec buttons), a Call button activates the selected preset, and the Alt button accesses alternate screens for editing. Everything does just what it says, which makes reviewers like me very happy.

Each preset can send one program change command on one MIDI channel, through one (or all) of the four MIDI output jacks. A preset can also send up to fourteen MIDI hex bytes, as well as a separate hex byte through the auxiliary port. Since the aux port is intended primarily for the EAC-8, you can think of this last 8-bit number as eight separate on/off controls for the EAC-8. The Manager even provides a screen that lets you program these eight bits individually as "hi" or "lo," perfectly suited to the EAC-8.

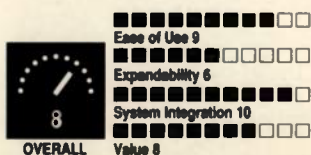
It's unfortunate that each preset can send only one program change on only one channel. However, there are two ways around this restriction. The first method uses the fourteen program-mable hex values, which can be sent through a different MIDI output from the program change message. These numbers could conceivably contain up to seven more program changes on independent MIDI channels, but of course this requires you to program MIDI messages in hex.

The other way to send more complex messages involves the *chain* function. You can connect up to sixteen internal presets within each chain and link chains together to create longer messages. The Manager will remember 128 chains and play them back like sequences. You can select chains with MIDI song select commands and play them with MIDI start messages—sounds like a job for the MIDI Director. When the Manager sends a chain, it does so almost instantaneously; however, there are times when I would like to send a chain slowly, and there is no simple way to do this.

All told, the Manager is a simple device, but it performs its duties well.

Product Summaries

- **PRODUCT:**
MIDI Director
TYPE:
MIDI remote control unit
FEATURES:
Sends MIDI program change, song select, start, and stop/continue
PRICE:
\$109
- **PRODUCT:**
MIDI Manager
TYPE:
MIDI system setup controller
FEATURES:
Translates MIDI program change commands into user-definable MIDI commands; 4 MIDI outs; controls EAC-8
PRICE:
\$399
- **PRODUCT:**
EAC-8
TYPE:
Relay-switching remote control unit
FEATURES:
Eight relays controlled by MIDI Manager
PRICE:
\$119
- **PRODUCT:**
Map 8x4
TYPE:
MIDI-programmable patch bay
FEATURES:
Routes four effects to eight mixer insert points; individual volume controls; effects can be chained; frequency response: 20 Hz - 20 kHz ± 1 dB; S/N ratio: 90 dB (effect), 100 dB (dry)
PRICE:
\$449
- **MANUFACTURER:**
Peavey Electronics
711 A St.
Meridian, MS 39301
tel. (601) 483-5365



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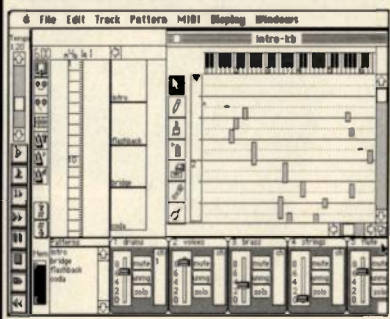
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EAC-8 EVENT AUTOMATION CONTROLLER

Now here's something unique. While the EAC-8 is just a box with eight relays, when combined with the MIDI Manager, it lets you control just about anything that has a footswitch jack or switch-style remote control. This includes tape transport remotes, effects unit bypass switches, pyrotechnics and lighting effects, switching between clean and grungy guitar preamps, synchronizing a slide show to music, etc.

One limitation is that the EAC-8's relays are rated at only 100V, 500 mA, so they cannot switch high-current power sources. For example, if you want to use the EAC-8 to control lights, you cannot switch those lights directly; you'll need a lighting controller that responds to external control, or you can use the EAC-8 to control relays whose contacts can handle higher power. However, this limitation still leaves plenty of uses for the EAC-8.

The EAC-8's control signals come from the MIDI Manager via the Man-

ager's DB9 Auxiliary Port. Unfortunately, the EAC-8 does not come with a suitable DB9 cable, so you will have to find one at a computer store or make one yourself. (Thank goodness they're standard!) Peavey really should include the cable since the EAC-8 is almost useless without it.

Eight tip-ring-sleeve, 1/4-inch phono jacks provide access to the single-pole, double-throw relays; the tip connects to the normally open relay contact, the ring to the normally closed contact, and the sleeve to common. By including both normally open and normally closed connections, the EAC-8 will work with standard and "reverse logic" switches—which should make many Yamaha owners happy. An LED next to each jack indicates whether its relay is activated.

MAP 8x4

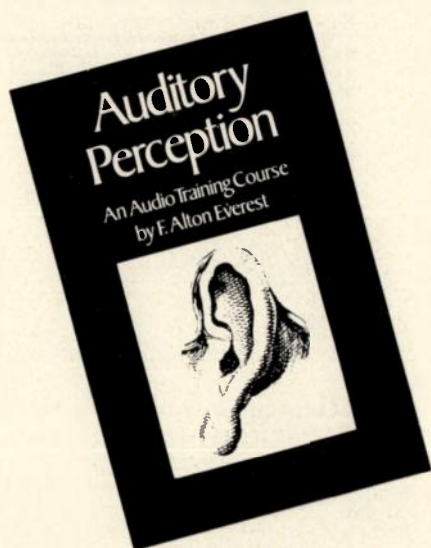
Here's yet *another* nifty automation device—a MIDI-controlled patch bay, optimized for patching effects into the insert points of a mixing board. It may not

be as flexible as a mechanical patch bay, but it's a lot more sexy.

The rear panel holds a total of 24 1/4-inch inputs and outputs. Eight inputs and eight outputs connect to the mixer's insert points (or any other signal that needs processing). The other four sends and four returns connect to external effects. Along with these audio connectors you will find MIDI in, out (used for sysex memory dumps), and thru, and a receptacle for the power transformer cord.

Like the other components in Peavey's MIDI system, the Map 8x4's front panel is intelligently arranged. Each of the four "effect stations" has its own single-character LED display, select button, and volume knob. A separate LED display tells you which preset you are in, and corresponding increment, decrement, store, and recall buttons let you navigate the preset memories. A bypass button disconnects all effects, letting the eight signals pass unhindered, and a power switch puts the Map to rest. This is actually a soft power switch—the unit

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Dear *Electronic Musician* reader,

We would like your input! To help us continue to give you the kind of magazine you want, each year we need to gather some information about you and your areas of interest. We would be very grateful if you would read and fill out the questionnaire below and return it to us by **October 31, 1989**.

ABOUT YOU:

1. What is your age? _____
2. Are you ☐ Male? ☐ Female?
3. What is the highest level of education you have currently attained?
 - ☐ Have not finished high school
 - ☐ High school graduate
 - ☐ Junior college graduate
 - ☐ College graduate
 - ☐ Postgraduate studies
 - ☐ Master's degree
 - ☐ Doctorate degree
4. What types of music are you most interested in? (Check all answers that apply.)
 - ☐ Classical
 - ☐ Jazz
 - ☐ Top 40
 - ☐ Rock
 - ☐ Heavy metal
 - ☐ R&B/Oldies
 - ☐ Funk/Soul
 - ☐ Technopop/Dance music
 - ☐ Experimental/Industrial
 - ☐ International/World beat
 - ☐ Rap/Hip hop
 - ☐ Country/Folk
 - ☐ Gospel/Religious
 - ☐ New age
 - ☐ Other: _____
5. Why are you involved in musical electronics? (Check all answers that apply.)
 - ☐ To make my job easier
 - ☐ To stay competitive
 - ☐ Interested in the technology
 - ☐ New way to use my computer
 - ☐ To make contemporary sounds
 - ☐ Other: _____
6. How are you using your MIDI equipment? (Check all answers that apply.)
 - ☐ Live performance
 - ☐ In a professional studio
 - ☐ In a home studio
 - ☐ For recreation/fun
 - ☐ For songwriting/scoring
 - ☐ For making records/CDs
 - ☐ Don't own MIDI equipment yet
 - ☐ Other: _____
7. What is your level of music interest?
 - ☐ Full-time/professional musician
 - ☐ Part-time/professional musician
 - ☐ Amateur or hobbyist musician
 - ☐ Potential musician
 - ☐ Music student
 - ☐ Other: _____

8. What is the ONE best description of your recording involvement?
 - ☐ Professional recording engineer or producer
 - ☐ Part-time professional engineer or producer
 - ☐ Sometimes or always record in a professional studio
 - ☐ Record in my home studio only
 - ☐ Have not recorded yet
9. Approximately how much time EACH WEEK do you dedicate to music? _____ hours
10. What percentage of the above time involves the use of computers? _____ %
11. Have you made a demo tape or record in the past year?
 - ☐ Yes ☐ No
12. Do you perform your music live?
 - ☐ Yes ☐ No

ABOUT ELECTRONIC MUSICIAN:

13. How did you first learn of *Electronic Musician*?
 - ☐ Through another publication
 - ☐ Through a friend
 - ☐ At a store or newsstand
 - ☐ Received offer in mail
 - ☐ Other: _____
14. Are you currently a subscriber to *Electronic Musician*?
 - ☐ Yes ☐ No
15. If not, where did you get this copy?
 - ☐ Through a friend
 - ☐ Newsstand
 - ☐ Bookstore
 - ☐ Record store
 - ☐ Retail music/pro audio store
 - ☐ Retail computer store
 - ☐ Other: _____
16. How many people, besides yourself, read your copy of *Electronic Musician*?
 - ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ more than 5
17. How much time, in total, do you spend reading or referring to a typical issue of *Electronic Musician*?
 - ☐ Less than one hour
 - ☐ 1 hour
 - ☐ 2 hours
 - ☐ 3 hours
 - ☐ 4 hours
 - ☐ 5 hours or more
18. When you are finished reading your copy of *EM*, what do you do with it?
 - ☐ Keep whole issue indefinitely
 - ☐ Cut out certain articles to save, throw away rest of issue
 - ☐ Pass issue on to a friend
 - ☐ Donate to a library or other institution
 - ☐ Throw issue away

EM READER SURVEY

19a. How important are the following topics to your reading of *Electronic Musician*?

	Very Important	Somewhat Important	Slightly Important	Not at all
Theory articles	☐	☐	☐	☐
General applications articles	☐	☐	☐	☐
Product reviews	☐	☐	☐	☐
Do-it-yourself projects/modifications	☐	☐	☐	☐
Articles on evaluating/choosing equipment	☐	☐	☐	☐
Recording techniques	☐	☐	☐	☐
Performance techniques	☐	☐	☐	☐
Program listings/DIY software	☐	☐	☐	☐
Interviews with behind-the-scenes people	☐	☐	☐	☐
Interviews with prominent personalities	☐	☐	☐	☐
Education-related topics	☐	☐	☐	☐
Current events/industry news	☐	☐	☐	☐
"What's New"	☐	☐	☐	☐
Record reviews	☐	☐	☐	☐

19b. Are there any topics not currently covered editorially in *Electronic Musician* that you would like to see covered?

20. On the list of magazines below, please indicate your involvement:

	Subscribe	Read Often	Read Occasionally	Never Read
<i>Aftertouch</i>	☐	☐	☐	☐
<i>Billboard</i>	☐	☐	☐	☐
<i>Byte</i>	☐	☐	☐	☐
<i>Computer Music Journal</i>	☐	☐	☐	☐
<i>Gig</i>	☐	☐	☐	☐
<i>Guitar Player</i>	☐	☐	☐	☐
<i>Guitar World</i>	☐	☐	☐	☐
<i>Home & Studio Recording</i>	☐	☐	☐	☐
<i>Keyboard</i>	☐	☐	☐	☐
<i>Mix</i>	☐	☐	☐	☐
<i>Modern Drummer</i>	☐	☐	☐	☐
<i>Musician</i>	☐	☐	☐	☐
<i>Music Technology</i>	☐	☐	☐	☐
<i>Pro Sound News</i>	☐	☐	☐	☐
<i>Recording Engineer/Producer</i>	☐	☐	☐	☐
<i>Roland User's Group</i>	☐	☐	☐	☐

ABOUT YOU AND YOUR BUYING HABITS:

21a. Which personal computers do you own? (Check all answers that apply.)

- ☐ Apple II series ☐ Apple Macintosh ☐ Atari ST ☐ Commodore Amiga
☐ Commodore 64/128 ☐ IBM PC/XT/AT or compatible ☐ Do not own one ☐ Other: _____

21b. Which ONE of the above computers is your main computer used for music?

- ☐ Apple II series ☐ Apple Macintosh ☐ Atari ST ☐ Commodore Amiga
☐ Commodore 64/128 ☐ IBM PC/XT/AT or compatible ☐ Do not use one ☐ Other: _____

22. Which of the following computer accessories do you currently own?

- ☐ Color monitor ☐ Dot matrix printer ☐ Hard drive ☐ Laser printer
☐ MIDI interface ☐ Modem ☐ Optical disk drive ☐ None of these

23. Do you subscribe to any of the following telecommunications networks? If so, which ones?

- ☐ PAN ☐ Esi Street ☐ CompuServe ☐ Source ☐ Do not subscribe ☐ Other: _____

24. Do you plan to subscribe to any of the following telecommunications networks?

- ☐ PAN ☐ Esi Street ☐ CompuServe ☐ Source ☐ Do not plan to subscribe ☐ Other: _____

EM READER SURVEY

25. Which of the following music-related products do you currently own, did you buy within the last 12 months, or do you plan to purchase?

	Currently Own	Number Owned	Bought Last Year	Plan to Purchase
Electronic pianos/organs	☐	_____	☐	☐
Keyboard synthesizers	☐	_____	☐	☐
Synthesizer expander modules	☐	_____	☐	☐
Guitar synthesizers	☐	_____	☐	☐
Drum machines	☐	_____	☐	☐
Electronic percussion kits	☐	_____	☐	☐
Acoustic pianos	☐	_____	☐	☐
Samplers	☐	_____	☐	☐
Electric guitars	☐	_____	☐	☐
Electric basses	☐	_____	☐	☐
MIDI wind controllers	☐	_____	☐	☐
Other electronic instruments	☐	_____	☐	☐

26. Which of the following music software products do you currently own, did you buy within the last 12 months, or do you plan to purchase?

	Currently Own	Bought Last Year	Plan to Purchase
Sequencers	☐	☐	☐
Patch librarian/editors	☐	☐	☐
Sample editors	☐	☐	☐
Transcription/scoring	☐	☐	☐
Educational	☐	☐	☐
Other: _____	☐	☐	☐

27. Which of the following audio and recording products do you currently own or plan to purchase within the next 12 months?

	Currently Own	Plan to Purchase
Stereo reel-to-reel recorders.	☐	☐
Multitrack reel-to-reel recorders.	☐	☐
Stereo cassette recorders.	☐	☐
Multitrack cassette recorders.	☐	☐
Mixers (8 tracks or under).	☐	☐
Mixers (over 8 tracks).	☐	☐
Instrument amplifiers.	☐	☐
Studio monitors/speakers.	☐	☐
Performance monitors/speakers.	☐	☐
Hardware sequencers.	☐	☐
Reverb/delay units.	☐	☐
Other signal processors.	☐	☐
Microphones.	☐	☐
Headphones.	☐	☐
Instrument cases/stands.	☐	☐
DAT recorders.	☐	☐
VCRs.	☐	☐
CD players.	☐	☐

28. How many of each of the following do you buy each month?

☐ Phonograph records ☐ Floppy disks ☐ Pre-recorded tapes ☐ Blank cassette tapes
☐ Reel-to-reel tapes ☐ Blank video tapes ☐ DAT tapes ☐ CDs

29a. What is your annual household income?

☐ Under \$15,000 ☐ \$15,000-\$25,000 ☐ \$25,000-\$35,000 ☐ \$35,000-\$50,000 ☐ Over \$50,000

29b. Approximately what is your annual expenditure for electronic music-related products?

☐ Under \$100 ☐ \$100-\$500 ☐ \$500-\$1,000 ☐ \$1,000-\$5,000 ☐ \$5,000-\$10,000 ☐ Over \$10,000

30. Where do you usually buy your electronic music products?

☐ Department store ☐ Computer store ☐ Music store ☐ Mail order ☐ Other: _____

EM READER SURVEY—PLEASE FILL OUT AND SEND.

31. When you are buying products, how much is your purchase influenced by the following factors?

	Very Much	Somewhat	Slightly	Not at all
Reviews of products in <i>Electronic Musician</i>	☐	☐	☐	☐
Advice from product retailers.....	☐	☐	☐	☐
Advice and recommendations from friends.....	☐	☐	☐	☐
Product advertising in <i>Electronic Musician</i>	☐	☐	☐	☐
Product literature from manufacturer.....	☐	☐	☐	☐
Editorial in other publications.....	☐	☐	☐	☐
Advertising in other publications.....	☐	☐	☐	☐

32. Which of the following factors influences your decision to buy a PARTICULAR BRAND of product?

	Very Much	Somewhat	Slightly	Not at all
Compatibility with instruments owned.....	☐	☐	☐	☐
Ease of use.....	☐	☐	☐	☐
Retailer support and service.....	☐	☐	☐	☐
Flexibility to upgrade.....	☐	☐	☐	☐
Price.....	☐	☐	☐	☐
Manufacturer's reputation.....	☐	☐	☐	☐
Documentation/training materials.....	☐	☐	☐	☐
Ease of maintenance/service.....	☐	☐	☐	☐
Previous experience with brand.....	☐	☐	☐	☐
Product's reputation for reliability.....	☐	☐	☐	☐
Ability of product to perform specific tasks.....	☐	☐	☐	☐
Artist's/professional's endorsement.....	☐	☐	☐	☐

33a. Do you read the advertisements in *Electronic Musician*?

☐ Yes ☐ No

33b. Have you ever used the EM Reader Service cards?

☐ Yes ☐ No

34. Have you ever taken any of the following actions as a DIRECT result of reading a product advertisement in *Electronic Musician*?

☐ Phoned the manufacturer ☐ Written to the manufacturer for information
☐ Visited retailer to demo product ☐ Purchased product

35. Do you have any other comments about *Electronic Musician*?

Thank you very much for supplying us with this information.

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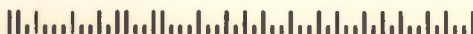
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WBH

will wake up if it receives any button presses or MIDI commands. Note that all signals are disconnected when the power is off.

To patch an effect to one of the eight channels, simply push the select button for the desired effect station, cycling through the eight insert channels until the display shows the desired channel number. You cannot patch more than one insert channel to the same effect, but you can patch several effects to one channel simply by setting them all to the same channel number. The unit will remember the order that you patched the effects and will link them in series, in that order.

I could go into more detail, but space is short. I like everything about the Map 8x4 except for one thing: I wish it had more effects sends. With stereo effects as common as they are now, you can use up four effects returns easily. The Map 8x4 has no simple way to expand its effect patching, so duplicate units couldn't share the same rack of effects. This limits the Map 8x4 to fairly small setups.

ARE THEY FOR YOU?

Each of these four MIDI products is functionally quite simple, and as a system they work well together and can do some unique tricks. The automation abilities would be welcome in a small home studio, although I think many studios would grow out of this system fairly quickly. I feel these products will be most valuable to performing MIDI guitarists and keyboardists with small setups. Simplicity can be a blessing in concert, and all of these components have a heavy, "gorilla-proof" feel to them that suggests road-worthiness.

A system like this is a bit tricky to evaluate solely on the merits of each component because it's the combination that makes things interesting. While the Manager may seem oversimplified, it really shines with the EAC-8; the MIDI Director doesn't do much, but it's very convenient to have around; and while the Map 8x4 is too small for most studio environments, it can be a real blessing in performance. I wouldn't hesitate to recommend this system to those needing to automate their rigs.

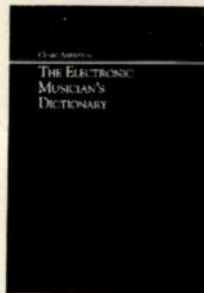
Robert Rich composes electronic music. He likes strange noises, especially wet ones. His latest album, *Rainforest*, has just been released on *Hearts of Space Records*.

New From EM Bookshelf



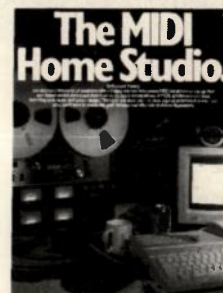
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by Howard Massey
Everything you need to know about setting up your own MIDI studio at home. It takes you through a step-by-step explanation of how MIDI works, how it can be utilized in a home studio environment, what the different components of the home studio are, and how they can be hooked up together and synchronized. Includes over fifty easy-to-follow illustrations.
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E-mu Proteus

Rumored to be the MIDI expander to end all MIDI expanders, E-mu's latest creation offers tremendous music-making power in a small, classy package.

By Bob O'Donnell



I can't believe all the buzz that built up around it."

"I heard they ran into some big problems in the development stage."

"Moments of inspired brilliance are scattered throughout, and even though there are a few disappointments, the overall impression is very positive. I'll bet it's going to be a huge hit."

Well, so much for the *Batman* review...As for Proteus, actually all the previous statements really do apply rather nicely. And there are other similarities: the very well-done demo song built into Proteus will probably generate as much excitement as the early *Batman* previews did, and Proteus's sounds are as clean and well-crafted as the movie's visuals. The only thing I haven't figured out is how Jack Nicholson fits into the Proteus story, but give me a few days, and I'll make this analogy work...

As with the movie, expectations were riding high; I've been looking forward to writing this review for a long time. What electronic musician wouldn't get excited about a 32-voice, sample-playback unit/synthesizer that sounds like a \$13,000 sampler and offers four megabytes of 16-bit samples burned into ROM, six individual outputs, and the ability to play up to sixteen timbres at once, all in a single rack space and all for under \$1,000? Granted, Proteus doesn't offer a new synthesis method or any

other conceptual breakthroughs, but when you put that much instrument into that small and affordable a package, it makes you pause and realize how far this industry has really come. I, for one, am impressed.

U'VE GOT THE LOOK

In addition to putting together a product that looks good on a spec sheet, E-mu obviously spent some time with the packaging and presentation of Proteus. Not only does it have a cool name and a stylish look, it comes with a well-written and illustrated manual that includes a lot of good reference information.

The lightweight, gray plastic box features a very clean design. The back panel includes six audio outputs, MIDI in/out/thru jacks, and a space for the detachable power cord. The slick-looking, grated front panel has four round, E-II-style buttons, two rotary dials, a 32-character backlit display, a power switch, and a very handy MIDI Activity LED (which verifies that MIDI info is reaching Proteus). One of the two main outputs also functions as a stereo headphone jack, although it isn't marked as such on the panel or mentioned in the manual.

Thankfully, the beautiful form does not come at the expense of function; Proteus's operation is very straightforward. You simply hit one of the two main

buttons (Master or Edit), dial in a page, cursor to the parameter you want, adjust the value, hit the appropriate button a second time to leave the edit mode, and you're done. That's the extent of the learning curve on Proteus.

Nice touches abound throughout the instrument. For example, the Data dial is detented, making it much easier to dial in the value that you want, and the unit's power supply will automatically reconfigure itself to work with either 110- or 220-volt systems. You don't have to worry about flipping some tiny switch if you travel with it overseas. Also, when you're playing a particular patch and switch to another sound, the previous one continues to play; it doesn't get abruptly cut off.

THE MIXMOBILE

Four of the six outputs (which are configured as two pairs of sub-outputs) have stereo jacks for use as effects send/returns. Thus, any signal input into them will be automatically mixed internally and sent to the master stereo outputs. So, for example, if you're suffering from the common affliction of "mixer input not-enoughitis," you can use stereo Y-cables in conjunction with dedicated outboard effects devices and create two stereo effects loops. This lets you individually process up to four sounds and only use two mixer inputs.

Unlike a real mixer, however, you can't adjust the level of the incoming signals—you'll have to adjust that and the mix level between the wet and dry sound at the effects unit—but hey, we're talkin' a thousand dollars here. The manual suggests that you can even use these "inputs" with the appropriate cables to submix audio from other sources if you're really stuck. For smaller MIDI studios (and even for bigger ones), these send/return loops are a godsend, not to mention particularly clever on an instrument that doesn't of-

fer built-in effects. Bravo, E-mu. Of course, if you don't want to use these features, just connect normal cables, and you can have up to six independent outputs or three sets of stereo pairs.

COMING SOON: THE SEQUEL

As if all this isn't enough, Proteus is also expandable. E-mu has announced that additional ROM boards will be available around the end of the year that double the onboard sample capacity to eight megabytes. The boards, which are expected to cost about \$500, will also be available in the form of other Proteuses (or is that Protei?). The unit currently available is officially called Proteus/1, with sequentially numbered units, featuring different onboard samples, to follow. According to E-mu's master plan, each instrument will be available both as a base unit and an expansion board, so you'll be able to mix and match the blocks of samples you want. Again, good (if not great) stuff.

HOLY BATSAMPLES...

Speaking of samples, let's talk about the Proteus voice structure. Basically, you have 125 instruments (i.e., waveforms) to choose from, and each patch can have up to two split or layered instruments. As expected, using two instruments instead of a single instrument

Product Summary

PRODUCT:

Proteus/1

TYPE:

Digital synthesizer/sample-playback expander

FEATURES:

32 voices, 4 megabytes of 16-bit samples, six audio outputs, 192 presets, MIDIPatch modulation routings

PRICE:

\$995

MANUFACTURER:

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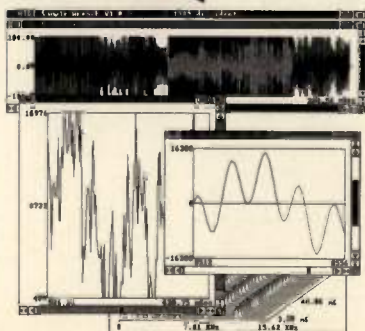
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● E-MU PROTEUS

halves the available number of voices.

For even more splits and bigger layers, up to four patches can be linked together as a single sound. One multipurpose auxiliary envelope, two LFOs, and two alternate amplitude envelopes are also available per patch. If you work with nonstandard tunings, each patch can be assigned one of six different tunings—Equal, Just C, Valotti, 19-tone, Gamelan (combining 5-tone Slendro and 7-tone Pelog), and a global, user-defined tuning that offers a slightly coarse resolution of 1.56 cents.

The 125 instruments include 69 sampled sounds (many of which are multisampled and a few of which feature stereo samples), 22 harmonic waveforms, 21 single-cycle waveforms, and 13 multicycle waveforms. Actually, 125 is a slightly misleading number because each instrument also includes a default amplitude envelope. While this helps keep the instrument's impressive signal-to-noise ratio (in excess of 90 dB) as high as it is, it also means that the same samples with different default envelopes are considered as different instruments. In addition, many of the shorter wave-

forms are quick snippets from the longer waveforms. Consequently, the real number of different instruments is closer to 50, but included in that is the drum and percussion instrument, which has 30 distinct samples.

JOKERS IN THE DECK

In general, the quality of the samples is outstanding; almost without exception they are incredibly clean and unbelievably quiet. Unfortunately, though, a few have questionable loops—one vocal sample has enough of a tremolo-like effect that I went to turn off the modulation routing, only to find that there wasn't one on—and a few are on the short side (the brass in particular). Basically, I think E-mu tried to squeeze a few more sounds into ROM than it should have. I would have preferred a smaller number of longer samples.

Another problem with some of the multisampled instruments (one which seems to be shared by just about all sample-based devices) is that there are several noticeable multisample points. The piano, strings, various saxophones, and other samples may cause problems

if you play slow solo lines through these junctions, so you need to be aware of them. In fairness, E-mu has done as good a job as its competitors, but I had hoped that Proteus would be better.

While I'm on the subject of limitations, the Proteus voice also lacks dynamic panning and filtering. You can use key number and velocity to modulate the initial stereo position of a sound and to affect a modulation destination simply labeled Tone (basically a static digital filter), but these changes cannot occur over time.

On the other hand, the Proteus's MIDIPatch modulation system permits you to adjust up to eight destinations (including crossfade)...in real time...via four MIDI continuous controllers, three MIDI switches, mono or polyphonic pressure, and all the various envelopes and LFOs, and an additional six destinations can be modulated by key position and velocity. Essentially, you have a multitude of modulation options with which to create movement in your sounds.

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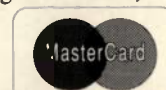
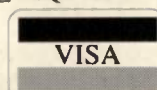
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192 sounds, only 64 of which are in RAM; the other 128 are burned into ROM. Included in that group are numerous acoustic piano, organ, string, choir, brass, wind, guitar, bass, percussion, tuned percussion, and synth sounds, with a few sound effects thrown in for good measure. All in all, it's a tremendous variety. The tuned percussion, guitar, drum, and surprisingly large number of synth sounds deserve particular mention, but to be honest, there isn't a bad or unusable sound in the bunch.

Some of Proteus's samples may sound familiar to owners of other E-mu instruments, because many come from the E-mu library—the grand piano, arco strings, and Strat, to name a few. The great thing about Proteus is that they sound like they're emanating from an E-III; the sound really is that clean. In fact, I'd go so far as to say that Proteus is generally one of, if not *the* cleanest, clearest, quietest electronic instruments I've ever heard. Weird loop points and bad multisamples occasionally interfere, and I did notice a little grunge in the lower octaves of the grand piano, but overall, this instrument sounds tremendous.

PROTEUS'S SECRET IDENTITY

Sure, it's a great musical instrument with fine individual sounds, but Proteus leads a double life as a multitimbral sound box just waiting to be hooked up to a sequencer. Creating a multitimbral setup requires a simple twist of the data dial to choose different sounds for different MIDI channels—that's all there is to it. I only wish there was memory allocated to save these setups.

Thirty-two dynamically allocated voices really come in handy when you start doing sequences; you can produce an amazing amount of music with Proteus alone. Of course you'll eventually run out of voices (I managed to get drums, bass, acoustic guitar arpeggiations, a synth pad, and a solo sax line going at once), and be forewarned that when you do, you'll run into a voice stealing algorithm that managed to occasionally steal just one side of the stereo sax patch I was using (and boy, was *that* weird). It's a relatively easy phenomenon to avoid, to be sure, but it would've been nice to have an option that guarantees a certain number of voices to a given part.

THE END

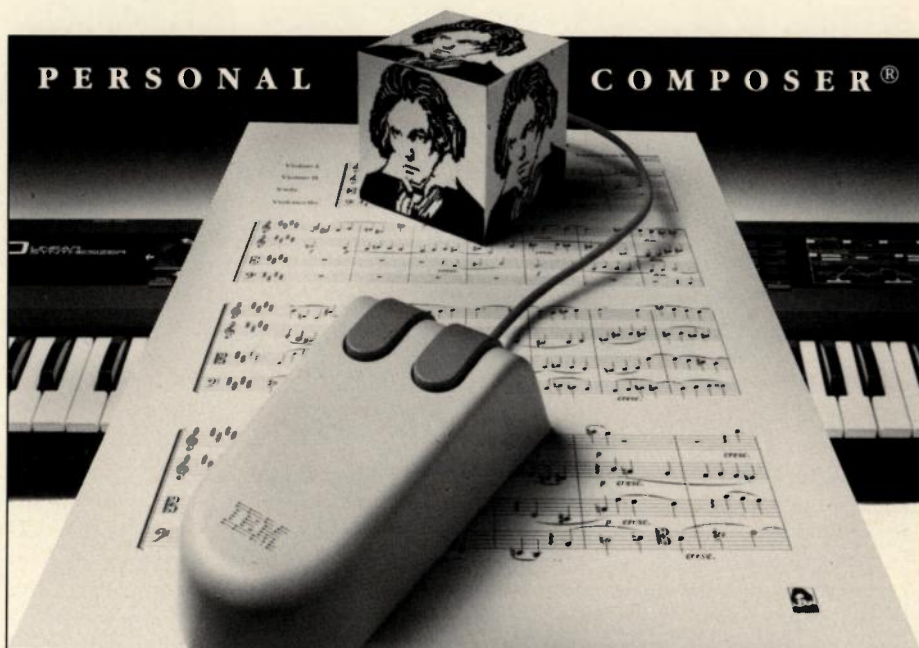
I've always wanted to write a review conclusion that said, "It's great. Buy one."

With Proteus, the temptation to do so is strong, but let's be objective: if you're looking for absolute, sonic perfection, with all the richness and depth of acoustic instruments, Proteus may not meet your needs. However, if you're looking for a excellent-sounding, affordable MIDI expander with lots of voices, a ton of great presets, a well-thought out design, a good manual, and lots of raw synthesis power to create sounds of your own, you can't go wrong with Proteus. It would make an excellent addition to just about any size MIDI studio, regardless of

the type of music you may be interested in playing or composing.

It's not perfect—nothing is—but Proteus has so many endearing qualities and features that I think you'll find yourself willing to overlook its few limitations. I'd call it the Batman of sample-playback synthesizers.

EM associate editor **Bob O'Donnell** graduated from the University of Notre Dame, where he studied the Great Books and played trombone in the marching, concert, and jazz bands.



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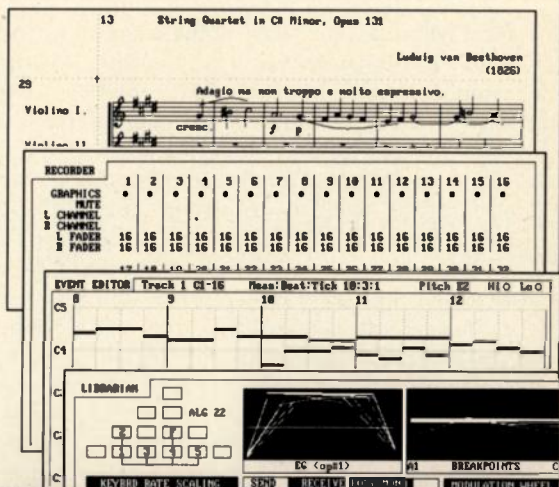
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Roland GR-50 Guitar Synthesizer System

Perfection still eludes the MIDI guitarist, but Roland's latest offering increases convenience while lowering tracking delay—and you don't need a special controller.

By Mike Sharp and Bill Burns



C.R. KING

The guitar synthesizer's potential has fascinated guitarists for many years, yet it's only been recently that the performance of these systems has started to match that potential. The modern breed of machines still requires careful setup and must be played with special attention to technique, but unlike earlier machines, they are fairly easy to play and give the guitarist access to a wide range of new and interesting possibilities for creative expression.

One of the most important features for a guitarist is the "feel" of the instrument, and for many who have become familiar and comfortable with their axes, it can be very difficult to adapt to the dedicated guitar-like controllers (some of which use all the same gauge string) that some synth systems use. With this in mind, Roland—one of the early pioneers of guitar synthesis—has designed the GR-50, which works with most standard metal-string guitars and also includes a dedicated sound module for easy setup.

SYSTEM BASICS AND INSTALLATION

The GK-2 pickup, which produces a separate output for each string and feeds into a small guitar-mounted con-

trol box, supplies the interface between guitar and sound module. (Roland G series guitars and guitars using the older GK-1 hex pickup are compatible with the GR-50 sound module via the BC-13 adapter, available separately.)

Mounting the GK-2 onto your favorite axe is not difficult and requires only simple hand tools, but if drilling holes in a vintage axe fills you with dread, find a competent guitar technician to do the job. Roland provides a slew of double-sided adhesive tapes and special spacers for temporary test-mounting of the pickup and control unit; once you are sure the pickup is in the right place (by testing the system with the GR-50), you're ready to drill three holes in the guitar's body and screw everything down, using springs under the pickup for later adjustment. We installed the GK-2 on an SG and on a Strat-style guitar with little trouble.

The GK-2 control unit, which connects to the hex pickup via a short cable, is a small, stylish, plastic box housing various controls and connectors. The unit provides an input for the direct signal from the guitar's own pickup and a switch to select guitar only, synth only, or a mix of the two. The controls com-

prise a small volume knob to adjust the level of the guitar synth sounds, and two programmable buttons that control various synth section parameters. A special multiconductor cable with locking DIN plugs carries the signal from the guitar to the single rack space GR-50 sound module. This may represent a possible source of trouble, for although the 16-foot-long cable is very flexible, the GK-2's locking connectors and plastic body might be damaged if the cable were pulled accidentally.

THE SOUND MODULE

The 1U rack-mount GR-50 contains not just a guitar-to-MIDI interface, but also a D-110 synthesizer section—no external sound modules are necessary, although if desired you can drive these via MIDI (more on this later). Using a dedicated sound module, aside from being convenient, also cuts down on the delay caused by analyzing the string and converting it to MIDI data because the guitar interfaces directly with the synth and bypasses MIDI altogether. Roland claims delays of eight milliseconds or less, roughly a third of the worst-case delay suffered by pitch-to-MIDI-based guitar synths. (Note that there is a delay

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● ROLAND GR-50

when driving external units via MIDI, the severity of which depends on the pitch of the note being analyzed; it's very noticeable on low strings, but not much of an issue in the upper registers.)

The GR-50's rear panel sports the usual MIDI in, out, and thru connectors; stereo/mono synth audio outs; a separate jack for the guitar direct signal; and a connector for the optional FC-100 foot controller. The front panel has the locking DIN connector, a standard 1/4-inch headphone jack, and a 16-character by 2-line illuminated display. Eighteen buttons access the unit's various functions and allow user programming similar to the D-110. A memory card slot and power switch complete the picture.

PLAYTIME!

So how about playing the GR-50? First of all, the unit provides a tuning function with a variable reference (A=440 to 459.9). It works with nonstandard tuning, too. Another function sets the sensitivity of each string to match your guitar and playing style. The two buttons on the GK-2 can control modulation on/



The GK-2 pickup and control unit mounted on a Strat-style guitar.

off, sostenuto, octave shift, or patch change, in either latching or momentary modes.

Patch changes can be programmed into one or more chains of up to 25 patches, with five of these chains held in memory. The switches on the guitar-mounted GK-2 can then switch patches up and down. Pressing both switches takes you to the first patch in the chain.

The GR-50's sounds have been de-

signed specifically for guitar use, so you can start playing right away without doing any programming. The internal sounds are on the whole excellent. There are several lifelike acoustic guitar sounds (steel, nylon, stereo, 12-string, Nashville-tuned, etc.). I (Mike Sharp) loved the various bass guitar sounds (slide bass, Fender-style thumb bass, solid rock bass, and so on); I have a conventional bass guitar, but being able to play bass on my regular electric where I have better chops was a whole new experience, and let me play better bass lines. Also, being able to use a guitar to load parts into my sequencer gave a far better feel; I find it much easier than using a keyboard.

Several useful string presets are included, as are the usual complement of synth brass and stabs, and there are also a few interesting overdrive lead guitar sounds, some of which include very effective harmonic sustains. While guitar sounds are, of course, obtainable with your regular guitar and an amp, the differences in sound and technique can quickly lead you into interesting new

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areas of expression. Going further afield, there are some excellent organ presets—try playing a B-3 or full church toccata and fugue from your Strat! Finally, the unit also includes a nice complement of modern “D series” Roland sounds and a complete set of percussion samples (playing drums from a guitar is really something).

A little “crunch” is evident if you listen hard to some of the presets, but that’s part of the sound’s character. The built-in programmable reverb unit is a welcome addition. While you may not like some of the GR-50’s built-in sounds, there’s enough selection so you’ll be taken care of until you get involved with MIDI expanders and/or programming your own sounds. Another good point is that the GR-50 accepts sound cards for the D-10/20/110, which immediately opens up a large sound library (although loading sounds from cards can be a little complicated).

The GR-50 allows two “tones” (sounds) to be assigned to each string. Further, there are several velocity-sensitive parameters (i.e., governed by how hard you play), including mix, crossfade, and switch. Crossfade lets you, for example, play a sound that fades to an octave up as you pick harder. You can

Product Summary

PRODUCT:

GR-50 and GK-2 hex pickup

TYPE:

Guitar synthesizer, hex pickup, and guitar-to-MIDI converter

PRICES:

GR-50 \$1,549.50;

GK-2 \$199.95;

BC-13 adapter \$120

MAIN FEATURES:

Fast tracking when used with supplied sound module; compatibility with previous Roland guitar synths with BC-13 adapter

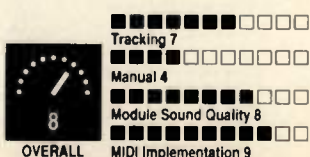
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● ROLAND GR-50

get another neat effect by programming velocity to switch between two different sounds, thus letting you play either sound just by varying your picking technique. Similar to keyboard split, the unit also lets you assign different sounds to each string or group of strings on the guitar, which, among other things, is great for playing two-part music. For example, I set up a bass on the bottom three strings and a flute on the top; within minutes I was playing a passable version of *Bourrée*.

As with all guitar synths you have to pay attention to your playing technique to avoid glitches from damped notes, unintentional hammer-ons, etc. Be aware that the GR-50 is limited in the number of notes it can play simultaneously (the limit is 32 partials—most patches use four partials per note); so if you set a nice bright acoustic guitar patch, then try to play Townshend-style strummed chords, the GR-50 will very quickly run out of partials to use for sustain, with unpredictable results.

However, don't forget that you can increase the apparent number of voices—and get some really great effects—by using the GR-50 to mix straight guitar with the synth sounds. You can set up excellent 12-string and other thickening patches, as well as, for example, adding a little touch of a string timbre to your rhythm chords, or bass doubling a lead for that really heavy sound.

THE MIDI CONNECTION

The GR-50 also presents a full complement of MIDI features. Both poly mode and mono mode (i.e., one MIDI channel to each string) are supported. If you have the channels available, mono mode best captures the nuances of playing individual string bends, chokes, and other effects. Mono also allows different sounds to be allocated to different strings if you use external sound module(s).

The GR-50 sends pitch bend messages—continuous or chromatic (discrete steps)—to indicate string bend. Being in poly mode and bending more than one string automatically selects chromatic mode. A branch structure enables the same signal to be sent with two different sets of MIDI parameters at the same time (for instance, poly mode data over MIDI channel 3, and mono mode data to 11 through 16). The GR-

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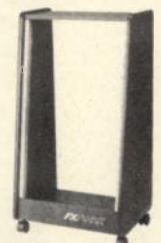
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50 may also be used as a MIDI-controlled multitimbral expander. Two timbres and the rhythm sounds may be used through three MIDI channels; if the guitar is used at the same time it will take precedence over the other timbres as necessary. I wish it would've offered eight different timbres like the D-110, but I guess you can't have everything.

DOCUMENTATION

We've talked about some of the things we like about the GR-50, so this is a good point to bring up our main gripe: the manual. Although most of the information you might want is there, it can be very difficult to find it, and then to understand exactly what to do. As with most of today's feature-packed synths and samplers, programming the unit is so complicated that writing a clear and concise manual is very difficult, though not impossible.

When we spoke to Roland, they agreed that the manual could be improved, and said they are working hard to that end. If you have trouble with Roland's manual, the best alternative is to get one of the after-market books on the D-110 (as far as we know, no one has a book on the GR-50 yet). This will give you a good background in programming and should make Roland's manual much easier to comprehend.

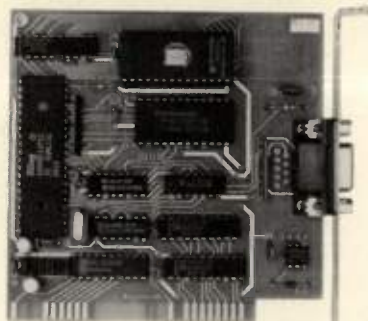
THE CONCLUSION

We could go on talking about the GR-50 at considerable length, but this should give you a good idea of its capabilities. The end result is the product of some hard choices, and some compromises. The unit is not perfect; wired-fret or ultrasonic units may track marginally better, for example, but it's hard to beat the GR-50's combination of built-in sounds, MIDI capabilities, and especially the ability to play from your own axe, switching and mixing straight guitar with the synth output. Roland's latest product opens the door to some very interesting possibilities for the creative guitarist.

Formerly technical director for the C.E. Hammond Group in England, Mike Sharp runs a medical electronics company by day but is more often found writing, playing, or "working" in the studio. Bill Burns is an electronics engineer. Formerly with the BBC in London, he now works as a computer systems designer and consultant.

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Hip Software's Harmony Grid

A newcomer to the Mac music marketplace offers a new—and entertaining—approach to music theory investigation.

By John Duesenberry

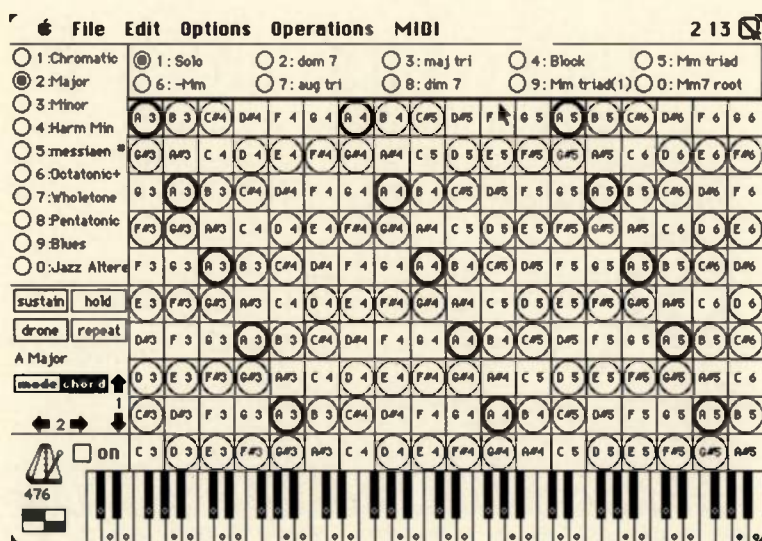


FIG. 1: A major scale on 2 x 1 grid.

As its authors describe it, *Harmony Grid* is a Macintosh "software instrument" targeted mainly at the musically untutored—"hobbyists and amateurs who never learned much harmony theory, who would learn it if it were more fun than practicing scales." While it doesn't fit easily into existing music software categories, the program has at least two functions: it's a tool/toy for the exploration of some aspects of harmony theory, and it's a vehicle for live improvisation.

The program presents musical elements—scales, modes, and chords—as visual patterns on a two-dimensional graph (the Harmony Grid) as well as on a conventional clavier. You can hear these elements by moving the mouse over the grid or the clavier, or by playing a MIDI keyboard. The tutorial value of this approach is that the grid shape of a given scale or chord type is uniform across all transpositions, as opposed to the multiplicity of spatial and fingering

patterns encountered on musical instruments.

VIEWING AND PLAYING SCALES

Should the user wish to use the program for purposes other than mouse-driven improvisation, then learning to read the grid is the first order of business. The grid can represent the interval pattern of a given scale or chord in many different ways; the exact way depends on how the user calibrates the X-Y axes of the grid in units of semitones. For example, suppose you want to play with a major scale. You first select this scale from the ten available types (you can also make up and save your own), then set up the grid. Fig. 1 shows an A major scale laid out on a 2 x 1 grid. Moving from left to right you read two semitones per square; moving from bottom to top you read one semitone per square. The members of the scale are circled, with the root of the scale heavily outlined. On the clavier, small circles appear on each key

that is a scale member, again with the roots marked by a dark circle.

The grid's visual pattern is the same in all octaves and remains the same if the scale root is changed. The main benefit of this particular grid is that it shows the whole-step/half-step makeup of the scale clearly in terms of location; therefore, it is useful for viewing most any traditional scale or mode.

Using a MIDI interface and synth, or the Mac's internal sound generator, you can play individual notes of the scale in real time by clicking on their squares or clavier keys, or traverse the scale by moving the mouse across the scale notes on the grid or clavier; you can also transpose the scale or select a new one. There are various other performance options, but the main point is that all such improvisation is restricted to the given scale's notes. Also, the performance can be rhythmically restricted to trigger only on ticks of the built-in metronome. This recalls the constrained drawing options of many paint programs, which allow artistic klutzes to draw perfectly straight lines. Now musical klutzes can improvise in a key without hitting any wrong notes or making timing errors.

VIEWING AND PLAYING CHORDS

This program isn't called Scale Grid, so where's the harmony? By clicking on one of a group of "buttons," you can select one of ten chord types (again, you can define and save your own). Suppose you select the minor chord type, while remaining within the A major scale. By moving the mouse across the grid or clavier, you find that Harmony Grid attempts to harmonize each scale degree by building and playing a partial or complete minor triad on that degree. If you trigger the notes B, C#, or F#, you hear the ii, iii, and vi chords of the scale; if you trigger an E, you hear only the fifth E-B. Why? The correct diatonic

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harmonization of the fifth degree of the major scale is a major triad; that is, one can't legally build a minor triad on E while using only the members of the A major scale—so Harmony Grid uses only the two notes that fit.

This does not imply that Harmony Grid has anything like a musician's knowledge base about diatonic harmonization (or any other sort). The software simply applies a filtering algorithm: given the selected pitch, build a chord of the current type, then filter out any chord members that are not also members of the current scale. This technique is applicable to any sort of pitch collection, not just to the privileged collections of traditional tonality. (Unfortunately, the pitch materials supplied with Harmony Grid and the way the program is described in the manual disguise the software's generality.)

Now, by choosing another grid setting, we can extract harmonic information about the scale. Fig. 2 shows A major again, this time laid out on a 4 x 3 grid. The interval from square to square, reading left-right, is four semitones (major third); from bottom to top, three semitones (minor third). This representation is obviously poor if we think of the scale linearly, but is somewhat richer from the standpoint of chord formation. Fig. 3 shows the grid when the minor

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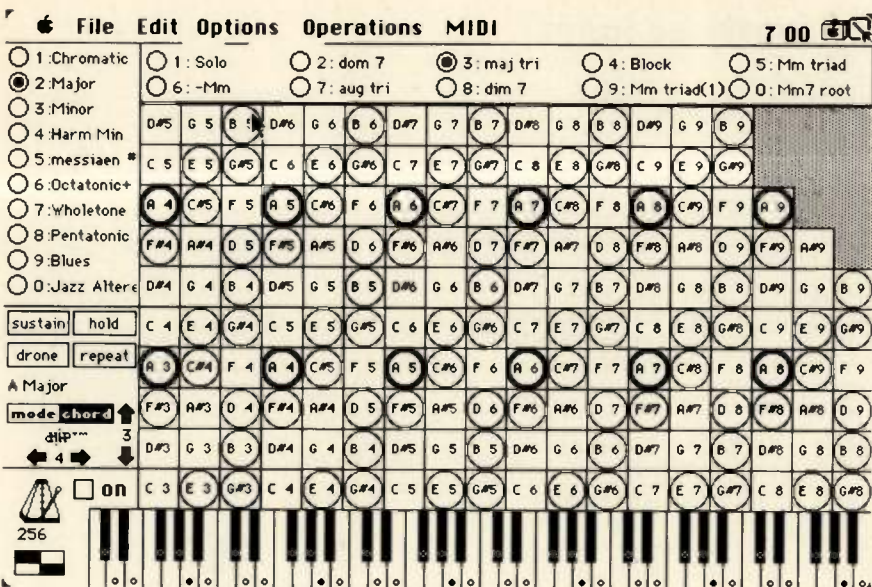


FIG. 2: A major scale on 4 x 3 grid.

triad is the selected chord type, and F# is triggered. The F# minor triad is outlined on the clavier and the grid.

Once you get used to reading the 4 x 3 grid, you can read the interval pattern (up one minor third, right one major third) fairly easily and can distinguish it from the patterns produced by other chord types. The 4 x 3 grid is useful for representing chords built on thirds (see Fig. 4), and is used extensively in the Harmony Grid manual and reference cards. This chord-building capability expands the possibilities for improvising. One might, for example, select a chord type

and trigger chords by playing single notes on a MIDI keyboard, while soloing on the grid with the mouse.

PERFORMANCE OPTIONS

The programmers have done a good job choosing the combinations of keyboard/mouse actions that control real-time performance parameters such as choice of scale/mode, chord type, transposition, metronome speed, sustain and articulation, and so on, even if (like many other music programs) some of the Mac interface guidelines are violated. Since the number of performance

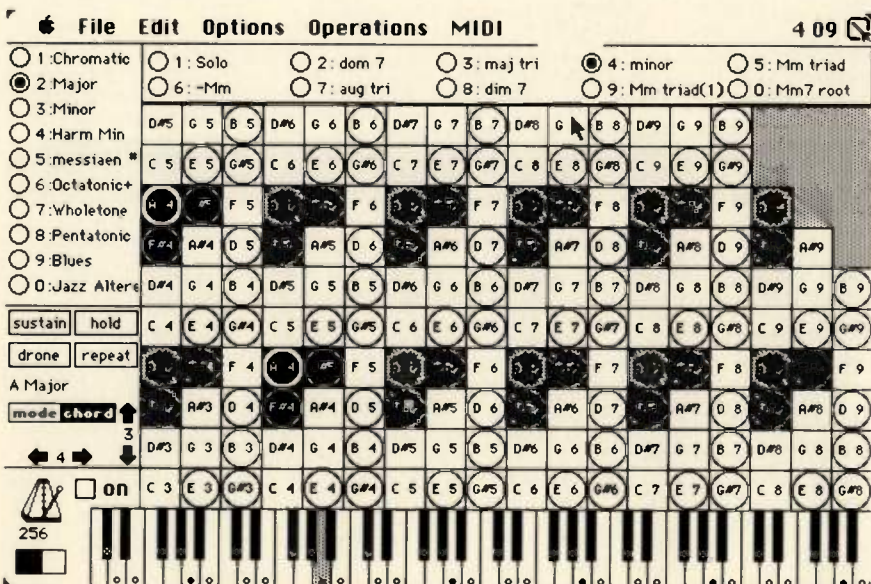


FIG. 3: Minor triad outlined on clavier and 4 x 3 grid.

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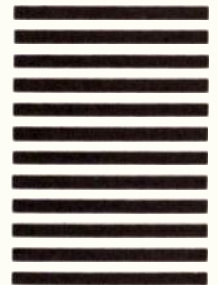
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• HARMONY GRID

options is limited, and they are summarized in a help window, the necessary keystrokes are quite easy to learn.

One sorely needed performance option is some way of capturing an im-

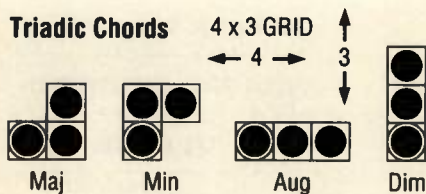


FIG. 4: The structure of some common chord types as displayed on a 4 x 3 grid. The interval between vertical squares is a minor third; the interval between horizontal squares is a major third.

provisation for later playback and editing. An audible metronome would also be nice; in my opinion, the current flash indicator by itself is of little use.

DOCUMENTATION

The main body of the manual consists of three short introductory/tutorial chapters, a reference manual, and a potpourri of various subjects entitled "Seeing Harmony Theory." The manual concludes with brief performing tips and summary material; there are also some reference cards concerning scales and performance options.

The manual seems to be the weakest part of the product. I have no problems with the reference section, but the opening tutorial—presumably aimed at users who are newcomers to both the program and to music theory in general—contains a great deal of undefined music vocabulary. A random scan of a couple of pages shows terms such as minor tonality, root, unstable pitch, harmonic suspense, and many others.

Some of this is cleared up in the chapter "Seeing Harmony Theory," but that comes about twenty pages too late. This chapter attempts to cover a huge amount of ground (everything from a history of pitch systems to Harmony Grid-specific topics) in 41 small pages, and the assumed reader's level of sophistication fluctuates wildly—from as basic as not knowing the difference between major and minor to as complex as following the derivation of 12-tone equal temperament.

Somebody really ought to take a hard look at this manual and decide for

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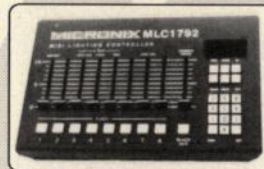


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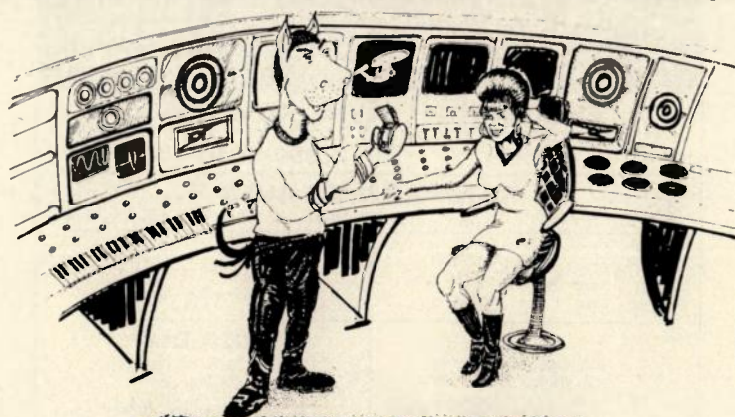
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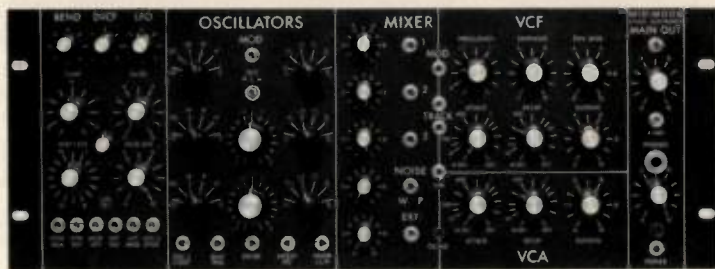
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whom it is intended. I think beginners will put the manual aside and go back to playing with the program, which is a lot more fun.

In the end, that's what I did, too. In fact, what I wound up doing with this program had nothing to do with harmony. Instead, I set up a percussion grid and loaded up my sampler with a variety of percussion sounds assigned to different keys. I then devised scales of percussion timbres and chords that would trigger different combinations of instruments. This produced some interesting results, especially if the metronome was set to absurdly high speeds. One other nice touch is that Version 1.5 added MIDI velocity control keys, which provide three user-settable accent levels to the various real-time controls.

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I feel Harmony Grid is good for exploring fuzzy, informal questions on the order of, "What does the X scale sound like?" in a nonrigorous way. It's hard to answer such questions verbally (try it!), yet with a tool like Harmony Grid, the unsophisticated user gets to try it out and have fun in the process. Its novel graphics approach will probably appeal more to untrained users than to those accustomed to traditional notation. Keep an eye on Hip Software; the people there don't think like other people in the business, and considering that Harmony Grid is their first commercial effort, I'm looking forward to what they come up with in the future.

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Drowning in a Sea of Packaging

As ecological considerations take center stage in the battle for the Earth's survival, musicians can—as with benefit concerts—set an example for the world to follow.

By Craig Anderton



I've been known to bemoan the lack of new ideas coming out of the record industry, but there are some notable exceptions. One case in point comes from Robert Simonds, an executive at Rykodisc, who wants more ecologically sound CD packages. According to an excellent piece on CD packaging by Pete Howard in the June 15 issue of *Rolling Stone* magazine, "Consumers presently pay about a dollar for the cardboard 'long boxes' that are taken home and immediately discarded. Besides being a waste of money, the packaging also looms as an environmental black eye." The blister packs (large, semisoft plastic packages that are difficult to open without scissors) for CDs are equally unsound from an environmental and cost standpoint, if not more so.

There are two reasons why the long boxes exist. One is to prevent shoplifting—it's hard to stick one of those under your shirt and walk out of the store. The other is that the long box presents an opportunity to include colorful graphics that catch the customer's eye, which is helpful in the competitive CD marketplace.

But luxuries such as fancy graphics may no longer be relevant in today's super-saturated, overpopulated, over-polluted planet. We need to collectively cut back on the ridiculous and excessive packaging that haunts almost all aspects of our lives. If you're too selfish to care about the environment, then consider your pocketbook: all that plastic and cardboard packaging you throw away costs *you* money.

Solutions abound. Simonds suggests placing the CDs in long, box-sized frames that, after the CD is purchased, would be recycled back into holding another CD. An alternative is already in place in several European countries; there, CD jewel cases are displayed with the inserts visible, but no CD inside. When you pay your money, you take the case to the clerk, who then gets the appropriate CD and puts it into the case. This eliminates the shoplifting problem and the excessive packaging, and seems like an idea worth pursuing in the American market.

Another possible solution is to just shrink-wrap the jewel case. Although this generates some garbage, it's less than a long box or blister pack. Or, use a cardboard slipcase for the CD, like the vinyl albums of yesteryear (remember those?). This would be used to store the CD, so you wouldn't throw it away. Granted the jewel box is cuter and sexier than a cardboard slipcase, but are we in a position where "cute and sexy" is more important than the environmental impact of excessive plastic?

The record industry is certainly not alone in creating excessive packaging. The software industry often wraps a simple disk and manual in a four-color, non-recyclable, plastic or cardboard package. I suppose this is done primarily because consumers resist the idea of shelling out a couple hundred dollars

for software and not getting a classy package, but again, I don't think we can afford this luxury any more. Or consider the plastic envelopes that held your synthesizer's manual when you unpacked it—is that envelope really necessary? Wouldn't you settle for a slightly wrinkled manual if it meant unleashing less garbage into the world?

We need to do something. Although many musicians have sold out their credibility to Madison Avenue, some musicians still command respect, and setting an example for a less polluted planet could have very positive results. Consumers need to turn their heads around, too, and realize that minimal packaging—not an overdose of plastic garbage—is what's *really* cool and hip.

We're far too swayed by form rather than content, and it's time for that to stop. A great software program shipped in a brown paper bag is far better than a mediocre program shipped in a fancy plastic package. People place way too much importance on image and not enough on substance; how else can you explain the fact that an actor who plays a cop, but doesn't have to deal with the messy reality of law enforcement, gets paid so much more than the police officers who genuinely put their lives on the line for us every day?

Anyway, back to CDs. Simonds suggests that if enough consumers call for the abolition of excessive packaging, it will happen; he recommends writing the various labels and letting them know what you think (just look on the CD insert for the address). Meanwhile, should you want to give Simonds your support, write him (preferably on recycled paper!) c/o Rykodisc, 200 North Third Ave., Minneapolis, MN 55401.



All Together Now.

If you're making music with MIDI and computers and are looking for a better way than ever to put it all together, look to Opcode. Our fully integrated system supports every aspect of music production from composing, recording and film scoring, to notation, and voice editing and organization. We understand what your needs are, because at Opcode we're musicians too.

Vision is the professionals' choice for sequencing. Grammy award winning composer and producer Jan Hammer says "If you compare all the really good sequencers on the Mac, Vision is more than the sum total of those sequencers." With the power of Vision, pros who haven't already, now have more reason than ever to switch from whatever program they've been using.

Keyboard Magazine called Opcode "the undisputed leader in **Editor/Librarian** technologies." Our programs support hundreds of synthesizers with innovative features and graphics. And with our new Desk Accessory Librarian, included free with every package, you can audition voices while Vision is playing! We've got programs for the E-mu Proteus, Korg M-1, and Kurzweil K1000 series, so you won't miss a beat while using today's hottest synths.

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a **Studio Plus Two** or **Professional Plus** MIDI interface and add SMPTE to your system later with the **Timecode Machine**.

Every system needs integration, and since we're the company who created MIDI Files, all our programs use the Standard MIDI File format, from **Editor/Librarians** and **CUE**, to **Vision**. And **Vision** also easily exchanges files directly with most popular notation programs, so you can print out your music or play back your written scores. We've even got our own notation program in the works.

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